Also A because "The company's data science team wants to query ingested data in near-real time" and kinesis data analytics is doing that. With option B we query the tada once is in the data lake.

upvoted 1 times

Dantecito 1 year, 4 months ago

Selected Answer: A

Option a for me. I will assume that the EC2 instance is responsible for placing the data into the correct database. Therefore, we just need to wait for the instance to reboot. Since we can store the data in Kinesis Data Streams for up to 7 days, we are covered in that regard.

For option B, we are forced to use Amazon Redshift, and the problem statement does not mention anything related to that. I would choose option A because it does not affect the original requirements, and it helps us avoid in-flight data loss as requested.

upvoted 1 times

wwwxxch 1 year, 5 months ago

Selected Answer: B

near-real time --> Kinesis Data Firehose
And retention day of Kinesis Data Streams cannot be longer than 365 days

upvoted 1 times

EllenLiu 1 year, 5 months ago

Selected Answer: A

A: focus on performing complex data processing without in-flight data lost, not mention data persistence
B: focus on data persist for later analysis

upvoted 1 times

LeonSauveterre 1 year, 6 months ago

Selected Answer: A

Instance store & ElasiCache are all temporary storages, which cannot address data loss. That rules out C & D.

B: Kinesis Data Firehose is optimized for batch processing rather than real-time querying. It can indeed deliver data to S3 or Redshift, but there's a good chance the delay between ingestion and query availability cannot meet the "near-real-time" requirement.

upvoted 1 times

Lin878 2 years ago

Selected Answer: B

https://aws.amazon.com/pm/kinesis/?gclid=CjwKCAjwvIWzBhAlEiwAHHWgvRQuJmBubZDnO2GasDWwc2iBapfVD6GBeIgj2JV6qkldm-K_CmMzmxoCdCwQAvD_BwE&trk=ee1218b7-7c10-4762-97df-274836a44566&sc_channel=ps&ef_id=CjwKCAjwvIWzBhAlEiwAHHWgvRQuJmBubZDnO2GasDWwc2iBapfVD6GBeIgj2JV6qkldm-K_CmMzmxoCdCwQAvD_BwE:G:s&s_kwcid=AL!4422!3!651510255264!p!!g!!kinesis%20stream!19836376690!149589222920

upvoted 2 times

ray320x 2 years, 4 months ago

Option A is actually correct. The question ask for minimal data loss and that query of data should be near real time, not the ingestion. Kinesis data analytics is near real time.

Recent changes to Redshift actually make B correct as well, but A is also correct.

upvoted 2 times

dkw2342 2 years, 3 months ago

Streaming ingestion provides low-latency, high-speed ingestion of stream data from Amazon Kinesis Data Streams and Amazon Managed Streaming for Apache Kafka into an Amazon Redshift provisioned or Amazon Redshift Serverless materialized view.[1]

Option B mentions Kinesis Data Firehose (now just Firehose), so this won't work.

Option A is the correct answer.

[1]https://docs.aws.amazon.com/redshift/latest/dg/materialized-view-streaming-ingestion.html

upvoted 2 times

farnamjam 2 years, 4 months ago

Selected Answer: A

Comparison to other options:

B. Kinesis Data Firehose with Redshift: While Redshift is scalable, it doesn't offer real-time querying capabilities. Data needs to be loaded into Redshif from Firehose, introducing latency.
C. EC2 instance store with Kinesis Data Firehose and S3: Storing data in an EC2 instance store is not persistent and data will be lost during reboots. EBS volumes are more appropriate for persistent storage, but the architecture becomes more complex.
D. EBS volume with ElastiCache and Redis: While ElastiCache offers fast in-memory storage, it's not designed for high-volume data ingestion like 1 MB/s. It might struggle with scalability and persistence.

upvoted 3 times

Firdous586 2 years, 5 months ago

I don't understand why people are giving wrong information
in the QUESTION its clearly mentioned near Real Time
Kinesis Data Streams is for Real time
Where are Kinesis Datafirehose is for Near real time there for answer is B only

upvoted 5 times

Marco_St 2 years, 6 months ago

Selected Answer: A

Read the question: near real-time querying of data.... it is more about real-time data query once the data is ingested, It does not mention how long time the data needs to be stored. A is better option. B introduces delay of data buffer before it can be queried in redshift

upvoted 1 times

practice_makes_perfect 2 years, 7 months ago

Selected Answer: B

A is not correct because Kinesis can only store data up to 1 year. The solution need to support querying ALL data instead of "recent" data.

upvoted 3 times

pentium75 2 years, 5 months ago

Says who? They want to "query ingested data in near-real time", it does not say anything about historical data.

upvoted 2 times

Ruffyit 2 years, 7 months ago

A: is the solution for the company's requirements. Publishing data to Amazon Kinesis Data Streams can support ingestion rates as high as 1 MB/s and provide real-time data processing. Kinesis Data Analytics can query the ingested data in real-time with low latency, and the solution can scale as needed to accommodate increases in ingestion rates or querying needs. This solution also ensures minimal data loss in the event of an EC2 instance reboot since Kinesis Data Streams has a persistent data store for up to 7 days by default.

upvoted 2 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: A

Publish data to Amazon Kinesis Data Streams, Use Kinesis Data Analytics to query the data

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: A

• Provide near-real-time data ingestion into Kinesis Data Streams with the ability to handle the 1 MB/s ingestion rate. Data would be stored redundantly across shards.
• Enable near-real-time querying of the data using Kinesis Data Analytics. SQL queries can be run directly against the Kinesis data stream.
• Minimize data loss since data is replicated across shards. If an EC2 instance is rebooted, the data stream is still accessible.
• Scale seamlessly to handle varying ingestion and query rates.

upvoted 4 times

Question #321 Topic 1

What should a solutions architect do to ensure that all objects uploaded to an Amazon S3 bucket are encrypted?

A. Update the bucket policy to deny if the PutObject does not have an s3:x-amz-acl header set.

B. Update the bucket policy to deny if the PutObject does not have an s3:x-amz-acl header set to private.

C. Update the bucket policy to deny if the PutObject does not have an aws:SecureTransport header set to true.

D. Update the bucket policy to deny if the PutObject does not have an x-amz-server-side-encryption header set. Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

bdp123 Highly Voted 2 years, 9 months ago

Selected Answer: D

https://aws.amazon.com/blogs/security/how-to-prevent-uploads-of-unencrypted-objects-to-amazon-s3/#:~:text=Solution%20overview

upvoted 16 times

Grace83 2 years, 8 months ago

Thank you!

upvoted 1 times

Sbbh Highly Voted 2 years, 8 months ago

Confusing question. It doesn't state clearly if the object needs to be encrypted at-rest or in-transit

upvoted 9 times

Guru4Cloud 2 years, 3 months ago

That's true

upvoted 2 times

awsgeek75 Most Recent 1 year, 11 months ago

Selected Answer: D

Related reading because (as of Jan 2023) S3 buckets have encryption enabled by default.
https://docs.aws.amazon.com/AmazonS3/latest/userguide/UsingServerSideEncryption.html

"If you require your data uploads to be encrypted using only Amazon S3 managed keys, you can use the following bucket policy. For example, the following bucket policy denies permissions to upload an object unless the request includes the x-amz-server-side-encryption header to request server-side encryption:"

upvoted 5 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: D

The x-amz-server-side-encryption header is used to specify the encryption method that should be used to encrypt objects uploaded to an Amazon S3 bucket. By updating the bucket policy to deny if the PutObject does not have this header set, the solutions architect can ensure that all objects uploaded to the bucket are encrypted.

upvoted 6 times

kruasan 2 years, 7 months ago

To encrypt an object at the time of upload, you need to add a header called x-amz-server-side-encryption to the request to tell S3 to encrypt the object using SSE-C, SSE-S3, or SSE-KMS. The following code example shows a Put request using SSE-S3.
https://aws.amazon.com/blogs/security/how-to-prevent-uploads-of-unencrypted-objects-to-amazon-s3/

upvoted 5 times

kruasan 2 years, 7 months ago

The other options would not enforce encryption:
A) Requiring an s3:x-amz-acl header does not mandate encryption. This header controls access permissions.

B) Requiring an s3:x-amz-acl header set to private also does not enforce encryption. It only enforces private access permissions.
C) Requiring an aws:SecureTransport header ensures uploads use SSL but does not specify that objects must be encrypted. Encryption is not required when using SSL transport.

upvoted 5 times

kruasan 2 years, 7 months ago

Selected Answer: D

To encrypt an object at the time of upload, you need to add a header called x-amz-server-side-encryption to the request to tell S3 to encrypt the object using SSE-C, SSE-S3, or SSE-KMS. The following code example shows a Put request using SSE-S3.
https://aws.amazon.com/blogs/security/how-to-prevent-uploads-of-unencrypted-objects-to-amazon-s3/

upvoted 2 times

Steve_4542636 2 years, 9 months ago

Selected Answer: D

I vote d

upvoted 2 times

LuckyAro 2 years, 9 months ago

Selected Answer: D

To ensure that all objects uploaded to an Amazon S3 bucket are encrypted, the solutions architect should update the bucket policy to deny any PutObject requests that do not have an x-amz-server-side-encryption header set. This will prevent any objects from being uploaded to the bucket unless they are encrypted using server-side encryption.

upvoted 4 times

jennyka76 2 years, 9 months ago

answer - D

upvoted 2 times

zTopic 2 years, 9 months ago

Selected Answer: D

Answer is D

upvoted 2 times

Neorem 2 years, 9 months ago

Selected Answer: D

https://docs.aws.amazon.com/AmazonS3/latest/userguide/amazon-s3-policy-keys.html

upvoted 2 times

Question #322 *Topic 1*

A solutions architect is designing a multi-tier application for a company. The application's users upload images from a mobile device. The application generates a thumbnail of each image and returns a message to the user to confirm that the image was uploaded successfully.

The thumbnail generation can take up to 60 seconds, but the company wants to provide a faster response time to its users to notify them that the original image was received. The solutions architect must design the application to asynchronously dispatch requests to the different application tiers.

What should the solutions architect do to meet these requirements?

A. Write a custom AWS Lambda function to generate the thumbnail and alert the user. Use the image upload process as an event source to invoke the Lambda function.

B. Create an AWS Step Functions workflow. Configure Step Functions to handle the orchestration between the application tiers and alert the user when thumbnail generation is complete.

C. Create an Amazon Simple Queue Service (Amazon SQS) message queue. As images are uploaded, place a message on the SQS queue for thumbnail generation. Alert the user through an application message that the image was received. **Most Voted**

D. Create Amazon Simple Notification Service (Amazon SNS) notification topics and subscriptions. Use one subscription with the application to generate the thumbnail after the image upload is complete. Use a second subscription to message the user's mobile app by way of a push notification after thumbnail generation is complete.

Correct Answer: C

Community vote distribution

C (94%) 6%

Comments

Steve_4542636 Highly Voted 2 years, 9 months ago

Selected Answer: C

I've noticed there are a lot of questions about decoupling services and SQS is almost always the answer.

upvoted 35 times

alain_maza 1 year ago

same that cloudfront

upvoted 3 times

Neha999 Highly Voted 2 years, 9 months ago

D

SNS fan out

upvoted 13 times

Manimgh Most Recent 1 year, 1 month ago

Selected Answer: C

C is correct -> take up to 60 seconds = SQS

upvoted 1 times

LoXoL 1 year, 10 months ago

They don't look like real answers from the official exam...

upvoted 1 times

awsgeek75 1 year, 10 months ago

Selected Answer: C

Each option is badly worded:
A: "generate the thumbnail and alert the user" doesn't sound sequential so could alert the user during, before or after the thumbnail generation whichever way you interpret it.
B: this is sequential and won't alert until the steps are complete
D: Could work without with the risk of notification loss so C is better but this is also ok
upvoted 2 times

awsgeek75 1 year, 11 months ago
Selected Answer: C
Safe answer is C but B is so badly worded that it can mean anything to confuse people. Step functions to use tiers. What if on of the step is to inform to the user and move on to next step. Anyway, I'll chose C for the exam as it is cleaner.
upvoted 3 times

wsdasdasdqwdaw 2 years, 1 month ago
... asynchronously dispatch ... => Amazon SQS
upvoted 8 times

TariqKipkemei 2 years, 2 months ago
Selected Answer: C
Asynchronous, Decoupling = Amazon Simple Queue Service
upvoted 6 times

Guru4Cloud 2 years, 3 months ago
Selected Answer: C
SQS is a fully managed message queuing service that can be used to decouple different parts of an application.
upvoted 3 times

Zox42 2 years, 8 months ago
Selected Answer: C
Answers B and D alert the user when thumbnail generation is complete. Answer C alerts the user through an application message that the image was received.
upvoted 5 times

Sbbh 2 years, 8 months ago
B:
Use cases for Step Functions vary widely, from orchestrating serverless microservices, to building data-processing pipelines, to defining a security-incident response. As mentioned above, Step Functions may be used for synchronous and asynchronous business processes.
upvoted 1 times

AlessandraSAA 2 years, 9 months ago
why not B?
upvoted 4 times

Wael216 2 years, 9 months ago
Selected Answer: C
Creating an Amazon Simple Queue Service (SQS) message queue and placing messages on the queue for thumbnail generation can help separate the image upload and thumbnail generation processes.
upvoted 2 times

vindahake 2 years, 9 months ago
C
The key here is "a faster response time to its users to notify them that the original image was received." i.e user needs to be notified when image was received and not after thumbnail was created.
upvoted 3 times

AlmeroSenior 2 years, 9 months ago
Selected Answer: C
A looks like the best way , but its essentially replacing the mentioned app , that's not the ask
upvoted 2 times

Mickey321 2 years, 9 months ago
Selected Answer: A
https://docs.aws.amazon.com/lambda/latest/dg/with-s3-tutorial.html
upvoted 2 times

bdp123 2 years, 9 months ago
Selected Answer: C
C is the only one that makes sense

upvoted 3 times

Question #323 | *Topic 1*

A company's facility has badge readers at every entrance throughout the building. When badges are scanned, the readers send a message over HTTPS to indicate who attempted to access that particular entrance.

A solutions architect must design a system to process these messages from the sensors. The solution must be highly available, and the results must be made available for the company's security team to analyze.

Which system architecture should the solutions architect recommend?

A. Launch an Amazon EC2 instance to serve as the HTTPS endpoint and to process the messages. Configure the EC2 instance to save the results to an Amazon S3 bucket.

B. Create an HTTPS endpoint in Amazon API Gateway. Configure the API Gateway endpoint to invoke an AWS Lambda function to process the messages and save the results to an Amazon DynamoDB table. **Most Voted**

C. Use Amazon Route 53 to direct incoming sensor messages to an AWS Lambda function. Configure the Lambda function to process the messages and save the results to an Amazon DynamoDB table.

D. Create a gateway VPC endpoint for Amazon S3. Configure a Site-to-Site VPN connection from the facility network to the VPC so that sensor data can be written directly to an S3 bucket by way of the VPC endpoint.

Correct Answer: B

Community vote distribution

B (100%)

Comments

kruasan Highly Voted 2 years, 1 month ago

Selected Answer: B

- Option A would not provide high availability. A single EC2 instance is a single point of failure.
- Option B provides a scalable, highly available solution using serverless services. API Gateway and Lambda can scale automatically, and DynamoDB provides a durable data store.
- Option C would expose the Lambda function directly to the public Internet, which is not a recommended architecture. API Gateway provides an abstraction layer and additional features like access control.
- Option D requires configuring a VPN to AWS which adds complexity. It also saves the raw sensor data to S3, rather than processing it and storing the results.

upvoted 22 times

TariqKipkemei Highly Voted 1 year, 8 months ago

Selected Answer: B

Highly available = Serverless
The readers send a message over HTTPS = HTTPS endpoint in Amazon API Gateway
Process these messages from the sensors = AWS Lambda function

upvoted 8 times

Guru4Cloud Most Recent 1 year, 9 months ago

Selected Answer: B

The correct answer is B. Create an HTTPS endpoint in Amazon API Gateway. Configure the API Gateway endpoint to invoke an AWS Lambda function to process the messages and save the results to an Amazon DynamoDB table.

Here are the reasons why:

API Gateway is a highly scalable and available service that can be used to create and expose RESTful APIs.
Lambda is a serverless compute service that can be used to process events and data.
DynamoDB is a NoSQL database that can be used to store data in a scalable and highly available way.

upvoted 4 times

Steve_4542636 2 years, 3 months ago

Selected Answer: B

I vote B

upvoted 2 times

KZM 2 years, 3 months ago

It is option "B"
Option "B" can provide a system with highly scalable, fault-tolerant, and easy to manage.

upvoted 2 times

LuckyAro 2 years, 3 months ago

Selected Answer: B

Deploy Amazon API Gateway as an HTTPS endpoint and AWS Lambda to process and save the messages to an Amazon DynamoDB table. This option provides a highly available and scalable solution that can easily handle large amounts of data. It also integrates with other AWS services, making it easier to analyze and visualize the data for the security team.

upvoted 4 times

zTopic 2 years, 3 months ago

Selected Answer: B

B is Correct

upvoted 4 times

Question #324 Topic 1

A company wants to implement a disaster recovery plan for its primary on-premises file storage volume. The file storage volume is mounted from an Internet Small Computer Systems Interface (iSCSI) device on a local storage server. The file storage volume holds hundreds of terabytes (TB) of data.

The company wants to ensure that end users retain immediate access to all file types from the on-premises systems without experiencing latency.

Which solution will meet these requirements with the LEAST amount of change to the company's existing infrastructure?

A. Provision an Amazon S3 File Gateway as a virtual machine (VM) that is hosted on premises. Set the local cache to 10 TB. Modify existing applications to access the files through the NFS protocol. To recover from a disaster, provision an Amazon EC2 instance and mount the S3 bucket that contains the files.

B. Provision an AWS Storage Gateway tape gateway. Use a data backup solution to back up all existing data to a virtual tape library. Configure the data backup solution to run nightly after the initial backup is complete. To recover from a disaster, provision an Amazon EC2 instance and restore the data to an Amazon Elastic Block Store (Amazon EBS) volume from the volumes in the virtual tape library.

C. Provision an AWS Storage Gateway Volume Gateway cached volume. Set the local cache to 10 TB. Mount the Volume Gateway cached volume to the existing file server by using iSCSI, and copy all files to the storage volume. Configure scheduled snapshots of the storage volume. To recover from a disaster, restore a snapshot to an Amazon Elastic Block Store (Amazon EBS) volume and attach the EBS volume to an Amazon EC2 instance.

D. Provision an AWS Storage Gateway Volume Gateway stored volume with the same amount of disk space as the existing file storage volume. Mount the Volume Gateway stored volume to the existing file server by using iSCSI, and copy all files to the storage volume. Configure scheduled snapshots of the storage volume. To recover from a disaster, restore a snapshot to an Amazon Elastic Block Store (Amazon EBS) volume and attach the EBS volume to an Amazon EC2 instance. **Most Voted**

Correct Answer: D

Community vote distribution

D (83%) C (17%)

Comments

Grace83 Highly Voted 3 years, 2 months ago

D is the correct answer

Volume Gateway CACHED Vs STORED
Cached = stores a subset of frequently accessed data locally
Stored = Retains the ENTIRE ("all file types") in on prem data centre

upvoted 30 times

netcj Highly Voted 2 years, 9 months ago

Selected Answer: D

"users retain immediate access to all file types"
immediate cannot be cached -> D

upvoted 6 times

dkw2342 Most Recent 2 years, 3 months ago

Bad question. No RTO/RPO, so impossible to properly answer. They probably want to hear option D.

Depending on RPO, option B is also an adequate solution (data remains immediately accessible without experiencing latency via existing infrastructure, backup to cloud for DR). Also, this option requires LESS changes to existing infra than A. Only argument against B is that VTLs are usually used for legacy DR solutions, not for new ones, where object storage such as S3 is usually supported natively.

upvoted 2 times

MrPCarrot 2 years, 3 months ago

Answer is C go argue somewhere.

upvoted 2 times

awsgeek75 2 years, 5 months ago

Selected Answer: D

A,B are wrong types of gateways for hundreds of TB of data that needs immediate access on-prem. C limits to 10TB. D provides access to all the files

upvoted 2 times

pentium75 2 years, 5 months ago

Selected Answer: D

"Immediate access to all file types from the on-premises systems without experiencing latency" requirement is not met by C. Also the solution is meant for DR purposes, the primary storage for the data should remain on premises.

upvoted 3 times

daniel1 2 years, 7 months ago

Selected Answer: C

From chatGPT4
Considering the requirements of minimal infrastructure change, immediate file access, and low-latency, Option C: Provisioning an AWS Storage Gateway Volume Gateway (cached volume) with a 10 TB local cache, seems to be the most fitting solution. This setup aligns with the existing iSCSI setup and provides a local cache for low-latency access, while also configuring scheduled snapshots for disaster recovery. In the event of a disaster, restoring a snapshot to an Amazon EBS volume and attaching it to an Amazon EC2 instance as described in this option would align with the recovery objective.

upvoted 1 times

pentium75 2 years, 5 months ago

ChatGPT is wrong. "Immediate access to all file types from the on-premises systems without experiencing latency" needs "stored volume" type. With "cached volume" not all data will be available locally.

upvoted 9 times

LoXoL 2 years, 4 months ago

pentium75 is right.

upvoted 2 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: D

End users retain immediate access to all file types = Volume Gateway stored volume

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

dddddddd

upvoted 3 times

alexandercamachop 3 years ago

Selected Answer: D

Correct answer is Volume Gateway Stored which keeps all data on premises.
To have immediate access to the data. Cached is for frequently accessed data only.

upvoted 3 times

omoakin 3 years ago

CCCCCCCCCCCCCCC

upvoted 1 times

24b2e9e 1 year, 11 months ago

The stored volume configuration stores the entire data set on-premises and asynchronously backs up the data to AWS. The cached volume configuration stores recently accessed data on-premises, and the remaining data is stored in Amazon S3
-that is why D is right

upvoted 2 times

lucdt4 3 years ago

Selected Answer: D

D is the correct answer
Volume Gateway CACHED Vs STORED
Cached = stores a data recentlly at local
Stored = Retains the ENTIRE ("all file types") in on prem data centre

upvoted 2 times

rushi0611 3 years, 1 month ago

Selected Answer: D

In the cached mode, your primary data is written to S3, while retaining your frequently accessed data locally in a cache for low-latency access.
In the stored mode, your primary data is stored locally and your entire dataset is available for low-latency access while asynchronously backed up to AWS.
Reference: https://aws.amazon.com/storagegateway/faqs/
Good luck.

upvoted 3 times

kruasan 3 years, 1 month ago

Selected Answer: D

It is stated the company wants to keep the data locally and have DR plan in cloud. It points directly to the volume gateway

upvoted 2 times

UnluckyDucky 3 years, 2 months ago

Selected Answer: D

"The company wants to ensure that end users retain immediate access to all file types from the on-premises systems "

D is the correct answer.

upvoted 3 times

CapJackSparrow 3 years, 2 months ago

Selected Answer: C

all file types, NOT all files. Volume mode can not cache 100TBs.

upvoted 3 times

eddie5049 3 years, 1 month ago

https://docs.aws.amazon.com/storagegateway/latest/vgw/StorageGatewayConcepts.html

Stored volumes can range from 1 GiB to 16 TiB in size and must be rounded to the nearest GiB. Each gateway configured for stored volumes can support up to 32 volumes and a total volume storage of 512 TiB (0.5 PiB).

upvoted 2 times

MssP 3 years, 2 months ago

all file types. Cached only save the most frecuently or lastest accesed. If you didn´t access any type for a long time, you will not cache it -> No immediate access

upvoted 4 times

pentium75 2 years, 5 months ago

Also the solution is meant for DR purposes, it's not like they need more storage or so.

upvoted 2 times

WherecanIstart 3 years, 2 months ago

Selected Answer: D

"The company wants to ensure that end users retain immediate access to all file types from the on-premises systems "

This points to stored volumes..

upvoted 2 times

Question #325 Topic 1

A company is hosting a web application from an Amazon S3 bucket. The application uses Amazon Cognito as an identity provider to authenticate users and return a JSON Web Token (JWT) that provides access to protected resources that are stored in another S3 bucket.

Upon deployment of the application, users report errors and are unable to access the protected content. A solutions architect must resolve this issue by providing proper permissions so that users can access the protected content.

Which solution meets these requirements?

A. Update the Amazon Cognito identity pool to assume the proper IAM role for access to the protected content. Most Voted

B. Update the S3 ACL to allow the application to access the protected content.

C. Redeploy the application to Amazon S3 to prevent eventually consistent reads in the S3 bucket from affecting the ability of users to access the protected content.

D. Update the Amazon Cognito pool to use custom attribute mappings within the identity pool and grant users the proper permissions to access the protected content.

Correct Answer: A

Community vote distribution

A (95%) 5%

Comments

alexandercamachop Highly Voted 2 years ago

Selected Answer: A

To resolve the issue and provide proper permissions for users to access the protected content, the recommended solution is:

A. Update the Amazon Cognito identity pool to assume the proper IAM role for access to the protected content.

Explanation:

Amazon Cognito provides authentication and user management services for web and mobile applications.
In this scenario, the application is using Amazon Cognito as an identity provider to authenticate users and obtain JSON Web Tokens (JWTs).
The JWTs are used to access protected resources stored in another S3 bucket.
To grant users access to the protected content, the proper IAM role needs to be assumed by the identity pool in Amazon Cognito.
By updating the Amazon Cognito identity pool with the appropriate IAM role, users will be authorized to access the protected content in the S3 bucket.

upvoted 17 times

alexandercamachop 2 years ago

Option B is incorrect because updating the S3 ACL (Access Control List) will only affect the permissions of the application, not the users accessing the content.

Option C is incorrect because redeploying the application to Amazon S3 will not resolve the issue related to user access permissions.

Option D is incorrect because updating custom attribute mappings in Amazon Cognito will not directly grant users the proper permissions to acces the protected content.

upvoted 12 times

LuckyAro Highly Voted 2 years, 3 months ago

Selected Answer: A

A is the best solution as it directly addresses the issue of permissions and grants authenticated users the necessary IAM role to access the protected content.

A suggests updating the Amazon Cognito identity pool to assume the proper IAM role for access to the protected content. This is a valid solution, as it would grant authenticated users the necessary permissions to access the protected content.

upvoted 6 times

Marco_St Most Recent 1 year, 6 months ago

Selected Answer: A

IAM role is assinged to IAM users or groups or assumed by AWS service. So IAM role is given to AWS Cognito service which provides temporary AWS credentials to authenticated users. so technically When a user is authenticated by Cognito, they receive temporary credentials based on the IAM role tied to the Cognito identity pool. If this IAM role has permissions to access certain S3 buckets or objects, the authenticated user will be able to access those resources as allowed by the role. This service is used under the hood by Cognito to provide these temporary credentials. The credentials are limited in time and scope based on the permissions defined in the IAM role.

upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: A

A. Update the Amazon Cognito identity pool to assume the proper IAM role for access to the protected content.

upvoted 3 times

Abrar2022 2 years ago

Selected Answer: A

Services access other services via IAM Roles. Hence why updating AWS Cognito identity pool to assume proper IAM Role is the right solution.

upvoted 2 times

shanwford 2 years, 2 months ago

Selected Answer: A

Amazon Cognito identity pools assign your authenticated users a set of temporary, limited-privilege credentials to access your AWS resources. The permissions for each user are controlled through IAM roles that you create. https://docs.aws.amazon.com/cognito/latest/developerguide/role-based-access-control.html

upvoted 3 times

Brak 2 years, 3 months ago

Selected Answer: D

A makes no sense - Cognito is not accessing the S3 resource. It just returns the JWT token that will be attached to the S3 request.

D is the right answer, using custom attributes that are added to the JWT and used to grant permissions in S3. See https://docs.aws.amazon.com/cognito/latest/developerguide/using-attributes-for-access-control-policy-example.html for an example.

upvoted 2 times

asoli 2 years, 2 months ago

A says "Identity Pool"
According to AWS: "With an identity pool, your users can obtain temporary AWS credentials to access AWS services, such as Amazon S3 and DynamoDB."
So, answer is A

upvoted 4 times

Abhineet9148232 2 years, 3 months ago

But even D requires setting up the permissions as bucket policy (as show in the shared example) which includes higher overhead than managing permissions attached to specific roles.

upvoted 3 times

Steve_4542636 2 years, 3 months ago

Selected Answer: A

Services access other services via IAM Roles.

upvoted 2 times

jennyka76 2 years, 3 months ago

ANSWER - A
https://docs.aws.amazon.com/cognito/latest/developerguide/tutorial-create-identity-pool.html
You have to create an custom role such as read-only

upvoted 5 times

zTopic 2 years, 3 months ago

Selected Answer: A

Answer is A

upvoted 3 times

Question #326 Topic 1

An image hosting company uploads its large assets to Amazon S3 Standard buckets. The company uses multipart upload in parallel by using S3 APIs and overwrites if the same object is uploaded again. For the first 30 days after upload, the objects will be accessed frequently. The objects will be used less frequently after 30 days, but the access patterns for each object will be inconsistent. The company must optimize its S3 storage costs while maintaining high availability and resiliency of stored assets.

Which combination of actions should a solutions architect recommend to meet these requirements? (Choose two.)

A. Move assets to S3 Intelligent-Tiering after 30 days. **Most Voted**

B. Configure an S3 Lifecycle policy to clean up incomplete multipart uploads. **Most Voted**

C. Configure an S3 Lifecycle policy to clean up expired object delete markers.

D. Move assets to S3 Standard-Infrequent Access (S3 Standard-IA) after 30 days.

E. Move assets to S3 One Zone-Infrequent Access (S3 One Zone-IA) after 30 days.

Correct Answer: AB

Community vote distribution

AB (67%) BD (23%) 9%

Comments

Neha999 Highly Voted 3 years, 3 months ago

AB
A : Access Pattern for each object inconsistent, Infrequent Access
B : Deleting Incomplete Multipart Uploads to Lower Amazon S3 Costs

upvoted 26 times

TungPham Highly Voted 3 years, 3 months ago

Selected Answer: AB

B because Abort Incomplete Multipart Uploads Using S3 Lifecycle => https://aws.amazon.com/blogs/aws-cloud-financial-management/discovering-and-deleting-incomplete-multipart-uploads-to-lower-amazon-s3-costs/
A because The objects will be used less frequently after 30 days, but the access patterns for each object will be inconsistent => random access => S3 Intelligent-Tiering

upvoted 16 times

LeonSauveterre Most Recent 1 year, 6 months ago

Selected Answer: AB

Inconsistent access patterns: A is correct, and D is out (could work but less ideal and costs more).

C: "object delete markers" only apply to versioned buckets.

upvoted 1 times

ChymKuBoy 1 year, 10 months ago

Selected Answer: AB

AB for sure

upvoted 3 times

bujuman 2 years, 3 months ago

Selected Answer: BD

If we consider these statements:
1. For the first 30 days after upload, the objects will be accessed frequently
2.The objects will be used less frequently after 30 days, but the access patterns for each object will be inconsistent
3.The company must optimize its S3 storage costs while maintaining high availability and resiliency of stored assets.
4.The company uses multipart upload in parallel by using S3 APIs and overwrites if the same object is uploaded again.
Statements 1 and 2 cloudl be completed with option D and not A because datas are infrequently accessed only after 30 days.

Due to usage of multipart upload, to meet requirement regarding cost optimization, option B will be used to clean up buckets uncompleted file parts(statements 3 & 4).

upvoted 4 times

NayeraB 2 years, 3 months ago

Selected Answer: AD

Because A & D address the main ask, there's no mention of cost optimization.

upvoted 2 times

NayeraB 2 years, 3 months ago

Facepalm It does ask for reducing the cost, A&B it is!

upvoted 3 times

NayeraB 2 years, 3 months ago

Selected Answer: AC

Because A & C address the main ask, there's no mention of cost optimization.

upvoted 1 times

NayeraB 2 years, 3 months ago

Not C ':D, I meant to say A&D. Added another vote for that one.

upvoted 2 times

awsgeek75 2 years, 5 months ago

Selected Answer: AB

A as the access pattern for each object is inconsistent so let AWS AWS do the handling.
B deals with multi-part duplication issues and saves money by deleting incomplete uploads
C No mention of deleted object so this is a distractor
D The objects will be accessed in unpredictable pattern so can't use this
E Not HA compliant

upvoted 4 times

awsgeek75 2 years, 5 months ago

Also, don't be confused by 30 days. The question has tricky wording: " The objects will be used less frequently after 30 days, but the access patterns for each object will be inconsistent"
It does NOT say that objects will be accessed less frequently after 30 days. It says the access is unpredictable which means it could go up or down. Don't make assumptions.

upvoted 5 times

pentium75 2 years, 5 months ago

Selected Answer: AB

C is nonsense
E does not meet the "high availability and resiliency" requirement
B is obvious (incomplete multipart uploads consume space -> cost money)

The tricky part is A vs. D. However, 'inconsistent access patterns' are the primary use case for Intelligent-Tiering. There are probably objects that will never be accessed and that would be moved to Glacier Instant Retrieval by Intelligent-Tiering, thus the overall cost would be lower than with D.

upvoted 4 times

osmk 2 years, 5 months ago

bd https://docs.aws.amazon.com/AmazonS3/latest/userguide/storage-class-intro.html#sc-infreq-data-access =>S3 Standard-IA objects are resilient to the loss of an Availability Zone. This storage class offers greater availability and resiliency than the S3 One Zone-IA class

upvoted 2 times

raymondfekry 2 years, 5 months ago

Selected Answer: AB

I wouldnt go with D since " the access patterns for each object will be inconsistent.", so we cannot move all assets to IA

upvoted 2 times

Marco_St 2 years, 6 months ago

Selected Answer: AB

incosistent access pattern brings more sense to use Intelligent-Tiering after 30 days which also covers infrequent access.

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: AB

A. Move assets to S3 Intelligent-Tiering after 30 days.
B. Configure an S3 Lifecycle policy to clean up incomplete multipart uploads.

upvoted 2 times

vini15 2 years, 10 months ago

should be A and B

upvoted 2 times

MrAWSAssociate 2 years, 11 months ago

Selected Answer: BD

Option A has not been mentioned for resiliency in S3, check the page: https://docs.aws.amazon.com/AmazonS3/latest/userguide/disaster-recovery-resiliency.html

Therefore, I am with B & D choices.

upvoted 2 times

pentium75 2 years, 5 months ago

Intelligent-Tiering just moves to Standard-IA or Glacier Instant Access based on access patterns. This does not affect resiliency.

upvoted 2 times

alexandercamachop 3 years ago

Selected Answer: AB

A. Move assets to S3 Intelligent-Tiering after 30 days.
B. Configure an S3 Lifecycle policy to clean up incomplete multipart uploads.

Explanation:

A. Moving assets to S3 Intelligent-Tiering after 30 days: This storage class automatically analyzes the access patterns of objects and moves them between frequent access and infrequent access tiers. Since the objects will be accessed frequently for the first 30 days, storing them in the frequent access tier during that period optimizes performance. After 30 days, when the access patterns become inconsistent, S3 Intelligent-Tiering will automatically move the objects to the infrequent access tier, reducing storage costs.

B. Configuring an S3 Lifecycle policy to clean up incomplete multipart uploads: Multipart uploads are used for large objects, and incomplete multipart uploads can consume storage space if not cleaned up. By configuring an S3 Lifecycle policy to clean up incomplete multipart uploads, unnecessary storage costs can be avoided.

upvoted 2 times

antropaws 3 years ago

Selected Answer: AD

AD.

B makes no sense because multipart uploads overwrite objects that are already uploaded. The question never says this is a problem.

upvoted 2 times

VellaDevil 2 years, 11 months ago

Questions says to optimize cost and if incomplete multiparts are not aborted it will still use capacity on S3 Bucket thus increase unnecessary cost.

upvoted 3 times

Question #327 *Topic 1*

A solutions architect must secure a VPC network that hosts Amazon EC2 instances. The EC2 instances contain highly sensitive data and run in a private subnet. According to company policy, the EC2 instances that run in the VPC can access only approved third-party software repositories on the internet for software product updates that use the third party's URL. Other internet traffic must be blocked.

Which solution meets these requirements?

A. Update the route table for the private subnet to route the outbound traffic to an AWS Network Firewall firewall. Configure domain list rule groups. **Most Voted**

B. Set up an AWS WAF web ACL. Create a custom set of rules that filter traffic requests based on source and destination IP address range sets.

C. Implement strict inbound security group rules. Configure an outbound rule that allows traffic only to the authorized software repositories on the internet by specifying the URLs.

D. Configure an Application Load Balancer (ALB) in front of the EC2 instances. Direct all outbound traffic to the ALB. Use a URL-based rule listener in the ALB's target group for outbound access to the internet.

Correct Answer: A

Community vote distribution

A (92%) 8%

Comments

Bhawesh Highly Voted 2 years, 9 months ago

Selected Answer: A

Correct Answer A. Send the outbound connection from EC2 to Network Firewall. In Network Firewall, create stateful outbound rules to allow certain domains for software patch download and deny all other domains.

https://docs.aws.amazon.com/network-firewall/latest/developerguide/suricata-examples.html#suricata-example-domain-filtering

upvoted 17 times

Guru4Cloud 2 years, 3 months ago

Option A uses a network firewall which is overkill for instance-level rules.

upvoted 2 times

UnluckyDucky Highly Voted 2 years, 9 months ago

Selected Answer: A

Can't use URLs in outbound rule of security groups. URL Filtering screams Firewall.

upvoted 13 times

TheFivePips Most Recent 1 year, 9 months ago

Selected Answer: A

Security Groups operate at the transport layer (Layer 4) of the OSI model and are primarily concerned with controlling traffic based on IP addresses, ports, and protocols. They do not have the capability to inspect or filter traffic based on URLs.
The solution to restrict outbound internet traffic based on specific URLs typically involves using a proxy or firewall that can inspect the application layer (Layer 7) of the OSI model, where URL information is available.
AWS Network Firewall operates at the network and application layers, allowing for more granular control, including the ability to inspect and filter traffic based on domain names or URLs.
By configuring domain list rule groups in AWS Network Firewall, you can specify which URLs are allowed for outbound traffic.
This option is more aligned with the requirement of allowing access to approved third-party software repositories based on their URLs.

upvoted 6 times

awsgeek75 1 year, 11 months ago

Selected Answer: A

https://aws.amazon.com/network-firewall/features/
"Web filtering:
AWS Network Firewall supports inbound and outbound web filtering for unencrypted web traffic. For encrypted web traffic, Server Name Indication (SNI) is used for blocking access to specific sites. SNI is an extension to Transport Layer Security (TLS) that remains unencrypted in the traffic flow and indicates the destination hostname a client is attempting to access over HTTPS. In addition, **AWS Network Firewall can filter fully qualified domain names (FQDN).**"
Always use an AWS product if the advertisement meets the use case.

upvoted 4 times

farnamjam 1 year, 11 months ago

Selected Answer: A

AWS Network Firewall
• Protect your entire Amazon VPC
• From Layer 3 to Layer 7 protection
• Any direction, you can inspect
Traffic filtering: Allow, drop, or alert for the traffic that matches the rules, • Active flow inspection to intrusion prevention

upvoted 2 times

Subhrangsu 1 year, 11 months ago

D not possible?

upvoted 1 times

awsgeek75 1 year, 11 months ago

ALB is for inbound traffic. D is not possible as it is suggesting to direct OUTBOUND traffic.

upvoted 4 times

Cyberkayu 1 year, 12 months ago

Selected Answer: A

AWS network firewall is stateful, providing control and visibility to Layer 3-7 network traffic, thus cover the application too

upvoted 2 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: A

Just tried on the console to set up an outbound rule, and URLs cannot be used as a destination. I will opt for A.

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: C

Implement strict inbound security group rules
Configure an outbound security group rule to allow traffic only to the approved software repository URLs
The key points:

Highly sensitive EC2 instances in private subnet that can access only approved URLs
Other internet access must be blocked
Security groups act as a firewall at the instance level and can control both inbound and outbound traffic.

upvoted 2 times

pentium75 1 year, 11 months ago

Security Groups work with CIDR ranges, not URLs.

upvoted 4 times

kelvintoys93 2 years, 5 months ago

Isnt private subnet not connectible to internet at all, unless with a NAT gateway?

upvoted 4 times

VeseljkoD 2 years, 9 months ago

Selected Answer: A

We can't specifu URL in outbound rule of security group. Create free tier AWS account and test it.

upvoted 2 times

Leo301 2 years, 9 months ago

Selected Answer: C

CCCCCCCCCCC

upvoted 1 times

pentium75 1 year, 11 months ago

Security Groups with IP ranges, not URLs

upvoted 2 times

Brak 2 years, 9 months ago

It can't be C. You cannot use URLs in the outbound rules of a security group.

upvoted 3 times

johnmcclane78 2 years, 9 months ago

Option C is the best solution to meet the requirements of this scenario. Implementing strict inbound security group rules that only allow traffic from approved sources can help secure the VPC network that hosts Amazon EC2 instances. Additionally, configuring an outbound rule that allows traffic only to the authorized software repositories on the internet by specifying the URLs will ensure that only approved third-party software repositories can be accessed from the EC2 instances. This solution does not require any additional AWS services and can be implemented using VPC security groups.

Option A is not the best solution as it involves the use of AWS Network Firewall, which may introduce additional operational overhead. While domain list rule groups can be used to block all internet traffic except for the approved third-party software repositories, this solution is more complex than necessary for this scenario.

upvoted 2 times

pentium75 1 year, 11 months ago

How do you use a Security Group to allow access to https://server.com/repoa while denying access to https://server.com/repob ? Security Groups work with IP ranges.

upvoted 2 times

Steve_4542636 2 years, 9 months ago

Selected Answer: C

In the security group, only allow inbound traffic originating from the VPC. Then only allow outbound traffic with a whitelisted IP address. The question asks about blocking EC2 instances, which is best for security groups since those are at the EC2 instance level. A network firewall is at the VPC level, which is not what the question is asking to protect.

upvoted 1 times

Theodorz 2 years, 9 months ago

Is Security Group able to allow a specific URL? According to https://docs.aws.amazon.com/vpc/latest/userguide/VPC_SecurityGroups.html, I cannot find such description.

upvoted 2 times

pentium75 1 year, 11 months ago

Security Groups work with IP ranges, not URLs.

upvoted 1 times

KZM 2 years, 9 months ago

I am confused that It seems both options A and C are valid solutions.

upvoted 3 times

Zohx 2 years, 9 months ago

Same here - why is C not a valid option?

upvoted 2 times

Karlos99 2 years, 9 months ago

Because in this case, the session is initialized from inside

upvoted 2 times

Karlos99 2 years, 9 months ago

And it is easier to do it at the level

upvoted 2 times

Karlos99 2 years, 9 months ago

And it is easier to do it at the VPC level

upvoted 2 times

Mia2009687 2 years, 5 months ago

I think C is in private subnet. Even with security group, it could not go public to download the software.

upvoted 1 times

ruqui 2 years, 6 months ago

C is not valid. Security groups can allow only traffic from specific ports and/or IPs, you can't use an URL. Correct answer is A

upvoted 3 times

jennyka76 2 years, 9 months ago

Answer - A

https://aws.amazon.com/premiumsupport/knowledge-center/ec2-al1-al2-update-yum-without-internet/

upvoted 6 times

asoli 2 years, 8 months ago
Although the answer is A, the link you provided here is not related to this question.
The information about "Network Firewall" and how it can help this issue is here:
https://docs.aws.amazon.com/network-firewall/latest/developerguide/suricata-examples.html#suricata-example-domain-filtering

(thanks to "@Bhawesh" to provide the link in their answer)
upvoted 4 times

Question #328 Topic 1

A company is hosting a three-tier ecommerce application in the AWS Cloud. The company hosts the website on Amazon S3 and integrates the website with an API that handles sales requests. The company hosts the API on three Amazon EC2 instances behind an Application Load Balancer (ALB). The API consists of static and dynamic front-end content along with backend workers that process sales requests asynchronously.

The company is expecting a significant and sudden increase in the number of sales requests during events for the launch of new products.

What should a solutions architect recommend to ensure that all the requests are processed successfully?

A. Add an Amazon CloudFront distribution for the dynamic content. Increase the number of EC2 instances to handle the increase in traffic.

B. Add an Amazon CloudFront distribution for the static content. Place the EC2 instances in an Auto Scaling group to launch new instances based on network traffic.

C. Add an Amazon CloudFront distribution for the dynamic content. Add an Amazon ElastiCache instance in front of the ALB to reduce traffic for the API to handle.

D. Add an Amazon CloudFront distribution for the static content. Add an Amazon Simple Queue Service (Amazon SQS) queue to receive requests from the website for later processing by the EC2 instances. **Most Voted**

Correct Answer: D

Community vote distribution

D (73%) B (27%)

Comments

Steve_4542636 Highly Voted 3 years, 3 months ago

Selected Answer: B

The auto-scaling would increase the rate at which sales requests are "processed", whereas a SQS will ensure messages don't get lost. If you were at a fast food restaurant with a long line with 3 cash registers, would you want more cash registers or longer ropes to handle longer lines? Same concept here.

upvoted 22 times

lizzard812 3 years, 2 months ago

Hell true: I'd rather combine the both options: a SQS + auto-scaled bound to the length of the queue.

upvoted 10 times

joechen2023 2 years, 12 months ago

As an architecture, it is not possible to add more backend workers (it is part of the HR and boss's job, not for architecture design the solution). So when the demand surge, the only correct choice is to buffer them using SQS so that workers can take their time to process it successfully

upvoted 2 times

rushi0611 3 years, 1 month ago

"ensure that all the requests are processed successfully?"
we want to ensure success not the speed, even in the auto-scaling, there is the chance for the failure of the request but not in SQS- if it is failed in sqs it is sent back to the queue again and new consumer will pick the request.

upvoted 19 times

Marco_St 2 years, 6 months ago

eeeee... In the restaurant also here, the request in the line will be dropped in case of high-sudden volume flooding in so it will bring risk of missing request. while with SQS, it can ensure no lost but still slow processing with only 3 processor though so it is based on what is the question's concern So I still voted for D.

upvoted 2 times

Abhineet9148232 Highly Voted 3 years, 1 month ago

Selected Answer: D

B doesn't fit because Auto Scaling alone does not guarantee that all requests will be processed successfully, which the question clearly asks for.

D ensures that all messages are processed.

upvoted 17 times

ChhatwaniB Most Recent 1 year, 2 months ago

Selected Answer: D

There is no scaling based on network traffic. since the backend worker works asynchronously SQS should be right here

upvoted 1 times

samadal 1 year, 8 months ago

The problem states that the application consists of "static and dynamic front-end content." Static content typically includes cacheable resources such as HTML, CSS, and image files. Therefore, from this statement, one can infer that caching static content using CloudFront would improve performance. In other words, the mention of "static content" in the problem itself leads to the conclusion that CloudFront should be added for static content.

Additionally, the problem mentions "asynchronously processed backend workers." Asynchronous processing is well-suited for services like SQS, which can improve efficiency by handling dynamic requests that do not require immediate processing. The mention of "successfully processing all requests" also suggests that SQS is needed to ensure that all requests are handled properly.

Therefore, the correct answer is D.

upvoted 3 times

Adinas_ 2 years, 3 months ago

Selected Answer: B

Important question to answer D. Can you connect the website with SQS directly? How do you control access to who can put messages to SQS? I have never seen such a situation it has to be at least behind API gateway. So that conclusion brings me to answer B, application also can process async everything without SQS.

upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: D

I chose D because I love SQS! These questions are hammering SQS in every solution as a "protagonist" that saves the day.
AC are clearly useless
B can work but D is better because of SQS being better than EC2 scaling. The other part is that backend workers process the request asynchronously therefore a queue is better.

upvoted 4 times

awsgeek75 2 years, 5 months ago

Selected Answer: D

A and C don't solve anything so ignore them.
Between B and D, D guarantees the scaling via SQS and order processing. B can also do that but it is not guaranteed that EC2 scaling will work to process the order.
As usual, I suspect that this "brain dump" may be missing critical wording to differentiate between the options so read carefully in the exam.

upvoted 4 times

pentium75 2 years, 5 months ago

Selected Answer: D

There are two components that we need
* Frontend: Hosted on S3, performance can be increased with CloudFront
* Backend: There's no reason to process all the orders instantly, so we should decouple the processing from the API which we do with SQS

Thus D, CloudFront + SQS

upvoted 8 times

pentium75 2 years, 5 months ago

And as others said, B might speed up the processing or reduce the number of lost orders, but we need to make sure that "ALL requests are processed successfully", NOT that "less requests are lost".

upvoted 3 times

Marco_St 2 years, 6 months ago

Selected Answer: D

I picked B before I read D option. Read the question again, it concerns:asynchronous processing of sales requests, Option D seems to align more closely with the requirements. So the requirement is ensuring all requests are processed successfully which means no request would be missed. So D is better option

upvoted 4 times

wsdasdasdqwdaw 2 years, 7 months ago

Amazon SQS will make sure that the requests are stored and didn't get lost. After that the workers asynchronously will process the requests. I would go for D

upvoted 4 times

TariqKipkemei 2 years, 8 months ago

Technically both option B and D would work. But, there's a need to process requests asynchronously, hence decoupling, hence Amazon SQS. I will settle with option D.

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

D is correct.

upvoted 3 times

antropaws 3 years ago

Selected Answer: D

D is correct.

upvoted 2 times

kruasan 3 years, 1 month ago

Selected Answer: D

An SQS queue acts as a buffer between the frontend (website) and backend (API). Web requests can dump messages into the queue at a high throughput, then the queue handles delivering those messages to the API at a controlled rate that it can sustain. This prevents the API from being overwhelmed.

upvoted 3 times

kruasan 3 years, 1 month ago

Options A and B would help by scaling out more instances, however, this may not scale quickly enough and still risks overwhelming the API. Caching parts of the dynamic content (option C) may help but does not provide the buffering mechanism that a queue does.

upvoted 1 times

seifshendy99 3 years, 1 month ago

Selected Answer: D

D make sens

upvoted 2 times

kraken21 3 years, 2 months ago

Selected Answer: D

D makes more sense

upvoted 2 times

kraken21 3 years, 2 months ago

There is no clarity on what the asynchronous process is but D makes more sense if we want to process all requests successfully. The way the questior is worded it looks like the msgs->SQS>ELB/Ec2. This ensures that the messages are processed but may be delayed as the load increases.

upvoted 1 times

Question #329 Topic 1

A security audit reveals that Amazon EC2 instances are not being patched regularly. A solutions architect needs to provide a solution that will run regular security scans across a large fleet of EC2 instances. The solution should also patch the EC2 instances on a regular schedule and provide a report of each instance's patch status.

Which solution will meet these requirements?

A. Set up Amazon Macie to scan the EC2 instances for software vulnerabilities. Set up a cron job on each EC2 instance to patch the instance on a regular schedule.

B. Turn on Amazon GuardDuty in the account. Configure GuardDuty to scan the EC2 instances for software vulnerabilities. Set up AWS Systems Manager Session Manager to patch the EC2 instances on a regular schedule.

C. Set up Amazon Detective to scan the EC2 instances for software vulnerabilities. Set up an Amazon EventBridge scheduled rule to patch the EC2 instances on a regular schedule.

D. Turn on Amazon Inspector in the account. Configure Amazon Inspector to scan the EC2 instances for software vulnerabilities. Set up AWS Systems Manager Patch Manager to patch the EC2 instances on a regular schedule. Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

elearningtakai Highly Voted 2 years, 8 months ago

Selected Answer: D

Amazon Inspector is a security assessment service that automatically assesses applications for vulnerabilities or deviations from best practices. It can be used to scan the EC2 instances for software vulnerabilities. AWS Systems Manager Patch Manager can be used to patch the EC2 instances on a regular schedule. Together, these services can provide a solution that meets the requirements of running regular security scans and patching EC2 instances on a regular schedule. Additionally, Patch Manager can provide a report of each instance's patch status.

upvoted 10 times

LuckyAro Highly Voted 2 years, 9 months ago

Selected Answer: D

Amazon Inspector is a security assessment service that helps improve the security and compliance of applications deployed on Amazon Web Service (AWS). It automatically assesses applications for vulnerabilities or deviations from best practices. Amazon Inspector can be used to identify security issues and recommend fixes for them. It is an ideal solution for running regular security scans across a large fleet of EC2 instances.

AWS Systems Manager Patch Manager is a service that helps you automate the process of patching Windows and Linux instances. It provides a simple, automated way to patch your instances with the latest security patches and updates. Patch Manager helps you maintain compliance with security policies and regulations by providing detailed reports on the patch status of your instances.

upvoted 5 times

awsgeek75 Most Recent 1 year, 11 months ago

Selected Answer: D

A handy reference page for such questions is:
https://aws.amazon.com/products/security/
Amazon Inspector = vulnerability detection = patching
https://aws.amazon.com/inspector/

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: D

dddddddddd

upvoted 2 times

Steve_4542636 2 years, 9 months ago

Selected Answer: D

Inspecter is for EC2 instances and network accessibility of those instances
https://portal.tutorialsdojo.com/forums/discussion/difference-between-security-hub-detective-and-inspector/

upvoted 2 times

TungPham 2 years, 9 months ago

Selected Answer: D

Amazon Inspector for EC2
https://aws.amazon.com/vi/inspector/faqs/?nc1=f_ls
Amazon system manager Patch manager for automates the process of patching managed nodes with both security-related updates and other types of updates.

http://webcache.googleusercontent.com/search?q=cache:FbFTc6XKycwJ:https://medium.com/aws-architech/use-case-aws-inspector-vs-guardduty-3662bf80767a&hl=vi&gl=kr&strip=1&vwsrc=0

upvoted 3 times

jennyka76 2 years, 9 months ago

answer - D
https://aws.amazon.com/inspector/faqs/

upvoted 3 times

Neha999 2 years, 9 months ago

D as AWS Systems Manager Patch Manager can patch the EC2 instances.

upvoted 2 times

Question #330 | *Topic 1*

A company is planning to store data on Amazon RDS DB instances. The company must encrypt the data at rest.

What should a solutions architect do to meet this requirement?

A. Create a key in AWS Key Management Service (AWS KMS). Enable encryption for the DB instances. **Most Voted**

B. Create an encryption key. Store the key in AWS Secrets Manager. Use the key to encrypt the DB instances.

C. Generate a certificate in AWS Certificate Manager (ACM). Enable SSL/TLS on the DB instances by using the certificate.

D. Generate a certificate in AWS Identity and Access Management (IAM). Enable SSL/TLS on the DB instances by using the certificate.

Correct Answer: A

Community vote distribution

A (100%)

Comments

awsgeek75 1 year, 11 months ago

Selected Answer: A

A: Enable encryption
B: KMS is for storage and doesn't directly integrate to DB without further work
C and D are for data encryption in transit not at rest

upvoted 4 times

awsgeek75 1 year, 10 months ago

Actually, D is total nonsense and no idea what it is saying

upvoted 1 times

robpalacios1 2 years ago

Selected Answer: A

KMS only generates and manages encryption keys. That's it. That's all it does. It's a fundamental service that you as well as other AWS Services (like Secrets Manager) use it to encrypt or decrypt.
Key Management Service. Secrets Manager is for database connection strings.
upvoted 3 times

upvoted 4 times

antropaws 2 years, 6 months ago

OK, but why not B???

upvoted 2 times

aaroncelestin 2 years, 3 months ago

KMS only generates and manages encryption keys. That's it. That's all it does. It's a fundamental service that you as well as other AWS Services (like Secrets Manager) use it to encrypt or decrypt.

Secrets Manager stores actual secrets like passwords, pass phrases, and anything else you want encrypted. SM uses KMS to encrypt its secrets, it would be circular to get an encryption key from KMS to use SM to encrypt the encryption key.

upvoted 5 times

SkyZeroZx 2 years, 7 months ago

Selected Answer: A

ANSWER - A

upvoted 1 times

datz 2 years, 8 months ago

Selected Answer: A

A for sure
upvoted 2 times

PRASAD180 2 years, 9 months ago
A is 100% Crt
upvoted 2 times

Steve_4542636 2 years, 9 months ago
Selected Answer: A
Key Management Service. Secrets Manager is for database connection strings.
upvoted 4 times

LuckyAro 2 years, 9 months ago
Selected Answer: A
A is the correct solution to meet the requirement of encrypting the data at rest.

To encrypt data at rest in Amazon RDS, you can use the encryption feature of Amazon RDS, which uses AWS Key Management Service (AWS KMS). With this feature, Amazon RDS encrypts each database instance with a unique key. This key is stored securely by AWS KMS. You can manage your own keys or use the default AWS-managed keys. When you enable encryption for a DB instance, Amazon RDS encrypts the underlying storage, including the automated backups, read replicas, and snapshots.
upvoted 4 times

bdp123 2 years, 9 months ago
Selected Answer: A
AWS Key Management Service (KMS) is used to manage the keys used to encrypt and decrypt the data.
upvoted 2 times

pbpally 2 years, 9 months ago
Selected Answer: A
Option A
upvoted 2 times

NolaHOla 2 years, 9 months ago
A. Create a key in AWS Key Management Service (AWS KMS). Enable encryption for the DB instances is the correct answer to encrypt the data at rest in Amazon RDS DB instances.

Amazon RDS provides multiple options for encrypting data at rest. AWS Key Management Service (KMS) is used to manage the keys used to encrypt and decrypt the data. Therefore, a solution architect should create a key in AWS KMS and enable encryption for the DB instances to encrypt the data at rest.
upvoted 2 times

jennyka76 2 years, 9 months ago
ANSWER - A
https://docs.aws.amazon.com/whitepapers/latest/efs-encrypted-file-systems/managing-keys.html
upvoted 2 times

Bhawesh 2 years, 9 months ago
Selected Answer: A
A. Create a key in AWS Key Management Service (AWS KMS). Enable encryption for the DB instances.

https://www.examtopics.com/discussions/amazon/view/80753-exam-aws-certified-solutions-architect-associate-saa-c02/
upvoted 3 times

Question #331 *Topic 1*

A company must migrate 20 TB of data from a data center to the AWS Cloud within 30 days. The company's network bandwidth is limited to 15 Mbps and cannot exceed 70% utilization.

What should a solutions architect do to meet these requirements?

A. Use AWS Snowball. **Most Voted**

B. Use AWS DataSync.

C. Use a secure VPN connection.

D. Use Amazon S3 Transfer Acceleration.

Correct Answer: A

Community vote distribution

A (93%) 7%

Comments

kruasan Highly Voted 2 years, 7 months ago

Selected Answer: A

Don't mix up between Mbps and Mbs.
The proper calculation is:

10 MB/s x 86,400 seconds per day x 30 days/8 = 3,402,000 MB or approximately 3.4 TB

upvoted 14 times

awsgeek75 Highly Voted 1 year, 11 months ago

Selected Answer: A

Honestly, the company has bigger problem with that slow connection :)
30 days is the first clue so you can get snowball shipped and sent back (5 days each way)

upvoted 8 times

cabta Most Recent 1 year, 11 months ago

Selected Answer: A

aws snowball은 대용량 데이터 이전하기 위한 것 입니다.

upvoted 1 times

wsdasdasdqwdaw 2 years, 1 month ago

(15/8) = 1.875 MB/s
1.875 MB/s x 0.7 = 1.3125 (70% NW utilization) MB/s
1.3125 MB/s x 3600 = 4725 MB (MB per 1 hour)
4725 x 24 = 113400 MB per 1 full day (24h)
113400 x 30 = 3402000 MB for 30 days
3402000 / 1024 = 3322.265625 GB for 30 days
3322.265625 / 1024 ~ 3.24 TB for 30 days => not enough for NW => Snowball which is A

upvoted 4 times

TariqKipkemei 2 years, 2 months ago

Selected Answer: A

I wont try to think to much about it, AWS Snowball was designed for this

upvoted 4 times

Guru4Cloud 2 years, 2 months ago

Selected Answer: A

° 15 Mbps bandwidth with 70% max utilization limits the effective bandwidth to 10.5 Mbps or 1.31 MB/s.
° 20 TB of data at 1.31 MB/s would take approximately 193 days to transfer over the network. ° This far exceeds the 30 day requirement.

° AWS Snowball provides a physical storage device that can be shipped to the data center. Up to 80 TB can be loaded onto a Snowball device and shipped back to AWS.
This allows the 20 TB of data to be transferred much faster by shipping rather than over the limited network bandwidth.
° Snowball uses tamper-resistant enclosures and 256-bit encryption to keep the data secure during transit.
° The data can be imported into Amazon S3 or Amazon Glacier once the Snowball is received by AWS.

upvoted 5 times

UnluckyDucky 2 years, 8 months ago

Selected Answer: B

10 MB/s x 86,400 seconds per day x 30 days = 25,920,000 MB or approximately 25.2 TB

That's how much you can transfer with a 10 Mbps link (roughly 70% of the 15 Mbps connection).

With a consistent connection of 8~ Mbps, and 30 days, you can upload 20 TB of data.

My math says B, my brain wants to go with A. Take your pick.

upvoted 3 times

Zox42 2 years, 8 months ago

15 Mbps * 0.7 = 1.3125 MB/s and 1.3125 * 86,400 * 30 = 3.402.000 MB
Answer A is correct.

upvoted 3 times

hozy_ 2 years, 4 months ago

How can 15 * 0.7 be 1.3125 LMAO

upvoted 1 times

hozy_ 2 years, 4 months ago

OMG it was Mbps! Not MBps. You are right! awesome!!!

upvoted 3 times

Zox42 2 years, 8 months ago

3,402,000

upvoted 2 times

Bilalazure 2 years, 9 months ago

Selected Answer: A

Aws snowball

upvoted 3 times

PRASAD180 2 years, 9 months ago

A is 100% Crt

upvoted 2 times

LuckyAro 2 years, 9 months ago

Selected Answer: A

AWS Snowball

upvoted 2 times

pbpally 2 years, 9 months ago

Selected Answer: A

Option a

upvoted 2 times

jennyka76 2 years, 9 months ago

ANSWER - A
https://docs.aws.amazon.com/snowball/latest/ug/whatissnowball.html

upvoted 2 times

AWSSHA1 2 years, 9 months ago

Selected Answer: A

option A

upvoted 4 times

Question #332 Topic 1

A company needs to provide its employees with secure access to confidential and sensitive files. The company wants to ensure that the files can be accessed only by authorized users. The files must be downloaded securely to the employees' devices.

The files are stored in an on-premises Windows file server. However, due to an increase in remote usage, the file server is running out of capacity.

.

Which solution will meet these requirements?

A. Migrate the file server to an Amazon EC2 instance in a public subnet. Configure the security group to limit inbound traffic to the employees' IP addresses.

B. Migrate the files to an Amazon FSx for Windows File Server file system. Integrate the Amazon FSx file system with the on-premises Active Directory. Configure AWS Client VPN. Most Voted

C. Migrate the files to Amazon S3, and create a private VPC endpoint. Create a signed URL to allow download.

D. Migrate the files to Amazon S3, and create a public VPC endpoint. Allow employees to sign on with AWS IAM Identity Center (AWS Single Sign-On).

Correct Answer: B

Community vote distribution

B (93%) 7%

Comments

elearningtakai Highly Voted 3 years, 2 months ago

Selected Answer: B

This solution addresses the need for secure access to confidential and sensitive files, as well as the increase in remote usage. Migrating the files to Amazon FSx for Windows File Server provides a scalable, fully managed file storage solution in the AWS Cloud that is accessible from on-premises and cloud environments. Integration with the on-premises Active Directory allows for a consistent user experience and centralized access control. AWS Client VPN provides a secure and managed VPN solution that can be used by employees to access the files securely.

upvoted 9 times

TariqKipkemei Highly Voted 2 years, 8 months ago

Selected Answer: B

Windows file server = Amazon FSx for Windows File Server file system
Files can be accessed only by authorized users = On-premises Active Directory

upvoted 5 times

JA2018 1 year, 6 months ago

"Data must be downloaded securely to users' devices" => AWS Client VPN

upvoted 2 times

NayeraB Most Recent 2 years, 3 months ago

Selected Answer: B

My money is on B, but it's still not mentioned that the customer used an on-prem Active Directory.

upvoted 3 times

pentium75 2 years, 5 months ago

Selected Answer: B

C has "signed URL", everyone who has the URL could download. Plus, only B ensure the "must be downloaded securely" part by using VPN.

upvoted 5 times

BrijMohan08 2 years, 9 months ago

Selected Answer: C

Remember: The file server is running out of capacity.
upvoted 2 times

awsgeek75 2 years, 4 months ago
But then how do you download the files to user's machine in a secure way?
upvoted 2 times

pentium75 2 years, 5 months ago
That's why we're using FSX for Windows File Server in AWS.

"Signed URL to allow download" would allow everyone who has the URL to download the files, but we must "ensure that the files can be accessed only by authorized users". Plus, the "private VPC endpoint" is not really of use here, it's still S3 and the users are not in AWS.
upvoted 4 times

SkyZeroZx 3 years, 1 month ago
Selected Answer: B
B is the correct answer
upvoted 2 times

LuckyAro 3 years, 3 months ago
Selected Answer: B
B is the best solution for the given requirements. It provides a secure way for employees to access confidential and sensitive files from anywhere using AWS Client VPN. The Amazon FSx for Windows File Server file system is designed to provide native support for Windows file system features such as NTFS permissions, Active Directory integration, and Distributed File System (DFS). This means that the company can continue to use their on-premises Active Directory to manage user access to files.
upvoted 4 times

Bilalazure 3 years, 3 months ago
B is the correct answer
upvoted 2 times

jennyka76 3 years, 3 months ago
Answer - B
1- https://docs.aws.amazon.com/fsx/latest/WindowsGuide/what-is.html
2- https://docs.aws.amazon.com/fsx/latest/WindowsGuide/managing-storage-capacity.html
upvoted 2 times

Neha999 3 years, 3 months ago
B
Amazon FSx for Windows File Server file system
upvoted 4 times

Question #333 Topic 1

A company's application runs on Amazon EC2 instances behind an Application Load Balancer (ALB). The instances run in an Amazon EC2 Auto Scaling group across multiple Availability Zones. On the first day of every month at midnight, the application becomes much slower when the month-end financial calculation batch runs. This causes the CPU utilization of the EC2 instances to immediately peak to 100%, which disrupts the application.

What should a solutions architect recommend to ensure the application is able to handle the workload and avoid downtime?

A. Configure an Amazon CloudFront distribution in front of the ALB.

B. Configure an EC2 Auto Scaling simple scaling policy based on CPU utilization.

C. Configure an EC2 Auto Scaling scheduled scaling policy based on the monthly schedule. **Most Voted**

D. Configure Amazon ElastiCache to remove some of the workload from the EC2 instances.

Correct Answer: C

Community vote distribution

C (100%)

Comments

TariqKipkemei Highly Voted 1 year, 8 months ago

Selected Answer: C

'On the first day of every month at midnight' = Scheduled scaling policy

upvoted 6 times

elearningtakai Most Recent 2 years, 2 months ago

Selected Answer: C

By configuring a scheduled scaling policy, the EC2 Auto Scaling group can proactively launch additional EC2 instances before the CPU utilization peaks to 100%. This will ensure that the application can handle the workload during the month-end financial calculation batch, and avoid any disruption or downtime.

Configuring a simple scaling policy based on CPU utilization or adding Amazon CloudFront distribution or Amazon ElastiCache will not directly address the issue of handling the monthly peak workload.

upvoted 4 times

Steve_4542636 2 years, 3 months ago

Selected Answer: C

If the scaling were based on CPU or memory, it requires a certain amount of time above that threshhold, 5 minutes for example. That would mean the CPU would be at 100% for five minutes.

upvoted 4 times

LuckyAro 2 years, 3 months ago

Selected Answer: C

C: Configure an EC2 Auto Scaling scheduled scaling policy based on the monthly schedule is the best option because it allows for the proactive scaling of the EC2 instances before the monthly batch run begins. This will ensure that the application is able to handle the increased workload without experiencing downtime. The scheduled scaling policy can be configured to increase the number of instances in the Auto Scaling group a few hours before the batch run and then decrease the number of instances after the batch run is complete. This will ensure that the resources are available when needed and not wasted when not needed.

The most appropriate solution to handle the increased workload during the monthly batch run and avoid downtime would be to configure an EC2 Auto Scaling scheduled scaling policy based on the monthly schedule.

upvoted 3 times

LuckyAro 2 years, 3 months ago

Scheduled scaling policies allow you to schedule EC2 instance scaling events in advance based on a specified time and date. You can use this feature to plan for anticipated traffic spikes or seasonal changes in demand. By setting up scheduled scaling policies, you can ensure that you have the right number of instances running at the right time, thereby optimizing performance and reducing costs.

To set up a scheduled scaling policy in EC2 Auto Scaling, you need to specify the following:

Start time and date: The date and time when the scaling event should begin.

Desired capacity: The number of instances that you want to have running after the scaling event.

Recurrence: The frequency with which the scaling event should occur. This can be a one-time event or a recurring event, such as daily or weekly.
upvoted 2 times

bdp123 2 years, 3 months ago

Selected Answer: C
C is the correct answer as traffic spike is known
upvoted 2 times

jennyka76 2 years, 3 months ago
ANSWER - C
https://docs.aws.amazon.com/autoscaling/ec2/userguide/ec2-auto-scaling-scheduled-scaling.html
upvoted 3 times

Neha999 2 years, 3 months ago
C as the schedule of traffic spike is known beforehand.
upvoted 2 times

Question #334 Topic 1

A company wants to give a customer the ability to use on-premises Microsoft Active Directory to download files that are stored in Amazon S3. The customer's application uses an SFTP client to download the files.

Which solution will meet these requirements with the LEAST operational overhead and no changes to the customer's application?

A. Set up AWS Transfer Family with SFTP for Amazon S3. Configure integrated Active Directory authentication. Most Voted

B. Set up AWS Database Migration Service (AWS DMS) to synchronize the on-premises client with Amazon S3. Configure integrated Active Directory authentication.

C. Set up AWS DataSync to synchronize between the on-premises location and the S3 location by using AWS IAM Identity Center (AWS Single Sign-On).

D. Set up a Windows Amazon EC2 instance with SFTP to connect the on-premises client with Amazon S3. Integrate AWS Identity and Access Management (IAM).

Correct Answer: A

Community vote distribution

A (100%)

Comments

Steve_4542636 Highly Voted 3 years, 3 months ago

Selected Answer: A

SFTP, FTP - think "Transfer" during test time

upvoted 18 times

wsdasdasdqwdaw Most Recent 2 years, 7 months ago

LEAST operational overhead => A, D is much more operational overhead

upvoted 3 times

JA2018 1 year, 6 months ago

1. AWS Transfer Family: Fully managed service that allows customers to transfer files over SFTP, FTPS, and FTP directly into and out of Amazon S3.

2. Eliminates the need to manage any infrastructure for file transfer, which reduces operational overhead.

3. You can also configure the service to use an existing Active Directory for authentication, =>>> no changes need to be made to the customer's application.

4. Rule of thumbs for AWS exams: Opt for AWS fully managed services. :p :p :p

upvoted 1 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: A

SFTP, No changes to the customer's application? = AWS Transfer Family

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Transfer family is used for SFTP

upvoted 3 times

live_reply_developers 2 years, 10 months ago

SFTP -> transfer family

upvoted 3 times

antropaws 3 years ago

Selected Answer: A

A no doubt. Why the system gives B as the correct answer?

upvoted 2 times

lht 3 years, 1 month ago

Selected Answer: A

just A

upvoted 2 times

LuckyAro 3 years, 3 months ago

Selected Answer: A

AWS Transfer Family

upvoted 3 times

LuckyAro 3 years, 3 months ago

AWS Transfer Family is a fully managed service that allows customers to transfer files over SFTP, FTPS, and FTP directly into and out of Amazon S3. It eliminates the need to manage any infrastructure for file transfer, which reduces operational overhead. Additionally, the service can be configured to use an existing Active Directory for authentication, which means that no changes need to be made to the customer's application.

upvoted 3 times

bdp123 3 years, 3 months ago

Selected Answer: A

Transfer family is used for SFTP

upvoted 2 times

TungPham 3 years, 3 months ago

Selected Answer: A

using AWS Batch to LEAST operational overhead
and have SFTP to no changes to the customer's application

https://aws.amazon.com/vi/blogs/architecture/managed-file-transfer-using-aws-transfer-family-and-amazon-s3/

upvoted 3 times

Bhawesh 3 years, 3 months ago

Selected Answer: A

A. Set up AWS Transfer Family with SFTP for Amazon S3. Configure integrated Active Directory authentication.

https://docs.aws.amazon.com/transfer/latest/userguide/directory-services-users.html

upvoted 4 times

Question #335 *Topic 1*

A company is experiencing sudden increases in demand. The company needs to provision large Amazon EC2 instances from an Amazon Machine Image (AMI). The instances will run in an Auto Scaling group. The company needs a solution that provides minimum initialization latency to meet the demand.

Which solution meets these requirements?

A. Use the aws ec2 register-image command to create an AMI from a snapshot. Use AWS Step Functions to replace the AMI in the Auto Scaling group.

B. Enable Amazon Elastic Block Store (Amazon EBS) fast snapshot restore on a snapshot. Provision an AMI by using the snapshot. Replace the AMI in the Auto Scaling group with the new AMI. **Most Voted**

C. Enable AMI creation and define lifecycle rules in Amazon Data Lifecycle Manager (Amazon DLM). Create an AWS Lambda function that modifies the AMI in the Auto Scaling group.

D. Use Amazon EventBridge to invoke AWS Backup lifecycle policies that provision AMIs. Configure Auto Scaling group capacity limits as an event source in EventBridge.

Correct Answer: B

Community vote distribution

B (95%) 3%

Comments

danielklein09 Highly Voted 3 years ago
readed the question 5 times, didn't understood a thing :(
upvoted 66 times

Guru4Cloud 2 years, 9 months ago
Me too
upvoted 4 times

lostmagnet001 2 years, 4 months ago
the same here!
upvoted 1 times

elmyth 1 year, 9 months ago
Me too(((terrible question
upvoted 2 times

bdp123 Highly Voted 3 years, 3 months ago
Selected Answer: B
Enabling Amazon Elastic Block Store (Amazon EBS) fast snapshot restore on a snapshot allows you to quickly create a new Amazon Machine Image (AMI) from a snapshot, which can help reduce the initialization latency when provisioning new instances. Once the AMI is provisioned, you can replace the AMI in the Auto Scaling group with the new AMI. This will ensure that new instances are launched from the updated AMI and are able to meet the increased demand quickly.
upvoted 16 times

LeonSauveterre Most Recent 1 year, 6 months ago
Selected Answer: B
The question focuses on reducing initialization latency when launching large EC2 instances in an Auto Scaling group during sudden demand spikes.

So B with fast snapshot is obviously the choice. But without the choices listed there, I would simply integrate a Warm Pool for the Auto Scaling group with pre-initialized instances based on an optimized AMI :)
upvoted 3 times

awsgeek75 2 years, 5 months ago

Selected Answer: B

The question wording is pretty weird but the only thing of value is latency during initialisation which makes B the correct option.
https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ebs-fast-snapshot-restore.html

A only helps with creating the AMI
C and D will probably work (ambiguous language) but won't handle initialising latency issues.

upvoted 6 times

farnamjam 2 years, 5 months ago

Selected Answer: B

Fast Snapshot Restore (FSR)
• Force full initialization of snapshot to have no
latency on the first use

upvoted 3 times

pentium75 2 years, 5 months ago

Selected Answer: B

"Fast snapshot restore" = pre-warmed snapshot
AMI from such a snapshot is pre-warmed AMI

upvoted 6 times

master9 2 years, 5 months ago

Selected Answer: D

Amazon Data Lifecycle Manager (DLM) is a feature of Amazon EBS that automates the creation, retention, and deletion of snapshots, which are used to back up your Amazon EBS volumes. With DLM, you can protect your data by implementing a backup strategy that aligns with your business requirements.

You can create lifecycle policies to automate snapshot management. Each policy includes a schedule of when to create snapshots, a retention rule with a defined period to retain each snapshot, and a set of Amazon EBS volumes to assign to the policy.

This service helps simplify the management of your backups, ensure compliance, and reduce costs.

upvoted 1 times

pentium75 2 years, 5 months ago

We're not asked to "simplify the management of our backups, ensure compliance, and reduce costs", we're asked to "provide minimum initializatior latency" for an auto-scaling group.

upvoted 2 times

master9 2 years, 5 months ago

Sorry, its "C" and not "D"

upvoted 1 times

Nisarg2121 2 years, 5 months ago

Selected Answer: B

b is correct

upvoted 1 times

meowruki 2 years, 6 months ago

B. Enable Amazon Elastic Block Store (Amazon EBS) fast snapshot restore on a snapshot. Provision an AMI by using the snapshot. Replace the AMI in the Auto Scaling group with the new AMI.

Here's the reasoning:

Amazon EBS Fast Snapshot Restore: This feature allows you to quickly create new EBS volumes (and subsequently AMIs) from snapshots. Fast Snapshot Restore optimizes the initialization process by pre-warming the snapshots, reducing the time it takes to create volumes from those snapshots.

Provision an AMI using the snapshot: By using fast snapshot restore, you can efficiently provision an AMI from the pre-warmed snapshot, minimizing the initialization latency.

Replace the AMI in the Auto Scaling group: This allows you to update the instances in the Auto Scaling group with the new AMI efficiently, ensuring that the new instances are launched with minimal delay.

upvoted 2 times

meowruki 2 years, 6 months ago

Option A (Use aws ec2 register-image command and AWS Step Functions): While this approach can be used to automate the creation of an AMI and update the Auto Scaling group, it may not offer the same level of optimization for initialization latency as Amazon EBS fast snapshot restore.

Option C (Enable AMI creation and define lifecycle rules in Amazon Data Lifecycle Manager, create a Lambda function): While Amazon DLM can help manage the lifecycle of your AMIs, it might not provide the same level of speed and responsiveness needed for sudden increases in demand.

Option D (Use Amazon EventBridge and AWS Backup): AWS Backup is primarily designed for backup and recovery, and it might not be as optimizec for quickly provisioning instances in response to sudden demand spikes. EventBridge can be used for event-driven architectures, but in this context, it might introduce unnecessary complexity.

upvoted 2 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: B

Enable Amazon Elastic Block Store (Amazon EBS) fast snapshot restore on a snapshot. Provision an AMI by using the snapshot. Replace the AMI in the Auto Scaling group with the new AMI

upvoted 2 times

kambarami 2 years, 8 months ago

Pleaw3 reword 5he question. Can not understand a thing!

upvoted 1 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

Enable EBS fast snapshot restore on a snapshot
Create an AMI from the snapshot
Replace the AMI used by the Auto Scaling group with this new AMI

The key points:

° Need to launch large EC2 instances quickly from an AMI in an Auto Scaling group
° Looking to minimize instance initialization latency

upvoted 3 times

antropaws 3 years ago

Selected Answer: B

B most def

upvoted 2 times

elearningtakai 3 years, 2 months ago

Selected Answer: B

B: "EBS fast snapshot restore": minimizes initialization latency. This is a good choice.

upvoted 3 times

Zox42 3 years, 2 months ago

Selected Answer: B

https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ebs-fast-snapshot-restore.html

upvoted 3 times

geekgirl22 3 years, 3 months ago

Keyword, minimize initilization latency == snapshot. A and B have snapshots in them, but B is the one that makes sense.
C has DLP that can create machines from AMI, but that does not talk about latency and snapshots.

upvoted 4 times

LuckyAro 3 years, 3 months ago

Selected Answer: B

Enabling Amazon Elastic Block Store (Amazon EBS) fast snapshot restore on a snapshot allows for rapid restoration of EBS volumes from snapshots. This reduces the time required to create an AMI from a snapshot, which is useful for quickly provisioning large Amazon EC2 instances.

Provisioning an AMI by using the fast snapshot restore feature is a fast and efficient way to create an AMI. Once the AMI is created, it can be replaced in the Auto Scaling group without any downtime or disruption to running instances.

upvoted 2 times

Question #336 *Topic 1*

A company hosts a multi-tier web application that uses an Amazon Aurora MySQL DB cluster for storage. The application tier is hosted on Amazon EC2 instances. The company's IT security guidelines mandate that the database credentials be encrypted and rotated every 14 days.

What should a solutions architect do to meet this requirement with the LEAST operational effort?

A. Create a new AWS Key Management Service (AWS KMS) encryption key. Use AWS Secrets Manager to create a new secret that uses the KMS key with the appropriate credentials. Associate the secret with the Aurora DB cluster. Configure a custom rotation period of 14 days. **Most Voted**

B. Create two parameters in AWS Systems Manager Parameter Store: one for the user name as a string parameter and one that uses the SecureString type for the password. Select AWS Key Management Service (AWS KMS) encryption for the password parameter, and load these parameters in the application tier. Implement an AWS Lambda function that rotates the password every 14 days.

C. Store a file that contains the credentials in an AWS Key Management Service (AWS KMS) encrypted Amazon Elastic File System (Amazon EFS) file system. Mount the EFS file system in all EC2 instances of the application tier. Restrict the access to the file on the file system so that the application can read the file and that only super users can modify the file. Implement an AWS Lambda function that rotates the key in Aurora every 14 days and writes new credentials into the file.

D. Store a file that contains the credentials in an AWS Key Management Service (AWS KMS) encrypted Amazon S3 bucket that the application uses to load the credentials. Download the file to the application regularly to ensure that the correct credentials are used. Implement an AWS Lambda function that rotates the Aurora credentials every 14 days and uploads these credentials to the file in the S3 bucket.

Correct Answer: A

Community vote distribution

A (100%)

Comments

elearningtakai Highly Voted 3 years, 2 months ago

Selected Answer: A

AWS Secrets Manager allows you to easily rotate, manage, and retrieve database credentials, API keys, and other secrets throughout their lifecycle. With this service, you can automate the rotation of secrets, such as database credentials, on a schedule that you choose. The solution allows you to create a new secret with the appropriate credentials and associate it with the Aurora DB cluster. You can then configure a custom rotation period of 14 days to ensure that the credentials are automatically rotated every two weeks, as required by the IT security guidelines. This approach requires the least amount of operational effort as it allows you to manage secrets centrally without modifying your application code or infrastructure.

upvoted 7 times

jennyka76 Highly Voted 3 years, 3 months ago

Answer is A
To implement password rotation lifecycles, use AWS Secrets Manager. You can rotate, manage, and retrieve database credentials, API keys, and other secrets throughout their lifecycle using Secrets Manager.
https://aws.amazon.com/blogs/security/how-to-use-aws-secrets-manager-rotate-credentials-amazon-rds-database-types-oracle/

upvoted 5 times

MJ45 Most Recent 1 year, 5 months ago

Selected Answer: A

Most questions related to rotating credentials - AWS Secrets Manager
AWS System Parameter Store CANNOT rotate credentials.

upvoted 1 times

LeonSauveterre 1 year, 6 months ago

Selected Answer: A

Well... I would simply rule out all the options with "Implement an AWS Lambda function" given that we need a solution with least operational effort. Why manually compose codes to rotate when you can achieve the same goal automatically?

upvoted 1 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: A

Create a new AWS Key Management Service (AWS KMS) encryption key. Use AWS Secrets Manager to create a new secret that uses the KMS key with the appropriate credentials. Associate the secret with the Aurora DB cluster. Configure a custom rotation period of 14 days

upvoted 3 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: A

Use AWS Secrets Manager to store the Aurora credentials as a secret
Encrypt the secret with a KMS key
Configure 14 day automatic rotation for the secret
Associate the secret with the Aurora DB cluster
The key points:

Aurora MySQL credentials must be encrypted and rotated every 14 days
Want to minimize operational effort

upvoted 3 times

elearningtakai 3 years, 2 months ago

Selected Answer: A

A: AWS Secrets Manager. Simply this supported rotate feature, and secure to store credentials instead of EFS or S3.

upvoted 2 times

Steve_4542636 3 years, 3 months ago

Selected Answer: A

Voting A

upvoted 2 times

LuckyAro 3 years, 3 months ago

Selected Answer: A

A proposes to create a new AWS KMS encryption key and use AWS Secrets Manager to create a new secret that uses the KMS key with the appropriate credentials. Then, the secret will be associated with the Aurora DB cluster, and a custom rotation period of 14 days will be configured. AWS Secrets Manager will automate the process of rotating the database credentials, which will reduce the operational effort required to meet the IT security guidelines.

upvoted 2 times

Neha999 3 years, 3 months ago

A
https://www.examtopics.com/discussions/amazon/view/59985-exam-aws-certified-solutions-architect-associate-saa-c02/

upvoted 2 times

Question #337 Topic 1

A company has deployed a web application on AWS. The company hosts the backend database on Amazon RDS for MySQL with a primary DB instance and five read replicas to support scaling needs. The read replicas must lag no more than 1 second behind the primary DB instance. The database routinely runs scheduled stored procedures.

As traffic on the website increases, the replicas experience additional lag during periods of peak load. A solutions architect must reduce the replication lag as much as possible. The solutions architect must minimize changes to the application code and must minimize ongoing operational overhead.

Which solution will meet these requirements?

A. Migrate the database to Amazon Aurora MySQL. Replace the read replicas with Aurora Replicas, and configure Aurora Auto Scaling. Replace the stored procedures with Aurora MySQL native functions. Most Voted

B. Deploy an Amazon ElastiCache for Redis cluster in front of the database. Modify the application to check the cache before the application queries the database. Replace the stored procedures with AWS Lambda functions.

C. Migrate the database to a MySQL database that runs on Amazon EC2 instances. Choose large, compute optimized EC2 instances for all replica nodes. Maintain the stored procedures on the EC2 instances.

D. Migrate the database to Amazon DynamoDB. Provision a large number of read capacity units (RCUs) to support the required throughput, and configure on-demand capacity scaling. Replace the stored procedures with DynamoDB streams.

Correct Answer: A

Community vote distribution

A (92%) 8%

Comments

fkie4 Highly Voted 3 years, 3 months ago

i hate this kind of question

upvoted 70 times

asoli Highly Voted 3 years, 2 months ago

Selected Answer: A

Using Cache required huge changes in the application. Several things need to change to use cache in front of the DB in the application. So, option B is not correct.
Aurora will help to reduce replication lag for read replica

upvoted 13 times

sheilawu Most Recent 2 years ago

Selected Answer: A

You need to read the question carefully.
The solutions architect must minimize changes to the application code = therefore A
If this question without this statement, B will be a better choice.

upvoted 4 times

JackyCCK 2 years, 2 months ago

minimize ongoing operational overhead = Not B
Using ElastiCache require app change

upvoted 4 times

awsgeek75 2 years, 5 months ago

Selected Answer: A

AWS Aurora and Native Functions are least application changes while providing better performance and minimum latency.
https://aws.amazon.com/rds/aurora/faqs/

B, C, D require lots of changes to the application so relatively speaking A is least code change and least maintenance/operational overhead.
upvoted 9 times

JA2018 1 year, 6 months ago
from the same URL:

Is Amazon Aurora MySQL compatible?
Amazon Aurora is drop-in compatible with existing MySQL open-source databases and adds support for new releases regularly. This means you car easily migrate MySQL databases to and from Aurora using standard import/export tools or snapshots. It also means that most of the code, applications, drivers, and tools you already use with MySQL databases today can be used with Aurora with little or no change. This makes it easy to move applications between the two engines.

You can see the current Amazon Aurora MySQL release compatibility information in the documentation.
upvoted 1 times

pentium75 2 years, 5 months ago
Selected Answer: A
A: Minimal changes to the application code, < 1 second lag
B: Does not address the replication lag issue at all, requires code changes and adds overhead
C: Moving from managed RDS to self-managed database on EC2 is ADDING, not minimizing, overhead, PLUS it does not address the replication lag issue
D: DynamoDB is a NoSQL DB, would require MASSIVE changes to application code and probably even application logic
upvoted 5 times

Murtadhaceit 2 years, 6 months ago
Selected Answer: A
imho, B is not valid because it involves extra coding and the question specifically mentions no more coding. Therefore, replacing the current db with another one is not considered as more coding.
upvoted 3 times

TariqKipkemei 2 years, 8 months ago
Selected Answer: A
Migrate the database to Amazon Aurora MySQL. Replace the read replicas with Aurora Replicas, and configure Aurora Auto Scaling. Replace the stored procedures with Aurora MySQL native functions
upvoted 2 times

Guru4Cloud 2 years, 9 months ago
Selected Answer: A
Migrate the RDS MySQL database to Amazon Aurora MySQL
Use Aurora Replicas for read scaling instead of RDS read replicas
Configure Aurora Auto Scaling to handle load spikes
Replace stored procedures with Aurora MySQL native functions
upvoted 2 times

MrAWSAssociate 2 years, 11 months ago
Selected Answer: A
First, Elasticache involves heavy change on application code. The question mentioned that "he solutions architect must minimize changes to the application code". Therefore B is not suitable and A is more appropriate for the question requirement.
upvoted 3 times

aaroncelestin 2 years, 9 months ago
... but migrating their ENTIRE prod database and its replicas to a new platform is not a heavy change?
upvoted 3 times

KMohsoe 3 years ago
Selected Answer: B
Why not B? Please explain to me.
upvoted 2 times

Terion 2 years, 8 months ago
It wouldn't have the most up to date info since it must no lag in relation to the main DB
upvoted 2 times

pentium75 2 years, 5 months ago
How would adding a cache "reduce the replication lag" between the primary instance and the read replicas? Plus, it would require "changes to the application code" that we want to avoid. The "AWS Lambda functions" would create "ongoing operational overhead" that we're also asked to avoid
upvoted 2 times

kaushald 3 years, 3 months ago

Option A is the most appropriate solution for reducing replication lag without significant changes to the application code and minimizing ongoing operational overhead. Migrating the database to Amazon Aurora MySQL allows for improved replication performance and higher scalability compared to Amazon RDS for MySQL. Aurora Replicas provide faster replication, reducing the replication lag, and Aurora Auto Scaling ensures that there are enough Aurora Replicas to handle the incoming traffic. Additionally, Aurora MySQL native functions can replace the stored procedures, reducing the load on the database and improving performance.

Option B is not the best solution since adding an ElastiCache for Redis cluster does not address the replication lag issue, and the cache may not have the most up-to-date information. Additionally, replacing the stored procedures with AWS Lambda functions adds additional complexity and may not improve performance.

upvoted 5 times

njufi 2 years, 2 months ago

I agree with your explanation. Additionally, considering the requirement that "the read replicas must lag no more than 1 second behind the primary DB instance," it's crucial to ensure that Elasticache for Redis also maintains this tight synchronization window. This implies that the main RDS instance would need to synchronize an additional database, potentially exacerbating lag during peak times rather than alleviating it.

upvoted 1 times

Nel8 3 years, 3 months ago

Selected Answer: B

By using ElastiCache you avoid a lot of common issues you might encounter. ElastiCache is a database caching solution. ElastiCache Redis per se, supports failover and Multi-AZ. And Most of all, ElastiCache is well suited to place in front of RDS.

Migrating a database such as option A, requires operational overhead.

upvoted 2 times

pentium75 2 years, 5 months ago

Database migration is one-time work, NOT "operational overhead". Plus, RDS for MySQL to Aurora with MySQL compatibility is not a big deal, and "minimizes changes to the application code" as requested.

upvoted 2 times

bdp123 3 years, 3 months ago

Selected Answer: A

Aurora can have up to 15 read replicas - much faster than RDS
https://aws.amazon.com/rds/aurora/

upvoted 5 times

ChrisG1454 3 years, 3 months ago

" As a result, all Aurora Replicas return the same data for query results with minimal replica lag. This lag is usually much less than 100 milliseconds after the primary instance has written an update "

Reference:
https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Replication.html

upvoted 3 times

ChrisG1454 3 years, 3 months ago

You can invoke an Amazon Lambda function from an Amazon Aurora MySQL-Compatible Edition DB cluster with the "native function"....

https://docs.amazonaws.cn/en_us/AmazonRDS/latest/AuroraUserGuide/AuroraMySQL.Integrating.Lambda.html

upvoted 1 times

jennyka76 3 years, 3 months ago

Answer - A
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_PostgreSQL.Replication.ReadReplicas.html
----
You can scale reads for your Amazon RDS for PostgreSQL DB instance by adding read replicas to the instance. As with other Amazon RDS database engines, RDS for PostgreSQL uses the native replication mechanisms of PostgreSQL to keep read replicas up to date with changes on the source DB. For general information about read replicas and Amazon RDS, see Working with read replicas.

upvoted 3 times

Question #338 Topic 1

A solutions architect must create a disaster recovery (DR) plan for a high-volume software as a service (SaaS) platform. All data for the platform is stored in an Amazon Aurora MySQL DB cluster.

The DR plan must replicate data to a secondary AWS Region.

Which solution will meet these requirements MOST cost-effectively?

A. Use MySQL binary log replication to an Aurora cluster in the secondary Region. Provision one DB instance for the Aurora cluster in the secondary Region.

B. Set up an Aurora global database for the DB cluster. When setup is complete, remove the DB instance from the secondary Region. Most Voted

C. Use AWS Database Migration Service (AWS DMS) to continuously replicate data to an Aurora cluster in the secondary Region. Remove the DB instance from the secondary Region.

D. Set up an Aurora global database for the DB cluster. Specify a minimum of one DB instance in the secondary Region.

Correct Answer: B

Community vote distribution

B (57%) D (37%) 6%

Comments

awsgeek75 Highly Voted 2 years, 5 months ago

Selected Answer: B

I originally went for D but now I think B is correct. D is active-active cluster so whereas B is active-passive (headless cluster) so it is cheaper than D.

https://aws.amazon.com/blogs/database/achieve-cost-effective-multi-region-resiliency-with-amazon-aurora-global-database-headless-clusters/

upvoted 20 times

jennyka76 Highly Voted 3 years, 3 months ago

Answer - A

https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/AuroraMySQL.Replication.CrossRegion.html

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Before you begin

Before you can create an Aurora MySQL DB cluster that is a cross-Region read replica, you must turn on binary logging on your source Aurora MySQL DB cluster. Cross-region replication for Aurora MySQL uses MySQL binary replication to replay changes on the cross-Region read replica DB cluster.

upvoted 9 times

ChrisG1454 3 years, 3 months ago

The question states " The DR plan must replicate data to a "secondary" AWS Region."

In addition to Aurora Replicas, you have the following options for replication with Aurora MySQL:

Aurora MySQL DB clusters in different AWS Regions.

You can replicate data across multiple Regions by using an Aurora global database. For details, see High availability across AWS Regions with Auror global databases

You can create an Aurora read replica of an Aurora MySQL DB cluster in a different AWS Region, by using MySQL binary log (binlog) replication. Each cluster can have up to five read replicas created this way, each in a different Region.

https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Replication.html

upvoted 1 times

ChrisG1454 3 years, 3 months ago

The question is asking for the most cost-effective solution.
Aurora global databases are more expensive.

https://aws.amazon.com/rds/aurora/pricing/
upvoted 1 times

leoattf 3 years, 3 months ago
On this same URL you provided, there is a note highlighted, stating the following:
"Replication from the primary DB cluster to all secondaries is handled by the Aurora storage layer rather than by the database engine, so lag time for replicating changes is minimal—typically, less than 1 second. Keeping the database engine out of the replication process means that the database engine is dedicated to processing workloads. It also means that you don't need to configure or manage the Aurora MySQL binlog (binary logging) replication."

So, answer should be A
upvoted 2 times

leoattf 3 years, 3 months ago
Correction: So, answer should be D
upvoted 3 times

Clouddon Most Recent 11 months ago
Selected Answer: D
D is correct
Set up an Aurora global database for the DB cluster. When setup is complete, remove the DB instance from the secondary Region.
The reason why B us incorrect!
Reason it's incorrect: Removing the DB instance from the secondary region would defeat the purpose of having a disaster recovery setup. Without a DB instance in the secondary region, you lose the ability to quickly failover in case of a regional outage, which is critical for DR.
upvoted 1 times

In2718 11 months, 1 week ago
Selected Answer: D
This makes the answer D - "a high-volume software as a service (SaaS) platform."
upvoted 1 times

Yak_Yeti 1 year, 1 month ago
Selected Answer: B
When you set up an Aurora Global Database, the storage layer (Aurora Storage) is continuously and automatically replicated across Regions — independent of whether you have DB instances (readers/writers) running in the secondary Region.
If you remove the DB instance from the secondary Region:
The replicated storage remains.
No compute (no DB instances) = No cost for compute, only storage costs.
BUT:
In a real failover event, you must promote the secondary cluster.
When you promote, you must first create a new DB instance in the secondary Region from the existing storage. This takes some time (typically a few minutes), because Aurora has to spin up the DB instance.
upvoted 1 times

tch 1 year, 3 months ago
Selected Answer: D
Answer could be D, as this is DR plan, answer B is ask to remove the DB instance which might not suitable for DR plan..... if you remove it... how to replicate data again? you setup again for next replication ?
upvoted 2 times

FlyingHawk 1 year, 4 months ago
Selected Answer: D
Giving the requirement is the DR plan for a high-volume software as a service (SaaS) platform, the RTO and RPO should be small, needs a quick recovery, D(warm standby) is a better choice than A (cold standby, pilot light)
upvoted 3 times

LeonSauveterre 1 year, 6 months ago
Selected Answer: D
A - No. Higher cost & higher latency.
B - No. A DB instance is mandatory in each Region for the global database to function.
C - Removing would make DR incomplete, because the secondary Region cannot process failovers or support read replicas.
D - Cost slightly higher, yes, but feasible & cost less than A.

Overall, I don't like this question ...
upvoted 2 times

theamachine 2 years ago

Selected Answer: B
Aurora Global Databases offer a cost-effective way to replicate data to a secondary region for disaster recovery. By removing the secondary DB instance after setup, you only pay for storage and minimal compute resources.
upvoted 2 times

thewalker 2 years, 4 months ago
Selected Answer: D
https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/aurora-global-database.html#aurora-global-database.advantages
upvoted 1 times

awsgeek75 2 years, 4 months ago
Wrong, while D will work, B is cheaper. This question is about DR, not cross region scaling
upvoted 2 times

upliftinghut 2 years, 5 months ago
Selected Answer: D
B is more cost-effective however because this is DR so when the region fails => still need a DB to fail over and if setting up a DB from snapshot at the time of failure will be risky => D is the answer
upvoted 3 times

pentium75 2 years, 5 months ago
Selected Answer: B
"Achieve cost-effective multi-Region resiliency with Amazon Aurora Global Database headless clusters" is exactly the topic here. "A headless secondary Amazon Aurora database cluster is one without a database instance. This type of configuration can lower expenses for an Aurora global database."

https://aws.amazon.com/blogs/database/achieve-cost-effective-multi-region-resiliency-with-amazon-aurora-global-database-headless-clusters/
upvoted 8 times

minagaboya 2 years, 7 months ago
shd be D i guess .. . Migrating the database to Amazon Aurora MySQL allows for improved replication performance and higher scalability compared to Amazon RDS for MySQL. Aurora Replicas provide faster replication, reducing the replication lag, and Aurora Auto Scaling ensures that there are enough Aurora Replicas to handle the incoming traffic. Additionally, Aurora MySQL native functions can replace the stored procedures, reducing the load on the database and improving performance.

Option B is not the best solution since adding an ElastiCache for Redis cluster does not address the replication lag issue, and the cache may not have the most up-to-date information. Additionally, replacing the stored procedures with AWS Lambda functions adds additional complexity and may not improve performance.
upvoted 2 times

pentium75 2 years, 5 months ago
This is about a different question
upvoted 4 times

TariqKipkemei 2 years, 8 months ago
Selected Answer: D
Set up an Aurora global database for the DB cluster. Specify a minimum of one DB instance in the secondary Region
upvoted 1 times

vini15 2 years, 10 months ago
should be B for most cost effective solution.
see the link - Achieve cost-effective multi-Region resiliency with Amazon Aurora Global Database headless clusters
https://aws.amazon.com/blogs/database/achieve-cost-effective-multi-region-resiliency-with-amazon-aurora-global-database-headless-clusters/
upvoted 2 times

luisgu 3 years ago
Selected Answer: B
MOST cost-effective --> B
See section "Creating a headless Aurora DB cluster in a secondary Region" on the link
https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Replication.html
"Although an Aurora global database requires at least one secondary Aurora DB cluster in a different AWS Region than the primary, you can use a headless configuration for the secondary cluster. A headless secondary Aurora DB cluster is one without a DB instance. This type of configuration can lower expenses for an Aurora global database. In an Aurora DB cluster, compute and storage are decoupled. Without the DB instance, you're not charged for compute, only for storage. If it's set up correctly, a headless secondary's storage volume is kept in-sync with the primary Aurora DB cluster."
upvoted 7 times

bsbs1234 2 years, 8 months ago
upvoted your message, but still think D is correct. Because the question is to design a DR plan.In case of DR, B need to create an instance in DR region manually.

upvoted 2 times

Abhineet9148232 3 years, 3 months ago

Selected Answer: D

D: With Amazon Aurora Global Database, you pay for replicated write I/Os between the primary Region and each secondary Region (in this case 1).

Not A because it achieves the same, would be equally costly and adds overhead.

upvoted 3 times

Question #339 Topic 1

A company has a custom application with embedded credentials that retrieves information from an Amazon RDS MySQL DB instance. Management says the application must be made more secure with the least amount of programming effort.

What should a solutions architect do to meet these requirements?

A. Use AWS Key Management Service (AWS KMS) to create keys. Configure the application to load the database credentials from AWS KMS. Enable automatic key rotation.

B. Create credentials on the RDS for MySQL database for the application user and store the credentials in AWS Secrets Manager. Configure the application to load the database credentials from Secrets Manager. Create an AWS Lambda function that rotates the credentials in Secret Manager.

C. Create credentials on the RDS for MySQL database for the application user and store the credentials in AWS Secrets Manager. Configure the application to load the database credentials from Secrets Manager. Set up a credentials rotation schedule for the application user in the RDS for MySQL database using Secrets Manager. Most Voted

D. Create credentials on the RDS for MySQL database for the application user and store the credentials in AWS Systems Manager Parameter Store. Configure the application to load the database credentials from Parameter Store. Set up a credentials rotation schedule for the application user in the RDS for MySQL database using Parameter Store.

Correct Answer: C

Community vote distribution

C (100%)

Comments

cloudbusting Highly Voted 3 years, 3 months ago

Parameter Store does not provide automatic credential rotation.

upvoted 16 times

Bhawesh Highly Voted 3 years, 3 months ago

Selected Answer: C

C. Create credentials on the RDS for MySQL database for the application user and store the credentials in AWS Secrets Manager. Configure the application to load the database credentials from Secrets Manager. Set up a credentials rotation schedule for the application user in the RDS for MySQL database using Secrets Manager.

https://www.examtopics.com/discussions/amazon/view/46483-exam-aws-certified-solutions-architect-associate-saa-c02/

upvoted 12 times

JA2018 Most Recent 1 year, 6 months ago

Selected Answer: C

Management says the application must be made more secure with the least amount of programming effort.

IMHO, this will rule out Lambda for this STEM.

upvoted 1 times

Gape4 1 year, 11 months ago

Selected Answer: C

credentials from Secrets Manager...

upvoted 3 times

d401c0d 2 years, 2 months ago

question is asking for "more secure with the least amount of programming effort." = Secrets Manager + Secretes Manager's built in rotation schedule instead of Lambda.

upvoted 2 times

awsgeek75 2 years, 5 months ago

Selected Answer: C

A KMS is for encryption keys specifically so this is a long way of doing the credentials storage
B is too much work for rotation
C exactly what secrets manager is designed for
D You can do that if C wasn't an option

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: C

Store the RDS credentials in Secrets Manager
Configure the application to retrieve the credentials from Secrets Manager
Use Secrets Manager's built-in rotation to rotate the RDS credentials automatically

upvoted 2 times

Hades2231 2 years, 9 months ago

Selected Answer: C

Secrets Manager can handle the rotation, so no need for Lambda to rotate the keys.

upvoted 2 times

chen0305_099 2 years, 9 months ago

WHY NOT B ?

upvoted 1 times

StacyY 2 years, 10 months ago

B, we need lambda for password rotation, confirmed!

upvoted 2 times

Nikki013 2 years, 9 months ago

It is not needed for certain types RDS, including MySQL as Secrets Manager has built-in rotation capabilities for it:
https://aws.amazon.com/blogs/security/rotate-amazon-rds-database-credentials-automatically-with-aws-secrets-manager/

upvoted 3 times

Abrar2022 3 years ago

Selected Answer: C

If you need your DB to store credentials then use AWS Secret Manager. System Manager Paramater Store is for CloudFormation (no rotation)

upvoted 2 times

AlessandraSAA 3 years, 3 months ago

why it's not A?

upvoted 4 times

MssP 3 years, 2 months ago

It is asking for credentials, not for encryption keys.

upvoted 7 times

PoisonBlack 3 years, 1 month ago

So credentials rotation is secrets manager and key rotation is KMS?

upvoted 3 times

bdp123 3 years, 3 months ago

Selected Answer: C

https://aws.amazon.com/blogs/security/rotate-amazon-rds-database-credentials-automatically-with-aws-secrets-manager/

upvoted 2 times

LuckyAro 3 years, 3 months ago

Selected Answer: C

C is a valid solution for securing the custom application with the least amount of programming effort. It involves creating credentials on the RDS for MySQL database for the application user and storing them in AWS Secrets Manager. The application can then be configured to load the database credentials from Secrets Manager. Additionally, the solution includes setting up a credentials rotation schedule for the application user in the RDS fo MySQL database using Secrets Manager, which will automatically rotate the credentials at a specified interval without requiring any programming effort.

upvoted 4 times

bdp123 3 years, 3 months ago

Selected Answer: C

https://docs.aws.amazon.com/secretsmanager/latest/userguide/create_database_secret.html

upvoted 3 times

jennyka76 3 years, 3 months ago

Answer - C

https://ws.amazon.com/blogs/security/rotate-amazon-rds-database-credentials-automatically-with-aws-secrets-manager/

upvoted 4 times

Question #340 Topic 1

A media company hosts its website on AWS. The website application's architecture includes a fleet of Amazon EC2 instances behind an Application Load Balancer (ALB) and a database that is hosted on Amazon Aurora. The company's cybersecurity team reports that the application is vulnerable to SQL injection.

How should the company resolve this issue?

A. Use AWS WAF in front of the ALB. Associate the appropriate web ACLs with AWS WAF. **Most Voted**

B. Create an ALB listener rule to reply to SQL injections with a fixed response.

C. Subscribe to AWS Shield Advanced to block all SQL injection attempts automatically.

D. Set up Amazon Inspector to block all SQL injection attempts automatically.

Correct Answer: A

Community vote distribution

A (100%)

Comments

Bhawesh Highly Voted 2 years, 3 months ago

Selected Answer: A

A. Use AWS WAF in front of the ALB. Associate the appropriate web ACLs with AWS WAF.

SQL Injection - AWS WAF
DDoS - AWS Shield

upvoted 27 times

jennyka76 Highly Voted 2 years, 3 months ago

Answer - A
https://aws.amazon.com/premiumsupport/knowledge-center/waf-block-common-attacks/#:~:text=To%20protect%20your%20applications%20against,%2C%20query%20string%2C%20or%20URI.

Protect against SQL injection and cross-site scripting
To protect your applications against SQL injection and cross-site scripting (XSS) attacks, use the built-in SQL injection and cross-site scripting engines. Remember that attacks can be performed on different parts of the HTTP request, such as the HTTP header, query string, or URI. Configure the AWS WAF rules to inspect different parts of the HTTP request against the built-in mitigation engines.

upvoted 8 times

wsdasdasdqwdaw Most Recent 1 year, 7 months ago

AWS WAF - for SQL Injection ---> A

AWS Shield - for DDOS
Amazon Inspector - for automated security assessment, like known vulnerability

upvoted 4 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: A

° Use AWS WAF in front of the Application Load Balancer
° Configure appropriate WAF web ACLs to detect and block SQL injection patterns
The key points:
° Website hosted on EC2 behind an ALB with Aurora database
° Application is vulnerable to SQL injection attacks
° AWS WAF is designed to detect and block SQL injection and other common web exploits. It can be placed in front of the ALB to inspect all incoming requests. WAF rules can identify malicious SQL patterns and block them.

upvoted 2 times

KMohsoe 2 years ago

Selected Answer: A

SQL injection -> WAF

upvoted 2 times

lexotan 2 years, 1 month ago

Selected Answer: A

WAF is the right one

upvoted 2 times

akram_akram 2 years, 2 months ago

Selected Answer: A

SQL Injection - AWS WAF
DDoS - AWS Shield

upvoted 2 times

movva12 2 years, 2 months ago

Answer C - Shield Advanced (WAF + Firewall Manager)

upvoted 1 times

fkie4 2 years, 3 months ago

Selected Answer: A

It is A. I am happy to see Amazon gives out score like this...

upvoted 2 times

LuckyAro 2 years, 3 months ago

Selected Answer: A

AWS WAF is a managed service that protects web applications from common web exploits that could affect application availability, compromise security, or consume excessive resources. AWS WAF enables customers to create custom rules that block common attack patterns, such as SQL injection attacks.

By using AWS WAF in front of the ALB and associating the appropriate web ACLs with AWS WAF, the company can protect its website application from SQL injection attacks. AWS WAF will inspect incoming traffic to the website application and block requests that match the defined SQL injection patterns in the web ACLs. This will help to prevent SQL injection attacks from reaching the application, thereby improving the overall security posture of the application.

upvoted 3 times

LuckyAro 2 years, 3 months ago

B, C, and D are not the best solutions for this issue. Replying to SQL injections with a fixed response
(B) is not a recommended approach as it does not actually fix the vulnerability, but only masks the issue. Subscribing to AWS Shield Advanced
(C) is useful to protect against DDoS attacks but does not protect against SQL injection vulnerabilities. Amazon Inspector
(D) is a vulnerability assessment tool and can identify vulnerabilities but cannot block attacks in real-time.

upvoted 3 times

pbpally 2 years, 3 months ago

Selected Answer: A

Bhawesh answers it perfect so I'm avoiding redundancy but agree on it being A.

upvoted 2 times

Question #341 Topic 1

A company has an Amazon S3 data lake that is governed by AWS Lake Formation. The company wants to create a visualization in Amazon QuickSight by joining the data in the data lake with operational data that is stored in an Amazon Aurora MySQL database. The company wants to enforce column-level authorization so that the company's marketing team can access only a subset of columns in the database.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon EMR to ingest the data directly from the database to the QuickSight SPICE engine. Include only the required columns.

B. Use AWS Glue Studio to ingest the data from the database to the S3 data lake. Attach an IAM policy to the QuickSight users to enforce column-level access control. Use Amazon S3 as the data source in QuickSight.

C. Use AWS Glue Elastic Views to create a materialized view for the database in Amazon S3. Create an S3 bucket policy to enforce column-level access control for the QuickSight users. Use Amazon S3 as the data source in QuickSight.

D. Use a Lake Formation blueprint to ingest the data from the database to the S3 data lake. Use Lake Formation to enforce column-level access control for the QuickSight users. Use Amazon Athena as the data source in QuickSight. Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

K0nAn Highly Voted 2 years, 9 months ago

Selected Answer: D

This solution leverages AWS Lake Formation to ingest data from the Aurora MySQL database into the S3 data lake, while enforcing column-level access control for QuickSight users. Lake Formation can be used to create and manage the data lake's metadata and enforce security and governance policies, including column-level access control. This solution then uses Amazon Athena as the data source in QuickSight to query the data in the S3 data lake. This solution minimizes operational overhead by leveraging AWS services to manage and secure the data, and by using a standard query service (Amazon Athena) to provide a SQL interface to the data.

upvoted 15 times

jennyka76 Highly Voted 2 years, 9 months ago

Answer - D
https://aws.amazon.com/blogs/big-data/enforce-column-level-authorization-with-amazon-quicksight-and-aws-lake-formation/

upvoted 10 times

Charlesvg Most Recent 11 months, 3 weeks ago

Selected Answer: D

Option A require operationnal overhead, B & D does not mention Athena to connect to QuickSight which is required.
So option D

upvoted 1 times

awsgeek75 1 year, 10 months ago

Selected Answer: D

https://docs.aws.amazon.com/lake-formation/latest/dg/workflows-about.html

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: D

Use a Lake Formation blueprint to ingest data from the Aurora database into the S3 data lake
Leverage Lake Formation to enforce column-level access control for the marketing team
Use Amazon Athena as the data source in QuickSight
The key points:

Need to join S3 data lake data with Aurora MySQL data
Require column-level access controls for marketing team in QuickSight
Minimize operational overhead

upvoted 4 times

LuckyAro 2 years, 9 months ago

Selected Answer: D

Using a Lake Formation blueprint to ingest the data from the database to the S3 data lake, using Lake Formation to enforce column-level access control for the QuickSight users, and using Amazon Athena as the data source in QuickSight. This solution requires the least operational overhead as it utilizes the features provided by AWS Lake Formation to enforce column-level authorization, which simplifies the process and reduces the need for additional configuration and maintenance.

upvoted 5 times

Bhawesh 2 years, 9 months ago

Selected Answer: D

D. Use a Lake Formation blueprint to ingest the data from the database to the S3 data lake. Use Lake Formation to enforce column-level access control for the QuickSight users. Use Amazon Athena as the data source in QuickSight.

https://www.examtopics.com/discussions/amazon/view/80865-exam-aws-certified-solutions-architect-associate-saa-c02/

upvoted 3 times

Question #342 Topic 1

A transaction processing company has weekly scripted batch jobs that run on Amazon EC2 instances. The EC2 instances are in an Auto Scaling group. The number of transactions can vary, but the baseline CPU utilization that is noted on each run is at least 60%. The company needs to provision the capacity 30 minutes before the jobs run.

Currently, engineers complete this task by manually modifying the Auto Scaling group parameters. The company does not have the resources to analyze the required capacity trends for the Auto Scaling group counts. The company needs an automated way to modify the Auto Scaling group's desired capacity.

Which solution will meet these requirements with the LEAST operational overhead?

A. Create a dynamic scaling policy for the Auto Scaling group. Configure the policy to scale based on the CPU utilization metric. Set the target value for the metric to 60%.

B. Create a scheduled scaling policy for the Auto Scaling group. Set the appropriate desired capacity, minimum capacity, and maximum capacity. Set the recurrence to weekly. Set the start time to 30 minutes before the batch jobs run.

C. Create a predictive scaling policy for the Auto Scaling group. Configure the policy to scale based on forecast. Set the scaling metric to CPU utilization. Set the target value for the metric to 60%. In the policy, set the instances to pre-launch 30 minutes before the jobs run. **Most Voted**

D. Create an Amazon EventBridge event to invoke an AWS Lambda function when the CPU utilization metric value for the Auto Scaling group reaches 60%. Configure the Lambda function to increase the Auto Scaling group's desired capacity and maximum capacity by 20%.

Correct Answer: C

Community vote distribution

C (67%) B (31%) 2%

Comments

fkie4 Highly Voted 3 years, 3 months ago

Selected Answer: C

B is NOT correct. the question said "The company does not have the resources to analyze the required capacity trends for the Auto Scaling group counts.".
answer B said "Set the appropriate desired capacity, minimum capacity, and maximum capacity".
how can someone set desired capacity if he has no resources to analyze the required capacity.
Read carefully Amigo

upvoted 29 times

jjcode 2 years, 3 months ago

works loads can vary, how can you predict something that is random?

upvoted 2 times

ealpuche 3 years, 1 month ago

But you can make a vague estimation according to the resources used; you don't need to make machine learning models to do that. You only need common sense.

upvoted 2 times

Murtadhaceit 2 years, 6 months ago

Your explanation is contradicting your answer. Since "the company does not have the resources to analyze the required capacity trend for the ASG", how come they can create and ASG based on a historic trend?
C doesn't make sense for me.

upvoted 4 times

omoakin 3 years ago

scheduled scaling....

upvoted 4 times

neverdie Highly Voted 3 years, 2 months ago

Selected Answer: B

A scheduled scaling policy allows you to set up specific times for your Auto Scaling group to scale out or scale in. By creating a scheduled scaling policy for the Auto Scaling group, you can set the appropriate desired capacity, minimum capacity, and maximum capacity, and set the recurrence to weekly. You can then set the start time to 30 minutes before the batch jobs run, ensuring that the required capacity is provisioned before the jobs run.

Option C, creating a predictive scaling policy for the Auto Scaling group, is not necessary in this scenario since the company does not have the resources to analyze the required capacity trends for the Auto Scaling group counts. This would require analyzing the required capacity trends for the Auto Scaling group counts to determine the appropriate scaling policy.

upvoted 8 times

MssP 3 years, 2 months ago

Look at fkie4 comment... no way to know desired capacity!!! -> B not correct

upvoted 1 times

Lalo 3 years ago

the text says

1.-"A transaction processing company has weekly scripted batch jobs", there is a Schedule

2.-" The company does not have the resources to analyze the required capacity trends for the Auto Scaling " Do not use

the answer is B

upvoted 4 times

[Removed] 3 years, 2 months ago

B is correct. "Predictive scaling uses machine learning to predict capacity requirements based on historical data from CloudWatch.", meaning the company does not have to analyze the capacity trends themselves. https://docs.aws.amazon.com/autoscaling/ec2/userguide/ec2-auto-scaling-predictive-scaling.html

upvoted 3 times

JustEugen Most Recent 9 months ago

Selected Answer: C

The reason why I picked C, and not B, is because: "The company does not have the resources to analyze the required capacity trends for the Auto Scaling group counts".

If you would use B, you have to manually analyze min and max to always have that window, with C it will be done by AWS + we already have a history, so should not be the problem

upvoted 1 times

Charlesvg 11 months, 3 weeks ago

Selected Answer: B

Predictive scaling is for irregular workload based on known history. (no history here)

Schedulked scaling is for the workload is know (60% baseline) and planned (here once a week)

upvoted 1 times

bora4motion 1 year, 1 month ago

Selected Answer: B

C is a bit weird - it says "predictive"while the question says "there's no budget to analyze".

upvoted 1 times

bora4motion 1 year, 1 month ago

Selected Answer: B

B is correct

You know the schedule: weekly jobs → exact match.

You don’t need to analyze utilization trends → simple config.

1-time setup → minimal overhead.

C is wrong, at first glance, looks great.

❌ BUT it requires historical data and analysis — which the company cannot support.

Setup and tuning introduce overhead.

❌ Breaks the “company lacks resources to analyze trends” condition.

❌ Incorrect.

upvoted 2 times

9820ad3 1 year, 2 months ago

Selected Answer: B

B is correct answer ....

upvoted 1 times

Denise123 1 year, 5 months ago

Selected Answer: C

Option A - Dynamic scaling policy is reactive not proactive.
Option B - Scheduled policy is not suitable as it says the work loads can vary
Option C - Correct
Option D - Irrelevant

upvoted 1 times

EllenLiu 1 year, 5 months ago

Selected Answer: C

"The company does not have the resources to analyze the required capacity trends for the Auto Scaling group counts. " ==> this means this company can take advantage of cloudwatch metric and auto scaling on AWS to set up a predictive scaling group, so that no need to have any resources for this kind of task any more.??

upvoted 1 times

studydue 1 year, 7 months ago

Answer should be B
Pref: https://docs.aws.amazon.com/autoscaling/ec2/userguide/ec2-auto-scaling-predictive-scaling.html
Pref: https://docs.aws.amazon.com/autoscaling/ec2/userguide/predictive-scaling-policy-overview.html
keyword for this question: "s. The company does not have
the resources to analyze the required capacity trends for the Auto Scaling group counts"

upvoted 2 times

Hkayne 2 years, 1 month ago

Selected Answer: C

B or C.
I think C because the company needs an automated way to modify the autoscaling desired capacity

upvoted 1 times

jjcode 2 years, 3 months ago

How does C works with : transactions can vary, clearly C is designed for workloads that are predictable, if the transactions can vary then predictive scaling will not work. The only one that will work is scheduled since its based on time not workload intensity.

upvoted 3 times

pentium75 2 years, 5 months ago

Selected Answer: C

C per https://docs.aws.amazon.com/autoscaling/ec2/userguide/predictive-scaling-create-policy.html.

B is out because it wants the company to 'set the desired/minimum/maximum capacity' but "the company does not have the resources to analyze the required capacity".

upvoted 6 times

Cyberkayu 2 years, 5 months ago

Lambda did not appear to take over scripting/batch job, what a surprise

upvoted 4 times

daniel1 2 years, 7 months ago

Selected Answer: B

From GPT4:
mong the provided options, creating a scheduled scaling policy (Option B) is the most direct and efficient way to ensure that the necessary capacity is provisioned 30 minutes before the weekly batch jobs run, with the least operational overhead. Here's a breakdown of Option B:

B. Create a scheduled scaling policy for the Auto Scaling group. Set the appropriate desired capacity, minimum capacity, and maximum capacity. Set the recurrence to weekly. Set the start time to 30 minutes before the batch jobs run.

Scheduled scaling allows you to change the desired capacity of your Auto Scaling group based on a schedule. In this case, setting the recurrence to weekly and adjusting the start time to 30 minutes before the batch jobs run will ensure that the necessary capacity is available when needed, without requiring manual intervention.

upvoted 5 times

TheFivePips 2 years, 3 months ago

yeah chatgpt told me this, so maybe dont take its word as gospel:

Upon reviewing the question again, it appears that the requirements emphasize the need to provision capacity 30 minutes before the batch jobs run and the company's constraint of not having resources to analyze capacity trends. In this context, the most suitable solution is C.

Predictive Scaling can use historical data to forecast future capacity needs.
Configuring the policy to scale based on CPU utilization with a target value of 60% aligns with the baseline CPU utilization mentioned in the scenario.
Setting instances to pre-launch 30 minutes before the jobs run provides the desired capacity just in time.

upvoted 2 times

TariqKipkemei 2 years, 8 months ago

Selected Answer: C

Predictive scaling: increases the number of EC2 instances in your Auto Scaling group in advance of daily and weekly patterns in traffic flows. If you have regular patterns of traffic increases use predictive scaling, to help you scale faster by launching capacity in advance of forecasted load. You don't have to spend time reviewing your application's load patterns and trying to schedule the right amount of capacity using scheduled scaling. Predictive scaling uses machine learning to predict capacity requirements based on historical data from CloudWatch. The machine learning algorithm consumes the available historical data and calculates capacity that best fits the historical load pattern, and then continuously learns based on new data to make future forecasts more accurate.

upvoted 2 times

bsbs1234 2 years, 8 months ago

should be C. Question does not say how long the job will run. don't know when to set the end time in the schedule policy.

upvoted 2 times

Question #343 *Topic 1*

A solutions architect is designing a company's disaster recovery (DR) architecture. The company has a MySQL database that runs on an Amazon EC2 instance in a private subnet with scheduled backup. The DR design needs to include multiple AWS Regions.

Which solution will meet these requirements with the LEAST operational overhead?

A. Migrate the MySQL database to multiple EC2 instances. Configure a standby EC2 instance in the DR Region. Turn on replication.

B. Migrate the MySQL database to Amazon RDS. Use a Multi-AZ deployment. Turn on read replication for the primary DB instance in the different Availability Zones.

C. Migrate the MySQL database to an Amazon Aurora global database. Host the primary DB cluster in the primary Region. Host the secondary DB cluster in the DR Region. **Most Voted**

D. Store the scheduled backup of the MySQL database in an Amazon S3 bucket that is configured for S3 Cross-Region Replication (CRR). Use the data backup to restore the database in the DR Region.

Correct Answer: C

Community vote distribution

C (100%)

Comments

LuckyAro Highly Voted 2 years, 9 months ago

C: Migrate MySQL database to an Amazon Aurora global database is the best solution because it requires minimal operational overhead. Aurora is a managed service that provides automatic failover, so standby instances do not need to be manually configured. The primary DB cluster can be hosted in the primary Region, and the secondary DB cluster can be hosted in the DR Region. This approach ensures that the data is always available and up-to-date in multiple Regions, without requiring significant manual intervention.

upvoted 11 times

AlessandraSAA Highly Voted 2 years, 9 months ago

Selected Answer: C

A. Multiple EC2 instances to be configured and updated manually in case of DR.
B. Amazon RDS=Multi-AZ while it asks to be multi-region
C. correct, see comment from LuckyAro
D. Manual process to start the DR, therefore same limitation as answer A

upvoted 11 times

gulmichamagaun5 Most Recent 1 year, 11 months ago

hello friends, question required: The DR design needs to include multiple AWS Regions, but the correct answer is B, how it comes, because the DR here is on AZ not Different Region so the i would go with D

upvoted 1 times

TariqKipkemei 2 years, 1 month ago

Selected Answer: C

LEAST operational overhead = Serverless = Amazon Aurora global database

upvoted 3 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: C

Amazon Aurora global database can span and replicate DB Servers between multiple AWS Regions. And also compatible with MySQL.

upvoted 2 times

GalileoEC2 2 years, 8 months ago

C, Why B? B is multi zone in one region, C is multi region as it was requested

upvoted 2 times

lucdt4 2 years, 6 months ago
" The DR design needs to include multiple AWS Regions."
with the requirement "DR SITE multiple AWS region" -> B is wrong, because it deploy multy AZ (this is not multi region)
upvoted 2 times

KZM 2 years, 9 months ago
Amazon Aurora global database can span and replicate DB Servers between multiple AWS Regions. And also compatible with MySQL.
upvoted 4 times

LuckyAro 2 years, 9 months ago
With dynamic scaling, the Auto Scaling group will automatically adjust the number of instances based on the actual workload. The target value for the CPU utilization metric is set to 60%, which is the baseline CPU utilization that is noted on each run, indicating that this is a reasonable level of utilization for the workload. This solution does not require any scheduling or forecasting, reducing the operational overhead.
upvoted 1 times

LuckyAro 2 years, 9 months ago
Sorry, Posted right answer to the wrong question, mistakenly clicked the next question, sorry.
upvoted 6 times

geekgirl22 2 years, 9 months ago
C is the answer as RDS is only multi-zone not multi region.
upvoted 2 times

bdp123 2 years, 9 months ago
Selected Answer: C
https://docs.aws.amazon.com/AmazonRDS/latest/AuroraUserGuide/Aurora.Replication.html
upvoted 2 times

SMAZ 2 years, 9 months ago
C
option A has operation overhead whereas option C not.
upvoted 2 times

alexman 2 years, 9 months ago
Selected Answer: C
C mentions multiple regions. Option B is within the same region
upvoted 4 times

jennyka76 2 years, 9 months ago
ANSWER - B ?? NOT SURE
upvoted 1 times

Question #344 Topic 1

A company has a Java application that uses Amazon Simple Queue Service (Amazon SQS) to parse messages. The application cannot parse messages that are larger than 256 KB in size. The company wants to implement a solution to give the application the ability to parse messages as large as 50 MB.

Which solution will meet these requirements with the FEWEST changes to the code?

A. Use the Amazon SQS Extended Client Library for Java to host messages that are larger than 256 KB in Amazon S3.
Most Voted

B. Use Amazon EventBridge to post large messages from the application instead of Amazon SQS.

C. Change the limit in Amazon SQS to handle messages that are larger than 256 KB.

D. Store messages that are larger than 256 KB in Amazon Elastic File System (Amazon EFS). Configure Amazon SQS to reference this location in the messages.

Correct Answer: A

Community vote distribution

A (100%)

Comments

LuckyAro Highly Voted 3 years, 3 months ago
Selected Answer: A
A. Use the Amazon SQS Extended Client Library for Java to host messages that are larger than 256 KB in Amazon S3.

Amazon SQS has a limit of 256 KB for the size of messages. To handle messages larger than 256 KB, the Amazon SQS Extended Client Library for Java can be used. This library allows messages larger than 256 KB to be stored in Amazon S3 and provides a way to retrieve and process them. Using this solution, the application code can remain largely unchanged while still being able to process messages up to 50 MB in size.
upvoted 17 times

Neha999 Highly Voted 3 years, 3 months ago
A
For messages > 256 KB, use Amazon SQS Extended Client Library for Java
https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/quotas-messages.html
upvoted 7 times

tonywang0418 Most Recent 1 year, 7 months ago
who would know this...
upvoted 5 times

Gape4 1 year, 11 months ago
Selected Answer: A
To send messages larger than 256 KiB, you can use the Amazon SQS Extended Client Library for Java...
upvoted 2 times

TariqKipkemei 2 years, 8 months ago
Selected Answer: A
The Amazon SQS Extended Client Library for Java enables you to manage Amazon SQS message payloads with Amazon S3. This is especially useful for storing and retrieving messages with a message payload size greater than the current SQS limit of 256 KB, up to a maximum of 2 GB.
upvoted 4 times

Guru4Cloud 2 years, 9 months ago
Selected Answer: A
The SQS Extended Client Library enables storing large payloads in S3 while referenced via SQS. The application code can stay almost entirely unchanged - it sends/receives SQS messages normally. The library handles transparently routing the large payloads to S3 behind the scenes

upvoted 2 times

james2033 2 years, 10 months ago

Selected Answer: A

Quote "The Amazon SQS Extended Client Library for Java enables you to manage Amazon SQS message payloads with Amazon S3." and "An extension to the Amazon SQS client that enables sending and receiving messages up to 2GB via Amazon S3." at https://github.com/awslabs/amazon sqs-java-extended-client-lib

upvoted 2 times

Abrar2022 3 years ago

Selected Answer: A

Amazon SQS has a limit of 256 KB for the size of messages.

To handle messages larger than 256 KB, the Amazon SQS Extended Client Library for Java can be used.

upvoted 2 times

gold4otas 3 years, 2 months ago

The Amazon SQS Extended Client Library for Java enables you to publish messages that are greater than the current SQS limit of 256 KB, up to a maximum of 2 GB.

https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-s3-messages.html

upvoted 2 times

bdp123 3 years, 3 months ago

Selected Answer: A

https://github.com/awslabs/amazon-sqs-java-extended-client-lib

upvoted 4 times

Arathore 3 years, 3 months ago

Selected Answer: A

To send messages larger than 256 KiB, you can use the Amazon SQS Extended Client Library for Java. This library allows you to send an Amazon SQS message that contains a reference to a message payload in Amazon S3. The maximum payload size is 2 GB.

upvoted 5 times

Question #345 Topic 1

A company wants to restrict access to the content of one of its main web applications and to protect the content by using authorization techniques available on AWS. The company wants to implement a serverless architecture and an authentication solution for fewer than 100 users. The solution needs to integrate with the main web application and serve web content globally. The solution must also scale as the company's user base grows while providing the lowest login latency possible.

Which solution will meet these requirements MOST cost-effectively?

A. Use Amazon Cognito for authentication. Use Lambda@Edge for authorization. Use Amazon CloudFront to serve the web application globally. **Most Voted**

B. Use AWS Directory Service for Microsoft Active Directory for authentication. Use AWS Lambda for authorization. Use an Application Load Balancer to serve the web application globally.

C. Use Amazon Cognito for authentication. Use AWS Lambda for authorization. Use Amazon S3 Transfer Acceleration to serve the web application globally.

D. Use AWS Directory Service for Microsoft Active Directory for authentication. Use Lambda@Edge for authorization. Use AWS Elastic Beanstalk to serve the web application globally.

Correct Answer: A

Community vote distribution

A (100%)

Comments

Lonojack Highly Voted 2 years, 9 months ago

Selected Answer: A

CloudFront=globally
Lambda@edge = Authorization/ Latency
Cognito=Authentication for Web apps

upvoted 17 times

Lin878 Most Recent 1 year, 6 months ago

Selected Answer: A

https://aws.amazon.com/blogs/networking-and-content-delivery/external-server-authorization-with-lambdaedge/

upvoted 2 times

Cyberkayu 1 year, 12 months ago

fewer than 100 users but scattered around the globe, lowest latency.

Should have do nothing, most cost effective.

upvoted 3 times

TariqKipkemei 2 years, 1 month ago

Selected Answer: A

Use Amazon Cognito for authentication. Use Lambda@Edge for authorization. Use Amazon CloudFront to serve the web application globally

upvoted 4 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: A

Amazon Cognito is a serverless authentication service that can be used to easily add user sign-up and authentication to web and mobile apps. It is a good choice for this scenario because it is scalable and can handle a small number of users without any additional costs.

Lambda@Edge is a serverless compute service that can be used to run code at the edge of the AWS network. It is a good choice for this scenario because it can be used to perform authorization checks at the edge, which can improve the login latency.

Amazon CloudFront is a content delivery network (CDN) that can be used to serve web content globally. It is a good choice for this scenario because it can cache web content closer to users, which can improve the performance of the web application.

upvoted 4 times

antropaws 2 years, 6 months ago

Selected Answer: A

A is perfect.

upvoted 2 times

kraken21 2 years, 8 months ago

Selected Answer: A

Lambda@Edge for authorization
https://aws.amazon.com/blogs/networking-and-content-delivery/adding-http-security-headers-using-lambdaedge-and-amazon-cloudfront/

upvoted 3 times

LuckyAro 2 years, 9 months ago

Selected Answer: A

Amazon CloudFront is a global content delivery network (CDN) service that can securely deliver web content, videos, and APIs at scale. It integrates with Cognito for authentication and with Lambda@Edge for authorization, making it an ideal choice for serving web content globally.

Lambda@Edge is a service that lets you run AWS Lambda functions globally closer to users, providing lower latency and faster response times. It can also handle authorization logic at the edge to secure content in CloudFront. For this scenario, Lambda@Edge can provide authorization for the web application while leveraging the low-latency benefit of running at the edge.

upvoted 3 times

bdp123 2 years, 9 months ago

Selected Answer: A

CloudFront to serve globally

upvoted 2 times

SMAZ 2 years, 9 months ago

A
Amazon Cognito for authentication and Lambda@Edge for authorizatioN, Amazon CloudFront to serve the web application globally provides low-latency content delivery

upvoted 4 times

Question #346 Topic 1

A company has an aging network-attached storage (NAS) array in its data center. The NAS array presents SMB shares and NFS shares to client workstations. The company does not want to purchase a new NAS array. The company also does not want to incur the cost of renewing the NAS array's support contract. Some of the data is accessed frequently, but much of the data is inactive.

A solutions architect needs to implement a solution that migrates the data to Amazon S3, uses S3 Lifecycle policies, and maintains the same look and feel for the client workstations. The solutions architect has identified AWS Storage Gateway as part of the solution.

Which type of storage gateway should the solutions architect provision to meet these requirements?

A. Volume Gateway

B. Tape Gateway

C. Amazon FSx File Gateway

D. Amazon S3 File Gateway Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

LuckyAro Highly Voted 2 years, 9 months ago

Selected Answer: D

Amazon S3 File Gateway provides on-premises applications with access to virtually unlimited cloud storage using NFS and SMB file interfaces. It seamlessly moves frequently accessed data to a low-latency cache while storing colder data in Amazon S3, using S3 Lifecycle policies to transition data between storage classes over time.

In this case, the company's aging NAS array can be replaced with an Amazon S3 File Gateway that presents the same NFS and SMB shares to the client workstations. The data can then be migrated to Amazon S3 and managed using S3 Lifecycle policies

upvoted 17 times

everfly Highly Voted 2 years, 9 months ago

Selected Answer: D

Amazon S3 File Gateway provides a file interface to objects stored in S3. It can be used for a file-based interface with S3, which allows the company to migrate their NAS array data to S3 while maintaining the same look and feel for client workstations. Amazon S3 File Gateway supports SMB and NFS protocols, which will allow clients to continue to access the data using these protocols. Additionally, Amazon S3 Lifecycle policies can be used tc automate the movement of data to lower-cost storage tiers, reducing the storage cost of inactive data.

upvoted 7 times

pentium75 Most Recent 1 year, 11 months ago

Selected Answer: D

A - provides virtual disk via iSCSI
B - provides virtual tape via iSCSI
C - provides access to FSx via SMB

upvoted 5 times

TariqKipkemei 2 years, 1 month ago

Selected Answer: D

The Amazon S3 File Gateway enables you to store and retrieve objects in Amazon Simple Storage Service (S3) using file protocols such as Network File System (NFS) and Server Message Block (SMB).

upvoted 4 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: D

It provides an easy way to lift-and-shift file data from the existing NAS to Amazon S3. The S3 File Gateway presents SMB and NFS file shares that client workstations can access just like the NAS shares.
Behind the scenes, it moves the file data to S3 storage, storing it durably and cost-effectively.
S3 Lifecycle policies can be used to transition less frequently accessed data to lower-cost S3 storage tiers like S3 Glacier.
From the client workstation perspective, access to files feels seamless and unchanged after migration to S3. The S3 File Gateway handles the underlying data transfers.
It is a simple, low-cost gateway option tailored for basic file share migration use cases.

upvoted 4 times

james2033 2 years, 4 months ago

Selected Answer: D

- Volume Gateway: https://aws.amazon.com/storagegateway/volume/ (Remove A, related iSCSI)

- Tape Gateway https://aws.amazon.com/storagegateway/vtl/ (Remove B)

- Amazon FSx File Gateway https://aws.amazon.com/storagegateway/file/fsx/ (C)

- Why not choose C? Because need working with Amazon S3. (Answer D, and it is correct answer) https://aws.amazon.com/storagegateway/file/s3/

upvoted 4 times

siyam008 2 years, 9 months ago

Selected Answer: D

https://aws.amazon.com/blogs/storage/how-to-create-smb-file-shares-with-aws-storage-gateway-using-hyper-v/

upvoted 3 times

bdp123 2 years, 9 months ago

Selected Answer: D

https://aws.amazon.com/about-aws/whats-new/2018/06/aws-storage-gateway-adds-smb-support-to-store-objects-in-amazon-s3/

upvoted 4 times

Question #347 Topic 1

A company has an application that is running on Amazon EC2 instances. A solutions architect has standardized the company on a particular instance family and various instance sizes based on the current needs of the company.

The company wants to maximize cost savings for the application over the next 3 years. The company needs to be able to change the instance family and sizes in the next 6 months based on application popularity and usage.

Which solution will meet these requirements MOST cost-effectively?

A. Compute Savings Plan Most Voted

B. EC2 Instance Savings Plan

C. Zonal Reserved Instances

D. Standard Reserved Instances

Correct Answer: A

Community vote distribution

A (84%) Other

Comments

AlmeroSenior Highly Voted 2 years, 9 months ago

Selected Answer: A

Read Carefully guys , They need to be able to change FAMILY , and although EC2 Savings has a higher discount , its clearly documented as not allowed >

EC2 Instance Savings Plans provide savings up to 72 percent off On-Demand, in exchange for a commitment to a specific instance family in a chosen AWS Region (for example, M5 in Virginia). These plans automatically apply to usage regardless of size (for example, m5.xlarge, m5.2xlarge, etc.), OS (for example, Windows, Linux, etc.), and tenancy (Host, Dedicated, Default) within the specified family in a Region.

upvoted 22 times

FFO 2 years, 7 months ago

Savings Plans are a flexible pricing model that offer low prices on Amazon EC2, AWS Lambda, and AWS Fargate usage, in exchange for a commitment to a consistent amount of usage (measured in $/hour) for a 1 or 3 year term. When you sign up for a Savings Plan, you will be chargec the discounted Savings Plans price for your usage up to your commitment.

The company wants savings over the next 3 years but wants to change the instance type in 6 months. This invalidates A

upvoted 5 times

FFO 2 years, 7 months ago

Disregard! found more information:

We recommend Savings Plans (over Reserved Instances). Like Reserved Instances, Savings Plans offer lower prices (up to 72% savings compared to On-Demand Instance pricing). In addition, Savings Plans offer you the flexibility to change your usage as your needs evolve. For example, with Compute Savings Plans, lower prices will automatically apply when you change from C4 to C6g instances, shift a workload from EU (Ireland) to EU (London), or move a workload from Amazon EC2 to AWS Fargate or AWS Lambda.

https://aws.amazon.com/ec2/pricing/reserved-instances/pricing/

upvoted 3 times

awsgeek75 Highly Voted 1 year, 10 months ago

Selected Answer: A

https://docs.aws.amazon.com/whitepapers/latest/cost-optimization-reservation-models/savings-plans.html

Compute Savings Plans provide the most flexibility and help to reduce your costs by up to 66% (just like Convertible RIs). These plans automatically apply to EC2 instance usage regardless of instance family...

EC2 Instance Savings Plans provide the lowest prices, offering savings up to 72% (just like Standard RIs) in exchange for commitment to usage of individual instance families

Instance Savings "locks" you in that instance family which is not desired by the company hence A is the best plan as they can change the instance family anytime

upvoted 10 times

awsgeek75 1 year, 10 months ago

Also, don't forget, the minimum commitment for both of these plans is 1 year and the company wants the ability to change in 6 months so it has tc be a plan which allows changing of instance within the commitment window (no refunds!)

upvoted 4 times

xBUGx Most Recent 1 year, 8 months ago

Selected Answer: A

EC2 Instance Savings Plans provide the lowest prices, offering savings up to 72% in exchange for commitment to usage of individual instance familie in a region (e.g. M5 usage in N. Virginia). This automatically reduces your cost on the selected instance family in that region regardless of AZ, size, O! or tenancy. ***EC2 Instance Savings Plans give you the flexibility to change your usage between instances within a family in that region.*** For example, you can move from c5.xlarge running Windows to c5.2xlarge running Linux and automatically benefit from the Savings Plans prices. https://aws.amazon.com/savingsplans/faq/#:~:text=EC2%20Instance%20Savings%20Plans%20give,from%20the%20Savings%20Plans%20prices.

upvoted 4 times

Mican07 1 year, 11 months ago

B is the definite answer

upvoted 1 times

pentium75 1 year, 11 months ago

Selected Answer: A

B does not allow changing the instance family, despite all the ChatGPT-based answers claiming the opposite

upvoted 4 times

meowruki 2 years ago

Selected Answer: A

While EC2 Instance Savings Plans also provide cost savings over On-Demand pricing, they offer less flexibility in terms of changing instance families. They provide a discount in excha

upvoted 3 times

hungta 2 years ago

Selected Answer: B

EC2 Instance Savings Plans is most saving. And it is enough for required flexibility
EC2 Instance Savings Plans provide the lowest prices, offering savings up to 72% (just like Standard RIs) in exchange for commitment to usage of individual instance families in a Region (for example, M5 usage in N. Virginia). This automatically reduces your cost on the selected instance family in that region regardless of AZ, size, operating system, or tenancy. EC2 Instance Savings Plans give you the flexibility to change your usage between instances within a family in that Region. For example, you can move from c5.xlarge running Windows to c5.2xlarge running Linux and automatically benefit from the Savings Plans prices.

upvoted 1 times

pentium75 1 year, 11 months ago

But it does not allow changing the instance family, which is a requirement here.

upvoted 3 times

Jazz888 1 year, 6 months ago

You voted against yourself. Did you mean to vote A?

upvoted 1 times

dilaaziz 2 years ago

Selected Answer: A

https://docs.aws.amazon.com/whitepapers/latest/cost-optimization-reservation-models/savings-plans.html

upvoted 2 times

EdenWang 2 years, 1 month ago

Selected Answer: B

The most cost-effective solution that meets the company's requirements would be B. EC2 Instance Savings Plan.

EC2 Instance Savings Plans provide significant cost savings, allowing the company to commit to a consistent amount of usage (measured in $/hour) for a 1- or 3-year term, and in return, receive a discount on the hourly rate for the instances that match the attributes of the plan.

With EC2 Instance Savings Plans, the company can benefit from the flexibility to change the instance family and sizes over the next 3 years, which aligns with their requirement to adjust based on application popularity and usage.

This option provides the best balance of cost savings and flexibility, making it the most suitable choice for the company's needs.

upvoted 2 times

pentium75 1 year, 11 months ago
"With EC2 Instance Savings Plans, the company can benefit from the flexibility to change the instance family" NO, this is simply wrong. Is this from ChatGPT?

"EC2 Instance Savings Plans provide savings up to 72 percent off On-Demand, in exchange for a commitment to a specific instance family (!) in a chosen AWS Region ... With an EC2 Instance Savings Plan, you can change your instance size within the instance family (!)".

https://docs.aws.amazon.com/savingsplans/latest/userguide/what-is-savings-plans.html
upvoted 4 times

TariqKipkemei 2 years, 1 month ago
Selected Answer: A
Change instance family = Compute Savings Plans
upvoted 4 times

Wayne23Fang 2 years, 3 months ago
Selected Answer: A
D is not right. D. Standard Reserved Instances. should be Convertible Reserved Instances if you need additional flexibility, such as the ability to use different instance families, operating systems.
upvoted 3 times

Guru4Cloud 2 years, 3 months ago
Selected Answer: B
The key factors are:

Need to maximize cost savings over 3 years
Ability to change instance family and sizes in 6 months
Standardized on a particular instance family for now
upvoted 2 times

awsgeek75 1 year, 10 months ago
"Ability to change instance family and sizes in 6 months" is not allowed in Instance Savings plan so B is wrong
upvoted 2 times

Kiki_Pass 2 years, 4 months ago
Why not C? Can do with Convertible Reserved Instance
https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/reserved-instances-types.html
upvoted 1 times

ITV2021 2 years, 4 months ago
Selected Answer: A
https://aws.amazon.com/savingsplans/compute-pricing/
upvoted 2 times

Mia2009687 2 years, 5 months ago
Selected Answer: A
EC2 Instance Savings Plan cannot change the family.
https://docs.aws.amazon.com/savingsplans/latest/userguide/what-is-savings-plans.html
upvoted 2 times

mattcl 2 years, 5 months ago
Anser D: You can use Standard Reserved Instances when you know that you need a specific instance type.
upvoted 1 times

kruasan 2 years, 7 months ago
Selected Answer: A
Savings Plans offer a flexible pricing model that provides savings on AWS usage. You can save up to 72 percent on your AWS compute workloads. Compute Savings Plans provide lower prices on Amazon EC2 instance usage regardless of instance family, size, OS, tenancy, or AWS Region. This alsc applies to AWS Fargate and AWS Lambda usage. SageMaker Savings Plans provide you with lower prices for your Amazon SageMaker instance usage, regardless of your instance family, size, component, or AWS Region.
https://docs.aws.amazon.com/savingsplans/latest/userguide/what-is-savings-plans.html
upvoted 3 times

kruasan 2 years, 7 months ago
With an EC2 Instance Savings Plan, you can change your instance size within the instance family (for example, from c5.xlarge to c5.2xlarge) or the operating system (for example, from Windows to Linux), or move from Dedicated tenancy to Default and continue to receive the discounted rate provided by your EC2 Instance Savings Plan.
https://docs.aws.amazon.com/savingsplans/latest/userguide/what-is-savings-plans.html
upvoted 1 times

kruasan 2 years, 7 months ago

The company needs to be able to change the instance family and sizes in the next 6 months based on application popularity and usage. Therefore EC2 Instance Savings Plan prerequisites are not fulfilled

upvoted 3 times

Question #348 *Topic 1*

A company collects data from a large number of participants who use wearable devices. The company stores the data in an Amazon DynamoDB table and uses applications to analyze the data. The data workload is constant and predictable. The company wants to stay at or below its forecasted budget for DynamoDB.

Which solution will meet these requirements MOST cost-effectively?

A. Use provisioned mode and DynamoDB Standard-Infrequent Access (DynamoDB Standard-IA). Reserve capacity for the forecasted workload.

B. Use provisioned mode. Specify the read capacity units (RCUs) and write capacity units (WCUs). **Most Voted**

C. Use on-demand mode. Set the read capacity units (RCUs) and write capacity units (WCUs) high enough to accommodate changes in the workload.

D. Use on-demand mode. Specify the read capacity units (RCUs) and write capacity units (WCUs) with reserved capacity.

Correct Answer: B

Community vote distribution

B (85%) A (15%)

Comments

everfly Highly Voted 2 years, 9 months ago

Selected Answer: B

The data workload is constant and predictable.

upvoted 11 times

hovnival Highly Voted 2 years, 1 month ago

Selected Answer: B

I think it is not possible to set Read Capacity Units(RCU)/Write Capacity Units(WCU) in on-demand mode.

upvoted 6 times

ThangAWS2022 Most Recent 3 months, 1 week ago

Selected Answer: A

Provisioned Mode and reserved capacity

upvoted 1 times

Clouddon 11 months ago

Selected Answer: A

Option A is preferred because it combines the benefits of provisioned mode, DynamoDB Standard-IA for lower storage costs, and reserved capacity for significant savings on provisioned throughput. This combination is ideal for your scenario with a constant and predictable workload, ensuring you stay within your forecasted budget.

upvoted 1 times

pentium75 1 year, 11 months ago

Selected Answer: B

C and D are impossible because you don't set or specify RCUs and WCUs in on-demand mode.

A is wrong because there is no indication of "infrequent access", and "the data workload is constant", there is no different between the current and the "forecasted" workload.

upvoted 4 times

wsdasdasdqwdaw 2 years, 1 month ago

predictable/constant => provisioned mode. On-demand mode is more suitable for workloads that are unpredictable and can vary widely from minute to minute.

The use case is not for Standard-IA which is described here: https://aws.amazon.com/dynamodb/standard-ia/

=> Option B
upvoted 3 times

TariqKipkemei 2 years, 1 month ago

Selected Answer: B

I rule out A because of this 'Standard-Infrequent Access ', clearly the company uses applications to analyze the data.
The data workload is constant and predictable making provisioned mode the best option.
upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: A

Option B lacks the cost benefits of Standard-IA.

Option C uses more expensive on-demand pricing.

Option D does not actually allow reserving capacity with on-demand mode.

So option A leverages provisioned mode, Standard-IA, and reserved capacity to meet the requirements in a cost-optimal way.
upvoted 1 times

MrAWSAssociate 2 years, 5 months ago

Selected Answer: A

A is correct!
upvoted 1 times

MrAWSAssociate 2 years, 5 months ago

Sorry, A will not work, since Reserved Capacity can only be used with DynamoDB Standard table class. So, B is right for this case.
upvoted 4 times

kayodea25 2 years, 9 months ago

Option C is the most cost-effective solution for this scenario. In on-demand mode, DynamoDB automatically scales up or down based on the current workload, so the company only pays for the capacity it uses. By setting the RCUs and WCUs high enough to accommodate changes in the workload, the company can ensure that it always has the necessary capacity without overprovisioning and incurring unnecessary costs. Since the workload is constant and predictable, using provisioned mode with reserved capacity (Options A and D) may result in paying for unused capacity during periods of low demand. Option B, using provisioned mode without reserved capacity, may result in throttling during periods of high demand if the provisioned capacity is not sufficient to handle the workload.
upvoted 3 times

Bofi 2 years, 8 months ago

Kayode olode..lol
upvoted 1 times

boxu03 2 years, 9 months ago

you forgot "The data workload is constant and predictable", should be B
upvoted 3 times

pentium75 1 year, 11 months ago

You can't 'set RCUs and WCUs' in on-demand mode.
upvoted 2 times

Steve_4542636 2 years, 9 months ago

"The data workload is constant and predictable."
https://docs.aws.amazon.com/wellarchitected/latest/serverless-applications-lens/capacity.html

"With provisioned capacity you pay for the provision of read and write capacity units for your DynamoDB tables. Whereas with DynamoDB on-demand you pay per request for the data reads and writes that your application performs on your tables."
upvoted 2 times

Charly0710 2 years, 9 months ago

Selected Answer: B

The data workload is constant and predictable, then, isn't on-demand mode.
DynamoDB Standard-IA is not necessary in this context
upvoted 2 times

Lonojack 2 years, 9 months ago

Selected Answer: B

The problem with (A) is: “Standard-Infrequent Access“. In the question, they say the company has to analyze the Data.
That’s why the Correct answer is (B)
upvoted 4 times

bdp123 2 years, 9 months ago

Selected Answer: A

workload is constant

upvoted 2 times

Lonojack 2 years, 9 months ago

The problem with (A) is: "Standard-Infrequent Access".
In the question, they say the company has to analyze the Data.
Correct answer is (B)

upvoted 3 times

Samuel03 2 years, 9 months ago

Selected Answer: B

As the numbers are already known

upvoted 4 times

Question #349 *Topic 1*

A company stores confidential data in an Amazon Aurora PostgreSQL database in the ap-southeast-3 Region. The database is encrypted with an AWS Key Management Service (AWS KMS) customer managed key. The company was recently acquired and must securely share a backup of the database with the acquiring company's AWS account in ap-southeast-3.

What should a solutions architect do to meet these requirements?

A. Create a database snapshot. Copy the snapshot to a new unencrypted snapshot. Share the new snapshot with the acquiring company's AWS account.

B. Create a database snapshot. Add the acquiring company's AWS account to the KMS key policy. Share the snapshot with the acquiring company's AWS account. **Most Voted**

C. Create a database snapshot that uses a different AWS managed KMS key. Add the acquiring company's AWS account to the KMS key alias. Share the snapshot with the acquiring company's AWS account.

D. Create a database snapshot. Download the database snapshot. Upload the database snapshot to an Amazon S3 bucket. Update the S3 bucket policy to allow access from the acquiring company's AWS account.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Abrar2022 Highly Voted 3 years ago

Selected Answer: B

A. - "So let me get this straight, with the current company the data is protected and encrypted. However, for the acquiring company the data is unencrypted? How is that fair?"

C - Wouldn't recommended this option because using a different AWS managed KMS key will not allow the acquiring company's AWS account to access the encrypted data.

D. - Don't risk it for a biscuit and get fired!!!! - by downloading the database snapshot and uploading it to an Amazon S3 bucket. This will increase the risk of data leakage or loss of confidentiality during the transfer process.

B - CORRECT

upvoted 16 times

njufi Most Recent 2 years, 2 months ago

I believe the reason why option C is not the correct answer is that adding the acquiring company's AWS account to the KMS key alias doesn't directly control access to the encrypted data. KMS key aliases are simply alternative names for KMS keys and do not affect access control. Access to encrypted data is goverened by KMS key policies, which define who can use the key for encryption and decryption.

upvoted 2 times

TariqKipkemei 2 years, 7 months ago

Selected Answer: B

Create a database snapshot. Add the acquiring company's AWS account to the KMS key policy. Share the snapshot with the acquiring company's AWS account.

upvoted 2 times

Vuuu 2 years, 10 months ago

Selected Answer: B

B. Create a database snapshot. Add the acquiring company's AWS account to the KMS key policy. Share the snapshot with the acquiring company's AWS account. Most Voted

upvoted 2 times

Abrar2022 3 years ago

Create a database snapshot of the encrypted. Add the acquiring company's AWS account to the KMS key policy. Share the snapshot with the acquiring company's AWS account.

upvoted 2 times

SkyZeroZx 3 years, 1 month ago

Selected Answer: B

To securely share a backup of the database with the acquiring company's AWS account in the same Region, a solutions architect should create a database snapshot, add the acquiring company's AWS account to the AWS KMS key policy, and share the snapshot with the acquiring company's AWS account.

Option A, creating an unencrypted snapshot, is not recommended as it will compromise the confidentiality of the data. Option C, creating a snapsho that uses a different AWS managed KMS key, does not provide any additional security and will unnecessarily complicate the solution. Option D, downloading the database snapshot and uploading it to an S3 bucket, is not secure as it can expose the data during transit.

Therefore, the correct option is B: Create a database snapshot. Add the acquiring company's AWS account to the KMS key policy. Share the snapshot with the acquiring company's AWS account.

upvoted 2 times

JA2018 1 year, 6 months ago

KMS key aliases are simply alternative names for KMS keys and do not affect access control.

Access to encrypted data is governed by KMS key policies, that define who can use the key for encryption and decryption.

upvoted 1 times

elearningtakai 3 years, 2 months ago

Selected Answer: B

Option B is the correct answer.
Option A is not recommended because copying the snapshot to a new unencrypted snapshot will compromise the confidentiality of the data.
Option C is not recommended because using a different AWS managed KMS key will not allow the acquiring company's AWS account to access the encrypted data.
Option D is not recommended because downloading the database snapshot and uploading it to an Amazon S3 bucket will increase the risk of data leakage or loss of confidentiality during the transfer process.

upvoted 3 times

Steve_4542636 3 years, 3 months ago

Selected Answer: B

https://docs.aws.amazon.com/kms/latest/developerguide/key-policy-modifying-external-accounts.html

upvoted 3 times

geekgirl22 3 years, 3 months ago

It is C, you have to create a new key. Read below
You can't share a snapshot that's encrypted with the default AWS KMS key. You must create a custom AWS KMS key instead. To share an encrypted Aurora DB cluster snapshot:

Create a custom AWS KMS key.
Add the target account to the custom AWS KMS key.
Create a copy of the DB cluster snapshot using the custom AWS KMS key. Then, share the newly copied snapshot with the target account.
Copy the shared DB cluster snapshot from the target account
https://aws.amazon.com/premiumsupport/knowledge-center/aurora-share-encrypted-snapshot/

upvoted 1 times

leoattf 3 years, 3 months ago

I also thought straight away that it could be C, however, the questions mentions that the database is encrypted with an AWS KMS custom key already. So maybe the letter B could be right, since it already has a custom key, not the default KMS Key.
What do you think?

upvoted 3 times

enzomv 3 years, 3 months ago

It is B.
There's no need to create another custom AWS KMS key.
https://aws.amazon.com/premiumsupport/knowledge-center/aurora-share-encrypted-snapshot/
Give target account access to the custom AWS KMS key within the source account
1. Log in to the source account, and go to the AWS KMS console in the same Region as the DB cluster snapshot.
2. Select Customer-managed keys from the navigation pane.
3. Select your custom AWS KMS key (ALREADY CREATED)
4. From the Other AWS accounts section, select Add another AWS account, and then enter the AWS account number of your target account.

Then:
Copy and share the DB cluster snapshot

upvoted 3 times

KZM 3 years, 3 months ago

Yes, as per the given information "The database is encrypted with an AWS Key Management Service (AWS KMS) customer managed key", it may not be the default AWS KMS key.

upvoted 1 times

KZM 3 years, 3 months ago

Yes, can't share a snapshot that's encrypted with the default AWS KMS key.
But as per the given information "The database is encrypted with an AWS Key Management Service (AWS KMS) customer managed key", it may not be the default AWS KMS key.

upvoted 3 times

enzomv 3 years, 3 months ago

I agree with KZM.
It is B.
There's no need to create another custom AWS KMS key.
https://aws.amazon.com/premiumsupport/knowledge-center/aurora-share-encrypted-snapshot/
Give target account access to the custom AWS KMS key within the source account
1. Log in to the source account, and go to the AWS KMS console in the same Region as the DB cluster snapshot.
2. Select Customer-managed keys from the navigation pane.
3. Select your custom AWS KMS key (ALREADY CREATED)
4. From the Other AWS accounts section, select Add another AWS account, and then enter the AWS account number of your target account.

Then:
Copy and share the DB cluster snapshot

upvoted 3 times

nyx12345 3 years, 3 months ago

Is it bad that in answer B the acquiring company is using the same KMS key? Should a new KMS key not be used?

upvoted 2 times

geekgirl22 3 years, 3 months ago

Yes, you are right, read my comment above.

upvoted 1 times

bsbs1234 2 years, 8 months ago

I think I would agree with you if option C say using a new "customer managed key" instead of AWS managed key

upvoted 1 times

bdp123 3 years, 3 months ago

Selected Answer: B

https://aws.amazon.com/premiumsupport/knowledge-center/aurora-share-encrypted-snapshot/

upvoted 3 times

jennyka76 3 years, 3 months ago

ANSWER - B

upvoted 2 times

Question #350 Topic 1

A company uses a 100 GB Amazon RDS for Microsoft SQL Server Single-AZ DB instance in the us-east-1 Region to store customer transactions. The company needs high availability and automatic recovery for the DB instance.

The company must also run reports on the RDS database several times a year. The report process causes transactions to take longer than usual to post to the customers' accounts. The company needs a solution that will improve the performance of the report process.

Which combination of steps will meet these requirements? (Choose two.)

A. Modify the DB instance from a Single-AZ DB instance to a Multi-AZ deployment. **Most Voted**

B. Take a snapshot of the current DB instance. Restore the snapshot to a new RDS deployment in another Availability Zone.

C. Create a read replica of the DB instance in a different Availability Zone. Point all requests for reports to the read replica. **Most Voted**

D. Migrate the database to RDS Custom.

E. Use RDS Proxy to limit reporting requests to the maintenance window.

Correct Answer: AC

Community vote distribution

AC (100%)

Comments

elearningtakai Highly Voted 3 years, 2 months ago

A and C are the correct choices.
B. It will not help improve the performance of the report process.
D. Migrating to RDS Custom does not address the issue of high availability and automatic recovery.
E. RDS Proxy can help with scalability and high availability but it does not address the issue of performance for the report process. Limiting the reporting requests to the maintenance window will not provide the required availability and recovery for the DB instance.

upvoted 8 times

rockyykrish Most Recent 1 year, 7 months ago

A and C

upvoted 2 times

TariqKipkemei 2 years, 7 months ago

Selected Answer: AC

Create a Multi-AZ deployment, create a read replica of the DB instance in the second Availability Zone, point all requests for reports to the read replica

upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: AC

The correct answers are A and C.

A. Modify the DB instance from a Single-AZ DB instance to a Multi-AZ deployment. This will provide high availability and automatic recovery for the DB instance. If the primary DB instance fails, the standby DB instance will automatically become the primary DB instance. This will ensure that the database is always available.

C. Create a read replica of the DB instance in a different Availability Zone. Point all requests for reports to the read replica. This will improve the performance of the report process by offloading the read traffic from the primary DB instance to the read replica. The read replica is a fully synchronized copy of the primary DB instance, so the reports will be accurate.

upvoted 4 times

elearningtakai 3 years, 2 months ago

Selected Answer: AC

A and C.

upvoted 3 times

WherecanIstart 3 years, 2 months ago

Selected Answer: AC

Options A & C...

upvoted 4 times

KZM 3 years, 3 months ago

Options A+C

upvoted 3 times

bdp123 3 years, 3 months ago

Selected Answer: AC

https://medium.com/awesome-cloud/aws-difference-between-multi-az-and-read-replicas-in-amazon-rds-60fe848ef53a

upvoted 4 times

jennyka76 3 years, 3 months ago

ANSWER - A & C

upvoted 4 times

Question #351 *Topic 1*

A company is moving its data management application to AWS. The company wants to transition to an event-driven architecture. The architecture needs to be more distributed and to use serverless concepts while performing the different aspects of the workflow. The company also wants to minimize operational overhead.

Which solution will meet these requirements?

A. Build out the workflow in AWS Glue. Use AWS Glue to invoke AWS Lambda functions to process the workflow steps.

B. Build out the workflow in AWS Step Functions. Deploy the application on Amazon EC2 instances. Use Step Functions to invoke the workflow steps on the EC2 instances.

C. Build out the workflow in Amazon EventBridge. Use EventBridge to invoke AWS Lambda functions on a schedule to process the workflow steps.

D. Build out the workflow in AWS Step Functions. Use Step Functions to create a state machine. Use the state machine to invoke AWS Lambda functions to process the workflow steps. Most Voted

Correct Answer: D

Community vote distribution

D (91%) 9%

Comments

Lonojack Highly Voted 2 years, 9 months ago

Selected Answer: D

This is why I'm voting D.....QUESTION ASKED FOR IT TO: use serverless concepts while performing the different aspects of the workflow. Is option D utilizing Serverless concepts?

upvoted 13 times

geekgirl22 Highly Voted 2 years, 9 months ago

It is D. Cannot be C because C is "scheduled"

upvoted 8 times

bujuman Most Recent 1 year, 8 months ago

Selected Answer: D

While considering this requirement: The architecture needs to be more distributed and to use serverless concepts while performing the different aspects of the workflow

And checking the following link : https://aws.amazon.com/step-functions/?nc1=h_ls, Answer D is the best for this use case

upvoted 3 times

TariqKipkemei 2 years, 1 month ago

Selected Answer: D

One of the use cases for step functions is to Automate extract, transform, and load (ETL) processes.

https://aws.amazon.com/step-functions/#:~:text=for%20modern%20applications.-,Use%20cases,-Automate%20extract%2C%20transform

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: D

AWS Step functions is serverless Visual workflows for distributed applications

https://aws.amazon.com/step-functions/

upvoted 2 times

TariqKipkemei 2 years, 7 months ago

Selected Answer: D

Step Functions is based on state machines and tasks. A state machine is a workflow. A task is a state in a workflow that represents a single unit of work that another AWS service performs. Each step in a workflow is a state.

Depending on your use case, you can have Step Functions call AWS services, such as Lambda, to perform tasks.
https://docs.aws.amazon.com/step-functions/latest/dg/welcome.html

upvoted 3 times

TariqKipkemei 2 years, 7 months ago

Answer is D.
Step Functions is based on state machines and tasks. A state machine is a workflow. A task is a state in a workflow that represents a single unit of work that another AWS service performs. Each step in a workflow is a state.
Depending on your use case, you can have Step Functions call AWS services, such as Lambda, to perform tasks.
https://docs.aws.amazon.com/step-functions/latest/dg/welcome.html

upvoted 3 times

Karlos99 2 years, 9 months ago

Selected Answer: C

There are two main types of routers used in event-driven architectures: event buses and event topics. At AWS, we offer Amazon EventBridge to build event buses and Amazon Simple Notification Service (SNS) to build event topics. https://aws.amazon.com/event-driven-architecture/

upvoted 1 times

pentium75 1 year, 11 months ago

How do you 'build out a workflow' in EventBridge?

upvoted 4 times

TungPham 2 years, 9 months ago

Selected Answer: D

Step 3: Create a State Machine
Use the Step Functions console to create a state machine that invokes the Lambda function that you created earlier in Step 1.
https://docs.aws.amazon.com/step-functions/latest/dg/tutorial-creating-lambda-state-machine.html
In Step Functions, a workflow is called a state machine, which is a series of event-driven steps. Each step in a workflow is called a state.

upvoted 3 times

Bilalazure 2 years, 9 months ago

Selected Answer: D

Distrubuted****

upvoted 2 times

Americo32 2 years, 9 months ago

Selected Answer: C

Vou de C, orientada a eventos

upvoted 2 times

MssP 2 years, 8 months ago

It is true that an Event-driven is made with EventBridge but with a Lambda on schedule??? It is a mismatch, isn´t it?

upvoted 2 times

kraken21 2 years, 8 months ago

Tricky question huh!

upvoted 3 times

bdp123 2 years, 9 months ago

Selected Answer: D

AWS Step functions is serverless Visual workflows for distributed applications
https://aws.amazon.com/step-functions/

upvoted 2 times

leoattf 2 years, 9 months ago

Besides, "Visualize and develop resilient workflows for EVENT-DRIVEN architectures."

upvoted 2 times

tellmenowwwww 2 years, 9 months ago

Could it be a C because it's event-driven architecture?

upvoted 3 times

SMAZ 2 years, 9 months ago

Option D..
AWS Step functions are used for distributed applications

upvoted 3 times

Question #352 *Topic 1*

A company is designing the network for an online multi-player game. The game uses the UDP networking protocol and will be deployed in eight AWS Regions. The network architecture needs to minimize latency and packet loss to give end users a high-quality gaming experience.

Which solution will meet these requirements?

A. Setup a transit gateway in each Region. Create inter-Region peering attachments between each transit gateway.

B. Set up AWS Global Accelerator with UDP listeners and endpoint groups in each Region. **Most Voted**

C. Set up Amazon CloudFront with UDP turned on. Configure an origin in each Region.

D. Set up a VPC peering mesh between each Region. Turn on UDP for each VPC.

Correct Answer: B

Community vote distribution

B (100%)

Comments

lucdt4 Highly Voted 3 years ago

Selected Answer: B

AWS Global Accelerator = TCP/UDP minimize latency

upvoted 13 times

mwwt2022 Highly Voted 2 years, 5 months ago

online game -> Global Accelerator
cloudfront is for static/dynamic content caching

upvoted 5 times

Danilus Most Recent 1 year, 7 months ago

Selected Answer: B

key-UDP networking protocol
key-minimize latency and packet loss
Transit gateway is to conect multiple vpc
The solution is B because its accept udp protocol and its s global .
The solution is not C because cloudfront just accept http and https protocols

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

Set up AWS Global Accelerator with UDP listeners and endpoint groups in each Region.

upvoted 3 times

TariqKipkemei 3 years, 1 month ago

Selected Answer: B

Connect to up to 10 regions within the AWS global network using the AWS Global Accelerator.

upvoted 2 times

TariqKipkemei 2 years, 7 months ago

UDP = Global Accelerator

upvoted 3 times

OAdekunle 3 years, 1 month ago

General
Q: What is AWS Global Accelerator?

A: AWS Global Accelerator is a networking service that helps you improve the availability and performance of the applications that you offer to your global users. AWS Global Accelerator is easy to set up, configure, and manage. It provides static IP addresses that provide a fixed entry point to your applications and eliminate the complexity of managing specific IP addresses for different AWS Regions and Availability Zones. AWS Global Accelerator always routes user traffic to the optimal endpoint based on performance, reacting instantly to changes in application health, your user's location, and policies that you configure. You can test the performance benefits from your location with a speed comparison tool. Like other AWS services, AWS Global Accelerator is a self-service, pay-per-use offering, requiring no long term commitments or minimum fees.

https://aws.amazon.com/global-accelerator/faqs/

upvoted 5 times

elearningtakai 3 years, 2 months ago

Selected Answer: B

Global Accelerator supports the User Datagram Protocol (UDP) and Transmission Control Protocol (TCP), making it an excellent choice for an online multi-player game using UDP networking protocol. By setting up Global Accelerator with UDP listeners and endpoint groups in each Region, the network architecture can minimize latency and packet loss, giving end users a high-quality gaming experience.

upvoted 5 times

Bofi 3 years, 3 months ago

Selected Answer: B

AWS Global Accelerator is a service that improves the availability and performance of applications with local or global users. Global Accelerator improves performance for a wide range of applications over TCP or UDP by proxying packets at the edge to applications running in one or more AWS Regions. Global Accelerator is a good fit for non-HTTP use cases, such as gaming (UDP), IoT (MQTT), or Voice over IP, as well as for HTTP use cases that specifically require static IP addresses or deterministic, fast regional failover. Both services integrate with AWS Shield for DDoS protection.

upvoted 2 times

K0nAn 3 years, 3 months ago

Selected Answer: B

Global Accelerator for UDP and TCP traffic

upvoted 3 times

bdp123 3 years, 3 months ago

Selected Answer: B

Global Accelerator

upvoted 2 times

Neha999 3 years, 3 months ago

B

Global Accelerator for UDP traffic

upvoted 3 times

Question #353 *Topic 1*

A company hosts a three-tier web application on Amazon EC2 instances in a single Availability Zone. The web application uses a self-managed MySQL database that is hosted on an EC2 instance to store data in an Amazon Elastic Block Store (Amazon EBS) volume. The MySQL database currently uses a 1 TB Provisioned IOPS SSD (io2) EBS volume. The company expects traffic of 1,000 IOPS for both reads and writes at peak traffic.

The company wants to minimize any disruptions, stabilize performance, and reduce costs while retaining the capacity for double the IOPS. The company wants to move the database tier to a fully managed solution that is highly available and fault tolerant.

Which solution will meet these requirements MOST cost-effectively?

A. Use a Multi-AZ deployment of an Amazon RDS for MySQL DB instance with an io2 Block Express EBS volume.

B. Use a Multi-AZ deployment of an Amazon RDS for MySQL DB instance with a General Purpose SSD (gp2) EBS volume.
Most Voted

C. Use Amazon S3 Intelligent-Tiering access tiers.

D. Use two large EC2 instances to host the database in active-passive mode.

Correct Answer: B

Community vote distribution

B (92%) 8%

Comments

AlmeroSenior Highly Voted 3 years, 3 months ago

Selected Answer: B

RDS does not support IO2 or IO2express . GP2 can do the required IOPS

RDS supported Storage >
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html
GP2 max IOPS >
https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/general-purpose.html#gp2-performance

upvoted 16 times

sophieb Highly Voted 2 years, 2 months ago

Selected Answer: B

RDS now supports io2 but it might still be an overkill given Gp2 is enough and we are looking for the most cost effective solution.

upvoted 9 times

EllenLiu Most Recent 1 year, 5 months ago

Selected Answer: B

gp2: 1,000 GiB => 3000 Baseline performance (IOPS) match the requirement which is 2000 only
Baseline IOPS performance : 100 ~16,000 at a rate of 3 IOPS per GiB of volume size.

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

RDS does not support IO2 or IO2express . GP2 can do the required IOPS

upvoted 4 times

Gooniegoogoo 2 years, 11 months ago

The Options is A only because it is sufficient.. Provisioned IOPS are available but overkill.. just want to make sure we understand why its A for the right reason

upvoted 1 times

dkw2342 2 years, 3 months ago
Provisioned IOPS are available, but not io2, just io1.
upvoted 1 times

Abrar2022 3 years ago
Simplified by Almero - thanks.

RDS does not support IO2 or IO2express . GP2 can do the required IOPS
upvoted 2 times

TariqKipkemei 3 years, 1 month ago
Selected Answer: B
I tried on the portal and only gp3 and i01 are supported.
This is 11 May 2023.
upvoted 4 times

ruqui 3 years ago
it doesn't matter whether or no io* is supported, using io2 is overkill, you only need 1K IOPS, B is the correct answer
upvoted 2 times

SimiTik 3 years, 1 month ago
A
Amazon RDS supports the use of Amazon EBS Provisioned IOPS (io2) volumes. When creating a new DB instance or modifying an existing one, you can select the io2 volume type and specify the amount of IOPS and storage capacity required. RDS also supports the newer io2 Block Express volumes, which can deliver even higher performance for mission-critical database workloads.
upvoted 2 times

TariqKipkemei 3 years, 1 month ago
Impossible. I just tried on the portal and only io1 and gp3 are supported.
upvoted 2 times

klayytech 3 years, 2 months ago
Selected Answer: B
he most cost-effective solution that meets the requirements is to use a Multi-AZ deployment of an Amazon RDS for MySQL DB instance with a General Purpose SSD (gp2) EBS volume. This solution will provide high availability and fault tolerance while minimizing disruptions and stabilizing performance. The gp2 EBS volume can handle up to 16,000 IOPS. You can also scale up to 64 TiB of storage.

Amazon RDS for MySQL provides automated backups, software patching, and automatic host replacement. It also provides Multi-AZ deployments that automatically replicate data to a standby instance in another Availability Zone. This ensures that data is always available even in the event of a failure.
upvoted 2 times

test_devops_aws 3 years, 2 months ago
Selected Answer: B
RDS does not support io2 !!!
upvoted 1 times

Maximus007 3 years, 2 months ago
B:gp3 would be the better option, but considering we have only gp2 option and such storage volume - gp2 will be the right choice
upvoted 4 times

Nel8 3 years, 2 months ago
Selected Answer: B
I thought the answer here is A. But when I found the link from Amazon website; as per AWS:

Amazon RDS provides three storage types: General Purpose SSD (also known as gp2 and gp3), Provisioned IOPS SSD (also known as io1), and magnetic (also known as standard). They differ in performance characteristics and price, which means that you can tailor your storage performance and cost to the needs of your database workload. You can create MySQL, MariaDB, Oracle, and PostgreSQL RDS DB instances with up to 64 tebibytes (TiB) of storage. You can create SQL Server RDS DB instances with up to 16 TiB of storage. For this amount of storage, use the Provisioned IOPS SSD and General Purpose SSD storage types.

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html
upvoted 1 times

JA2018 1 year, 6 months ago
[updated - 21 Nove 2024]

Amazon RDS provides three storage types: Provisioned IOPS SSD (also known as io1 and io2 Block Express), General Purpose SSD (also known as gp2 and gp3), and magnetic (also known as standard). They differ in performance characteristics and price, which means that you can tailor your storage performance and cost to the needs of your database workload. You can create Db2, MySQL, MariaDB, Oracle, SQL Server, and PostgreSQL RDS DB instances with up to 64 tebibytes (TiB) of storage. RDS for Db2 doesn't support the gp2 and magnetic storage types.

upvoted 1 times

Steve_4542636 3 years, 3 months ago

Selected Answer: B

for DB instances between 1 TiB and 4 TiB, storage is striped across four Amazon EBS volumes providing burst performance of up to 12,000 IOPS.

from "https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html"

upvoted 2 times

TungPham 3 years, 3 months ago

Selected Answer: B

https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html
Amazon RDS provides three storage types: General Purpose SSD (also known as gp2 and gp3), Provisioned IOPS SSD (also known as io1), and magnetic (also known as standard)
B - MOST cost-effectively

upvoted 4 times

KZM 3 years, 3 months ago

The baseline IOPS performance of gp2 volumes is 3 IOPS per GB, which means that a 1 TB gp2 volume will have a baseline performance of 3,000 IOPS. However, the volume can also burst up to 16,000 IOPS for short periods, but this burst performance is limited and may not be sustained for long durations.
So, I am more prefer option A.

upvoted 1 times

KZM 3 years, 3 months ago

If a 1 TB gp3 EBS volume is used, the maximum available IOPS according to calculations is 3000. This means that the storage can support a requirement of 1000 IOPS, and even 2000 IOPS if the requirement is doubled.
I am confusing between choosing A or B.

upvoted 2 times

mark16dc 3 years, 3 months ago

Selected Answer: A

Option A is the correct answer. A Multi-AZ deployment provides high availability and fault tolerance by automatically replicating data to a standby instance in a different Availability Zone. This allows for seamless failover in the event of a primary instance failure. Using an io2 Block Express EBS volume provides the needed IOPS performance and capacity for the database. It is also designed for low latency and high durability, which makes it a good choice for a database tier.

upvoted 1 times

CapJackSparrow 3 years, 2 months ago

How will you select io2 when RDS only offers io1....magic?

upvoted 1 times

bdp123 3 years, 3 months ago

Selected Answer: B

Correction - hit wrong answer button - meant 'B'
Amazon RDS provides three storage types: General Purpose SSD (also known as gp2 and gp3), Provisioned IOPS SSD (also known as io1)
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/CHAP_Storage.html

upvoted 2 times

Question #354 Topic 1

A company hosts a serverless application on AWS. The application uses Amazon API Gateway, AWS Lambda, and an Amazon RDS for PostgreSQL database. The company notices an increase in application errors that result from database connection timeouts during times of peak traffic or unpredictable traffic. The company needs a solution that reduces the application failures with the least amount of change to the code.

What should a solutions architect do to meet these requirements?

A. Reduce the Lambda concurrency rate.

B. Enable RDS Proxy on the RDS DB instance. Most Voted

C. Resize the RDS DB instance class to accept more connections.

D. Migrate the database to Amazon DynamoDB with on-demand scaling.

Correct Answer: B

Community vote distribution

B (100%)

Comments

TariqKipkemei Highly Voted 3 years, 1 month ago

Selected Answer: B

Many applications, including those built on modern serverless architectures, can have a large number of open connections to the database server and may open and close database connections at a high rate, exhausting database memory and compute resources. Amazon RDS Proxy allows applications to pool and share connections established with the database, improving database efficiency and application scalability. With RDS Proxy, failover times for Aurora and RDS databases are reduced by up to 66%.

https://aws.amazon.com/rds/proxy/

upvoted 11 times

Murtadhaceit Highly Voted 2 years, 6 months ago

Selected Answer: B

A. Reduce the Lambda concurrency rate? Has nothing to do with decreasing connections timeout.
B. Enable RDS Proxy on the RDS DB instance. Correct answer
C. Resize the RDS DB instance class to accept more connections? More connections means worse performance. Therefore, not correct.
D. Migrate the database to Amazon DynamoDB with on-demand scaling? DynamoDB is a noSQL database. Not correct.

upvoted 6 times

JA2018 Most Recent 1 year, 6 months ago

Selected Answer: B

Keys found in STEM:

1. The company notices an increase in application errors that result from database connection timeouts during times of peak traffic or unpredictable traffic.

2. The company needs a solution that reduces the application failures with the least amount of change to the code.

upvoted 1 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

RDS Proxy is a fully managed, highly available, and scalable proxy for Amazon Relational Database Service (RDS) that makes it easy to connect to your RDS instances from applications running on AWS Lambda. RDS Proxy offloads the management of connections to the database, which can help to improve performance and reliability.

upvoted 4 times

elearningtakai 3 years, 2 months ago

Selected Answer: B

To reduce application failures resulting from database connection timeouts, the best solution is to enable RDS Proxy on the RDS DB instance

upvoted 2 times

WherecanIstart 3 years, 2 months ago

Selected Answer: B

RDS Proxy

upvoted 4 times

nder 3 years, 3 months ago

Selected Answer: B

RDS Proxy will pool connections, no code changes need to be made

upvoted 2 times

bdp123 3 years, 3 months ago

Selected Answer: B

RDS proxy

upvoted 2 times

Neha999 3 years, 3 months ago

B RDS Proxy
https://aws.amazon.com/rds/proxy/

upvoted 3 times

Question #355 Topic 1

A company is migrating an old application to AWS. The application runs a batch job every hour and is CPU intensive. The batch job takes 15 minutes on average with an on-premises server. The server has 64 virtual CPU (vCPU) and 512 GiB of memory.

Which solution will run the batch job within 15 minutes with the LEAST operational overhead?

A. Use AWS Lambda with functional scaling.

B. Use Amazon Elastic Container Service (Amazon ECS) with AWS Fargate.

C. Use Amazon Lightsail with AWS Auto Scaling.

D. Use AWS Batch on Amazon EC2. Most Voted

Correct Answer: D

Community vote distribution

D (98%) 2%

Comments

NolaHOla Highly Voted 3 years, 3 months ago

The amount of CPU and memory resources required by the batch job exceeds the capabilities of AWS Lambda and Amazon Lightsail with AWS Auto Scaling, which offer limited compute resources. AWS Fargate offers containerized application orchestration and scalable infrastructure, but may require additional operational overhead to configure and manage the environment. AWS Batch is a fully managed service that automatically provisions the required infrastructure for batch jobs, with options to use different instance types and launch modes.

Therefore, the solution that will run the batch job within 15 minutes with the LEAST operational overhead is D. Use AWS Batch on Amazon EC2. AWS Batch can handle all the operational aspects of job scheduling, instance management, and scaling while using Amazon EC2 injavascript:void(0)stance with the right amount of CPU and memory resources to meet the job's requirements.

upvoted 22 times

everfly Highly Voted 3 years, 3 months ago

Selected Answer: D

AWS Batch is a fully-managed service that can launch and manage the compute resources needed to execute batch jobs. It can scale the compute environment based on the size and timing of the batch jobs.

upvoted 13 times

FlyingHawk Most Recent 1 year, 4 months ago

Selected Answer: D

Lambda has a hard limit of 10 GB of memory and 6 vCPUs per function execution, Amazon Light sail has the limit of 8vCPUs and 32GB memory, so A and C out, B and D can meet vCPU and memory requirements, but B is more expensive than D, so correct answer is D.

upvoted 1 times

Danilus 1 year, 7 months ago

Selected Answer: D

key-The batch job takes 15 minutes on average
key-LEAST operational overhead
key-the batch job
is not A because lambda execution takes 15 max and lambda dont support 512 GIB of memory
the answer is D batch it is designed to run large-scale batch jobs an automatically manages the scaling of resources also batch allow jobs to be distributed accross multiple instances because supports parallel execution

upvoted 3 times

Ramdi1 2 years, 8 months ago

Selected Answer: D

The question needs to be phrased differently. I assume at first it was Lambda, because it says 15 minutes in the question which can be done. Yes it also does say CPU intensive however they go on with a full stop and then give you the server specs. It does not say it uses that much of the specs so they need to really rephrase the questions.

upvoted 4 times

allanm 10 months, 1 week ago

The 15 mins requirement is a distraction - lambda cannot process the requirement given the memory size of the job

upvoted 1 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: D

The main reasons are:

AWS Batch can easily schedule and run batch jobs on EC2 instances. It can scale up to the required vCPUs and memory to match the on-premises server.
Using EC2 provides full control over the instance type to meet the resource needs.
No servers or clusters to manage like with ECS/Fargate or Lightsail. AWS Batch handles this automatically.
More cost effective and operationally simple compared to Lambda which is not ideal for long running batch jobs.

upvoted 7 times

BrijMohan08 2 years, 9 months ago

Selected Answer: A

On-Prem was avg 15 min, but target state architecture is expected to finish within 15 min

upvoted 1 times

pentium75 2 years, 5 months ago

How? The on-prem server has 64 CPUs and 512 GB RAM, Lambda offers much less. And even on-prem it takes 15 minutes ON AVERAGE, sometime more.

upvoted 5 times

jayce5 2 years, 10 months ago

Selected Answer: D

Not Lambda, "average 15 minutes" means there are jobs with running more and less than 15 minutes. Lambda max is 15 minutes.

upvoted 3 times

Gooniegoogoo 2 years, 11 months ago

This is for certain a tough one. I do see that they have thrown a curve ball in making it Lambda Functional scaling, however what we dont know is if this application has many request or one large one.. looks like Lambda can scale and use the same lambda env.. seems intensive tho so will go with D

upvoted 5 times

TariqKipkemei 3 years, 1 month ago

Selected Answer: D

AWS Batch

upvoted 3 times

JLII 3 years, 3 months ago

Selected Answer: D

Not A because: "AWS Lambda now supports up to 10 GB of memory and 6 vCPU cores for Lambda Functions." https://aws.amazon.com/about-aws/whats-new/2020/12/aws-lambda-supports-10gb-memory-6-vcpu-cores-lambda-functions/ vs. "The server has 64 virtual CPU (vCPU) and 512 GiB of memory" in the question.

upvoted 7 times

geekgirl22 3 years, 3 months ago

A is the answer. Lambda is known that has a limit of 15 minutes. So for as long as it says "within 15 minutes" that should be a clear indication it is Lambda

upvoted 2 times

nder 3 years, 3 months ago

Wrong, the job takes "On average 15 minutes" and requires more cpu and ram than lambda can deal with. AWS Batch is correct in this scenario

upvoted 4 times

geekgirl22 3 years, 3 months ago

read the rest of the question which gives the answer:
"Which solution will run the batch job within 15 minutes with the LEAST operational overhead?"
Keyword "Within 15 minutes"

upvoted 3 times

Lonojack 3 years, 3 months ago

What happens if it EXCEEDS the 15 min AVERAGE?
Average = possibly can be more than 15min.
The safer bet would be option D: AWS Batch on EC2

upvoted 8 times

Terion 2 years, 8 months ago

I think what he means is that it takes on average 15 min on prem only

upvoted 2 times

awsgeek75 2 years, 4 months ago

How are you going to get 64 vCPUS to a Lambda function?

upvoted 2 times

bdp123 3 years, 3 months ago

Selected Answer: D

AWS batch on EC2

upvoted 2 times

Question #356 *Topic 1*

A company stores its data objects in Amazon S3 Standard storage. A solutions architect has found that 75% of the data is rarely accessed after 30 days. The company needs all the data to remain immediately accessible with the same high availability and resiliency, but the company wants to minimize storage costs.

Which storage solution will meet these requirements?

A. Move the data objects to S3 Glacier Deep Archive after 30 days.

B. Move the data objects to S3 Standard-Infrequent Access (S3 Standard-IA) after 30 days. **Most Voted**

C. Move the data objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) after 30 days.

D. Move the data objects to S3 One Zone-Infrequent Access (S3 One Zone-IA) immediately.

Correct Answer: B

Community vote distribution

B (100%)

Comments

Lonojack Highly Voted 3 years, 3 months ago

Selected Answer: B

Needs immediate accessibility after 30days, IF they need to be accessed.

upvoted 6 times

Piccalo Highly Voted 3 years, 2 months ago

Highly available so One Zone IA is out the question
Glacier Deep archive isn't immediately accessible 12-48 hours
B is the answer.

upvoted 5 times

Danilus Most Recent 1 year, 7 months ago

Selected Answer: B

Key - high availability and resiliency
Key-the data to remain immediately accessible

Its not C or D because its One zone
its no B even if glacier is cheaper the problem is that data must be immediately accesible so the answer is B

upvoted 2 times

Aru_1994 1 year, 10 months ago

Option B

upvoted 2 times

Apexakil1996 2 years, 5 months ago

One -zone -infrequent access cannot be the answer because it requires high availability so standard infrequent access should be the answer

upvoted 5 times

TariqKipkemei 2 years, 7 months ago

Selected Answer: B

high availability, resiliency = multi AZ
75% of the data is rarely accessed but remain immediately accessible = Standard-Infrequent Access

upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

The correct answer is B.

S3 Standard-IA is a storage class that is designed for infrequently accessed data. It offers lower storage costs than S3 Standard, but it has a retrieval latency of 1-5 minutes.

upvoted 4 times

elearningtakai 3 years, 2 months ago

Selected Answer: B

S3 Glacier Deep Archive is intended for data that is rarely accessed and can tolerate retrieval times measured in hours. Moving data to S3 One Zone-IA immediately would not meet the requirement of immediate accessibility with the same high availability and resiliency.

upvoted 2 times

KS2020 3 years, 2 months ago

The answer should be C.
S3 One Zone-IA is for data that is accessed less frequently but requires rapid access when needed. Unlike other S3 Storage Classes which store data in a minimum of three Availability Zones (AZs), S3 One Zone-IA stores data in a single AZ and costs 20% less than S3 Standard-IA.

https://aws.amazon.com/s3/storage-classes/#:~:text=S3%20One%20Zone%2DIA%20is,less%20than%20S3%20Standard%2DIA.

upvoted 1 times

shanwford 3 years, 2 months ago

The Question emphasises to kepp same high availability class - S3 One Zone-IA doesnt support multiple Availability Zone data resilience model like S3 Standard-Infrequent Access.

upvoted 3 times

bdp123 3 years, 3 months ago

Selected Answer: B

S3 Standard-Infrequent Access after 30 days

upvoted 3 times

NolaHOla 3 years, 3 months ago

B
Option B - Move the data objects to S3 Standard-Infrequent Access (S3 Standard-IA) after 30 days - will meet the requirements of keeping the data immediately accessible with high availability and resiliency, while minimizing storage costs. S3 Standard-IA is designed for infrequently accessed data and it provides a lower storage cost than S3 Standard, while still offering the same low latency, high throughput, and high durability as S3 Standard.

upvoted 5 times

Question #357 *Topic 1*

A gaming company is moving its public scoreboard from a data center to the AWS Cloud. The company uses Amazon EC2 Windows Server instances behind an Application Load Balancer to host its dynamic application. The company needs a highly available storage solution for the application. The application consists of static files and dynamic server-side code.

Which combination of steps should a solutions architect take to meet these requirements? (Choose two.)

A. Store the static files on Amazon S3. Use Amazon CloudFront to cache objects at the edge. **Most Voted**

B. Store the static files on Amazon S3. Use Amazon ElastiCache to cache objects at the edge.

C. Store the server-side code on Amazon Elastic File System (Amazon EFS). Mount the EFS volume on each EC2 instance to share the files.

D. Store the server-side code on Amazon FSx for Windows File Server. Mount the FSx for Windows File Server volume on each EC2 instance to share the files. **Most Voted**

E. Store the server-side code on a General Purpose SSD (gp2) Amazon Elastic Block Store (Amazon EBS) volume. Mount the EBS volume on each EC2 instance to share the files.

Correct Answer: AD

Community vote distribution

AD (100%)

Comments

Steve_4542636 Highly Voted 3 years, 3 months ago

Selected Answer: AD

A because Elasticache, despite being ideal for leaderboards per Amazon, doesn't cache at edge locations. D because FSx has higher performance for low latency needs.

https://www.techtarget.com/searchaws/tip/Amazon-FSx-vs-EFS-Compare-the-AWS-file-services

"FSx is built for high performance and submillisecond latency using solid-state drive storage volumes. This design enables users to select storage capacity and latency independently. Thus, even a subterabyte file system can have 256 Mbps or higher throughput and support volumes up to 64 TB."

upvoted 7 times

Nel8 3 years, 2 months ago

Just to add, ElastiCache is use in front of AWS database.

upvoted 2 times

baba365 2 years, 8 months ago

Why not EFS?

upvoted 1 times

Guru4Cloud Highly Voted 2 years, 9 months ago

Selected Answer: AD

The reasons are:

Storing static files in S3 with CloudFront provides durability, high availability, and low latency by caching at edge locations.
FSx for Windows File Server provides a fully managed Windows native file system that can be accessed from the Windows EC2 instances to share server-side code. It is designed for high availability and scales up to 10s of GBPS throughput.
EFS and EBS volumes can be attached to a single AZ. FSx and S3 are replicated across AZs for high availability.

upvoted 6 times

Danilus Most Recent 1 year, 7 months ago

Selected Answer: AD

Key: Static files and dynamic server-side code.
Key: Windows Server.

The answer is A because S3 is for static files, and CloudFront is a CDN.

The second answer is D because FSx works for Windows File Server.

It's not C because EFS is just for Linux.
It's not E because EBS doesn't share files between EC2 instances.
upvoted 3 times

rodrigoleoncio 2 years ago

Selected Answer: AD

A because Elasticache doesn't cache at edge locations. D because FSx has higher performance for low latency needs.
upvoted 2 times

awsgeek75 2 years, 4 months ago

The question and options are badly worded. How does (D) storing server side code on a file server makes it executable?
upvoted 2 times

4fad2f8 2 years, 5 months ago

you can't mount efs on windows
upvoted 3 times

WherecanIstart 3 years, 2 months ago

Selected Answer: AD

A & D for sure
upvoted 5 times

KZM 3 years, 3 months ago

It is obvious that A and D.
upvoted 2 times

bdp123 3 years, 3 months ago

Selected Answer: AD

both A and D seem correct
upvoted 2 times

NolaHOla 3 years, 3 months ago

A and D seems correct
upvoted 2 times

Question #358 *Topic 1*

A social media company runs its application on Amazon EC2 instances behind an Application Load Balancer (ALB). The ALB is the origin for an Amazon CloudFront distribution. The application has more than a billion images stored in an Amazon S3 bucket and processes thousands of images each second. The company wants to resize the images dynamically and serve appropriate formats to clients.

Which solution will meet these requirements with the LEAST operational overhead?

A. Install an external image management library on an EC2 instance. Use the image management library to process the images.

B. Create a CloudFront origin request policy. Use the policy to automatically resize images and to serve the appropriate format based on the User-Agent HTTP header in the request.

C. Use a Lambda@Edge function with an external image management library. Associate the Lambda@Edge function with the CloudFront behaviors that serve the images. **Most Voted**

D. Create a CloudFront response headers policy. Use the policy to automatically resize images and to serve the appropriate format based on the User-Agent HTTP header in the request.

Correct Answer: C

Community vote distribution

C (89%) 11%

Comments

NolaHOla Highly Voted 2 years, 9 months ago

Use a Lambda@Edge function with an external image management library. Associate the Lambda@Edge function with the CloudFront behaviors tha serve the images.

Using a Lambda@Edge function with an external image management library is the best solution to resize the images dynamically and serve appropriate formats to clients. Lambda@Edge is a serverless computing service that allows running custom code in response to CloudFront events, such as viewer requests and origin requests. By using a Lambda@Edge function, it's possible to process images on the fly and modify the CloudFront response before it's sent back to the client. Additionally, Lambda@Edge has built-in support for external libraries that can be used to process images This approach will reduce operational overhead and scale automatically with traffic.

upvoted 21 times

TariqKipkemei Highly Voted 2 years, 1 month ago

Selected Answer: C

The moment there is a need to implement some logic at the CDN think Lambda@Edge.

upvoted 12 times

Guru4Cloud Most Recent 2 years, 3 months ago

Selected Answer: C

The correct answer is C.

A Lambda@Edge function is a serverless function that runs at the edge of the CloudFront network. This means that the function is executed close to the user, which can improve performance.
An external image management library can be used to resize images and to serve the appropriate format.
Associating the Lambda@Edge function with the CloudFront behaviors that serve the images ensures that the function is executed for all requests that are served by those behaviors.

upvoted 4 times

BrijMohan08 2 years, 3 months ago

Selected Answer: B

If the user asks for the most optimized image format (JPEG,WebP, or AVIF) using the directive format=auto, CloudFront Function will select the best format based on the Accept header present in the request.

Latest documentation: https://aws.amazon.com/blogs/networking-and-content-delivery/image-optimization-using-amazon-cloudfront-and-aws-lambda/

upvoted 3 times

pentium75 1 year, 11 months ago

But a policy alone cannot resize images.

upvoted 1 times

bdp123 2 years, 9 months ago

Selected Answer: C

https://aws.amazon.com/cn/blogs/networking-and-content-delivery/resizing-images-with-amazon-cloudfront-lambdaedge-aws-cdn-blog/

upvoted 5 times

everfly 2 years, 9 months ago

Selected Answer: C

https://aws.amazon.com/cn/blogs/networking-and-content-delivery/resizing-images-with-amazon-cloudfront-lambdaedge-aws-cdn-blog/

upvoted 3 times

Question #359 Topic 1

A hospital needs to store patient records in an Amazon S3 bucket. The hospital's compliance team must ensure that all protected health information (PHI) is encrypted in transit and at rest. The compliance team must administer the encryption key for data at rest.

Which solution will meet these requirements?

A. Create a public SSL/TLS certificate in AWS Certificate Manager (ACM). Associate the certificate with Amazon S3. Configure default encryption for each S3 bucket to use server-side encryption with AWS KMS keys (SSE-KMS). Assign the compliance team to manage the KMS keys.

B. Use the aws:SecureTransport condition on S3 bucket policies to allow only encrypted connections over HTTPS (TLS). Configure default encryption for each S3 bucket to use server-side encryption with S3 managed encryption keys (SSE-S3). Assign the compliance team to manage the SSE-S3 keys.

C. Use the aws:SecureTransport condition on S3 bucket policies to allow only encrypted connections over HTTPS (TLS). Configure default encryption for each S3 bucket to use server-side encryption with AWS KMS keys (SSE-KMS). Assign the compliance team to manage the KMS keys. Most Voted

D. Use the aws:SecureTransport condition on S3 bucket policies to allow only encrypted connections over HTTPS (TLS). Use Amazon Macie to protect the sensitive data that is stored in Amazon S3. Assign the compliance team to manage Macie.

Correct Answer: C

Community vote distribution

C (89%) 9%

Comments

NolaHOla Highly Voted 2 years, 9 months ago

Option C is correct because it allows the compliance team to manage the KMS keys used for server-side encryption, thereby providing the necessary control over the encryption keys. Additionally, the use of the "aws:SecureTransport" condition on the bucket policy ensures that all connections to th S3 bucket are encrypted in transit.

option B might be misleading but using SSE-S3, the encryption keys are managed by AWS and not by the compliance team

upvoted 31 times

Lonojack 2 years, 9 months ago

Perfect explanation. I Agree

upvoted 5 times

pentium75 Highly Voted 1 year, 11 months ago

Selected Answer: C

Not A, Certificate Manager has nothing to do with S3

Not B, SSE-S3 does not allow compliance team to manage the key

Not D, Macie is for identifying sensitive data, not protecting it

upvoted 13 times

Guru4Cloud Most Recent 2 years, 3 months ago

Selected Answer: C

Macie does not encrypt the data like the question is asking

https://docs.aws.amazon.com/macie/latest/user/what-is-macie.html

Also, SSE-S3 encryption is fully managed by AWS so the Compliance Team can't administer this.

upvoted 3 times

Yadav_Sanjay 2 years, 6 months ago

Selected Answer: C

D - Can't be because - Amazon Macie is a data security service that uses machine learning (ML) and pattern matching to discover and help protect your sensitive data.
Macie discovers sensitive information, can help in protection but can't protect

upvoted 3 times

TariqKipkemei 2 years, 7 months ago

Selected Answer: C

B can work if they do not want control over encryption keys.

upvoted 2 times

Russs99 2 years, 8 months ago

Selected Answer: A

Option A proposes creating a public SSL/TLS certificate in AWS Certificate Manager and associating it with Amazon S3. This step ensures that data is encrypted in transit. Then, the default encryption for each S3 bucket will be configured to use server-side encryption with AWS KMS keys (SSE-KMS), which will provide encryption at rest for the data stored in S3. In this solution, the compliance team will manage the KMS keys, ensuring that they control the encryption keys for data at rest.

upvoted 1 times

Shrestwt 2 years, 7 months ago

ACM cannot be integrated with Amazon S3 bucket directly.

upvoted 3 times

pentium75 1 year, 11 months ago

ACM is for website certificates, has nothing to do with S3.

upvoted 3 times

Bofi 2 years, 8 months ago

Selected Answer: C

Option C seems to be the correct answer, option A is also close but ACM cannot be integrated with Amazon S3 bucket directly, hence, u can not attached TLS to S3. You can only attached TLS certificate to ALB, API Gateway and CloudFront and maybe Global Accelerator but definitely NOT EC2 instance and S3 bucket

upvoted 2 times

CapJackSparrow 2 years, 8 months ago

Selected Answer: C

D makes no sense.

upvoted 3 times

Dody 2 years, 9 months ago

Selected Answer: C

Correct Answer is "C"
"D" is not correct because Amazon Macie securely stores your data at rest using AWS encryption solutions. Macie encrypts data, such as findings, using an AWS managed key from AWS Key Management Service (AWS KMS). However, in the question there is a requirement that the compliance team must administer the encryption key for data at rest.
https://docs.aws.amazon.com/macie/latest/user/data-protection.html

upvoted 3 times

cegama543 2 years, 9 months ago

Selected Answer: C

Option C will meet the requirements.

Explanation:

The compliance team needs to administer the encryption key for data at rest in order to ensure that protected health information (PHI) is encrypted in transit and at rest. Therefore, we need to use server-side encryption with AWS KMS keys (SSE-KMS). The default encryption for each S3 bucket can be configured to use SSE-KMS to ensure that all new objects in the bucket are encrypted with KMS keys.

Additionally, we can configure the S3 bucket policies to allow only encrypted connections over HTTPS (TLS) using the aws:SecureTransport condition. This ensures that the data is encrypted in transit.

upvoted 2 times

Karlos99 2 years, 9 months ago

Selected Answer: C

We must provide encrypted in transit and at rest. Macie is needed to discover and recognize any PII or Protected Health Information. We already know that the hospital is working with the sensitive data) so protect them witn KMS and SSL. Answer D is unnecessary

upvoted 2 times

Steve_4542636 2 years, 9 months ago

Selected Answer: C

Macie does not encrypt the data like the question is asking
https://docs.aws.amazon.com/macie/latest/user/what-is-macie.html

Also, SSE-S3 encryption is fully managed by AWS so the Compliance Team can't administer this.

upvoted 3 times

Abhineet9148232 2 years, 9 months ago

Selected Answer: C

C [Correct]: Ensures Https only traffic (encrypted transit), Enables compliance team to govern encryption key.
D [Incorrect]: Misleading; PHI is required to be encrypted not discovered. Maice is a discovery service. (https://aws.amazon.com/macie/)

upvoted 5 times

Nel8 2 years, 9 months ago

Selected Answer: D

Correct answer should be D. "Use Amazon Macie to protect the sensitive data..."
As requirement says "The hospitals's compliance team must ensure that all protected health information (PHI) is encrypted in transit and at rest."

Macie protects personal record such as PHI. Macie provides you with an inventory of your S3 buckets, and automatically evaluates and monitors the buckets for security and access control. If Macie detects a potential issue with the security or privacy of your data, such as a bucket that becomes publicly accessible, Macie generates a finding for you to review and remediate as necessary.

upvoted 4 times

Drayen25 2 years, 9 months ago

Option C should be

upvoted 3 times

Question #360 *Topic 1*

A company uses Amazon API Gateway to run a private gateway with two REST APIs in the same VPC. The BuyStock RESTful web service calls the CheckFunds RESTful web service to ensure that enough funds are available before a stock can be purchased. The company has noticed in the VPC flow logs that the BuyStock RESTful web service calls the CheckFunds RESTful web service over the internet instead of through the VPC. A solutions architect must implement a solution so that the APIs communicate through the VPC.

Which solution will meet these requirements with the FEWEST changes to the code?

A. Add an X-API-Key header in the HTTP header for authorization.

B. Use an interface endpoint. Most Voted

C. Use a gateway endpoint.

D. Add an Amazon Simple Queue Service (Amazon SQS) queue between the two REST APIs.

Correct Answer: B

Community vote distribution

B (94%) 6%

Comments

everfly Highly Voted 2 years, 9 months ago

Selected Answer: B

an interface endpoint is a horizontally scaled, redundant VPC endpoint that provides private connectivity to a service. It is an elastic network interface with a private IP address that serves as an entry point for traffic destined to the AWS service. Interface endpoints are used to connect VPCs with AWS services

upvoted 23 times

lucdt4 Highly Voted 2 years, 6 months ago

Selected Answer: B

C. Use a gateway endpoint is wrong because gateway endpoints only support for S3 and dynamoDB, so B is correct

upvoted 12 times

zdi561 Most Recent 1 year, 4 months ago

Selected Answer: B

AWS PrivateLink connects resources within one VPC (the consumer) to services within another VPC (the provider) using private IP addresses. But since it uses API gateway, you have to use VPC endpoint to keep private communication though there are other options to avoid using VPC endpoint

upvoted 1 times

Rcosmos 1 year, 4 months ago

Selected Answer: B

B. Usar um endpoint de interface:

Embora os endpoints de interface também possam facilitar a comunicação privada, eles são projetados para integrar serviços AWS ou serviços baseados em VPC endpoints. Para o caso de APIs REST privadas do API Gateway, o endpoint de gateway é mais apropriado.

upvoted 1 times

Rcosmos 1 year, 4 months ago

Selected Answer: B

Por que o endpoint de gateway é a solução ideal?
Comunicação por meio da VPC:Um endpoint de gateway no Amazon API Gateway permite que os serviços na mesma VPC se comuniquem diretamente, sem sair da rede interna ou passar pela Internet pública.
Menor impacto no código:O uso de um endpoint de gateway no API Gateway requer apenas a reconfiguração do endpoint na VPC, sem mudanças significativas no código das APIs.
Segurança:Com um endpoint de gateway, o tráfego é mantido dentro da rede privada da VPC, reduzindo os riscos de segurança associados ao tráfego pela Internet.

Custo-benefício:Um endpoint de gateway é econômico, pois utiliza a infraestrutura existente da VPC e evita custos adicionais relacionados ao tráfego pela Internet.

upvoted 1 times

meowruki 2 years ago

Selected Answer: B

B. Use an interface endpoint.

Here's the reasoning:

Interface Endpoint (Option B): An interface endpoint (also known as VPC endpoint) allows communication between resources in your VPC and services without traversing the public internet. In this case, you can create an interface endpoint for API Gateway in your VPC. This enables the communication between the BuyStock and CheckFunds RESTful web services within the VPC, and it doesn't require significant changes to the code.

X-API-Key header (Option A): Adding an X-API-Key header for authorization doesn't address the issue of ensuring that the APIs communicate through the VPC. It's more related to authentication and authorization mechanisms.

upvoted 5 times

liux99 2 years, 1 month ago

The question here is that the BuyStock RESTful web service calls the CheckFunds RESTful web service through API gateway (internet), not directly. How does API gateway connect the services BuyStock and CheckFunds? It connects the Interface Endpoint of the services through Privatelink. The interface endpoints provide direct connection between services within the same private subnet. Answer B is correct.

upvoted 3 times

youdelin 2 years, 2 months ago

how is it even possible, I mean if it's private and both are in the same VPC then we shouldn't even have such an issue right?

upvoted 4 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: B

B. Use an interface endpoint.

upvoted 2 times

envest 2 years, 6 months ago

Answer B (from abylead)

With API GW, you can create multiple prv REST APIs, only accessible with an interface VPC endpt. To allow/ deny simple or cross acc access to your API from selected VPCs & its endpts, you use resource plcys. In addition, you can also use DX for a connection between onprem network to VPC or your prv API.

API GW to VPC: https://docs.aws.amazon.com/apigateway/latest/developerguide/apigateway-private-apis.html

Less correct & incorrect (infeasible & inadequate) answers:

A)X-API-Key in HTTP header for authorization needs auto-process fcts & changes: inadequate.

C)VPC GW endpts for S3 or DynamDB aren't for RESTful svcs: infeasible.

D)SQS que between 2 REST APIs needs endpts & some changes: inadequate.

upvoted 2 times

aqmdla2002 2 years, 6 months ago

Selected Answer: C

I select C because it's the solution with the " FEWEST changes to the code"

upvoted 1 times

awsgeek75 1 year, 10 months ago

Fewest changes to the code doesn't mean break the code by doing something irrelevant. Gateway endpoint is for S3 and DynamoDB

upvoted 2 times

pentium75 1 year, 11 months ago

Gateway Endpoint can provide access to S3 or DynamoDB, not to API Gateway

upvoted 3 times

TariqKipkemei 2 years, 7 months ago

Selected Answer: B

An interface endpoint is powered by PrivateLink, and uses an elastic network interface (ENI) as an entry point for traffic destined to the service

upvoted 3 times

kprakashbehera 2 years, 9 months ago

Selected Answer: B

BBBBBB

upvoted 2 times

siyam008 2 years, 9 months ago

Selected Answer: C

https://www.linkedin.com/pulse/aws-interface-endpoint-vs-gateway-alex-chang

upvoted 1 times

siyam008 2 years, 9 months ago

Correct answer is B. Incorrectly selected C

upvoted 3 times

DASBOL 2 years, 9 months ago

Selected Answer: B

https://docs.aws.amazon.com/apigateway/latest/developerguide/apigateway-private-apis.html

upvoted 5 times

Sherif_Abbas 2 years, 9 months ago

Selected Answer: C

The only time where an Interface Endpoint may be preferable (for S3 or DynamoDB) over a Gateway Endpoint is if you require access from on-premises, for example you want private access from your on-premise data center

upvoted 2 times

Steve_4542636 2 years, 9 months ago

The RESTful services is neither an S3 or DynamDB service, so a VPC Gateway endpoint isn't available here.

upvoted 6 times

bdp123 2 years, 9 months ago

Selected Answer: B

fewest changes to code and below link:
https://gkzz.medium.com/what-is-the-differences-between-vpc-endpoint-gateway-endpoint-ae97bfab97d8

upvoted 3 times

PoisonBlack 2 years, 7 months ago

This really helped me understand the difference between the two. Thx

upvoted 2 times

Question #361 Topic 1

A company hosts a multiplayer gaming application on AWS. The company wants the application to read data with sub-millisecond latency and run one-time queries on historical data.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use Amazon RDS for data that is frequently accessed. Run a periodic custom script to export the data to an Amazon S3 bucket.

B. Store the data directly in an Amazon S3 bucket. Implement an S3 Lifecycle policy to move older data to S3 Glacier Deep Archive for long-term storage. Run one-time queries on the data in Amazon S3 by using Amazon Athena.

C. Use Amazon DynamoDB with DynamoDB Accelerator (DAX) for data that is frequently accessed. Export the data to an Amazon S3 bucket by using DynamoDB table export. Run one-time queries on the data in Amazon S3 by using Amazon Athena. **Most Voted**

D. Use Amazon DynamoDB for data that is frequently accessed. Turn on streaming to Amazon Kinesis Data Streams. Use Amazon Kinesis Data Firehose to read the data from Kinesis Data Streams. Store the records in an Amazon S3 bucket.

Correct Answer: C

Community vote distribution

C (100%)

Comments

lexotan Highly Voted 3 years, 1 month ago

Selected Answer: C

would be nice to have an explanation on why examtopic selects its answers.

upvoted 13 times

ale_brd_111 2 years, 5 months ago

exam topic does not select anything, these are questions from the free forum topics, the only thing exam topic does is to aggregate them all under one single point of view and if you pay you get to see them all aggregated else you can still scroll topic by topic for free

upvoted 7 times

TariqKipkemei Highly Voted 2 years, 7 months ago

Selected Answer: C

DAX delivers up to a 10 times performance improvement—from milliseconds to microseconds.
Using DynamoDB export to S3, you can export data from an Amazon DynamoDB table to an Amazon S3 bucket. This feature enables you to perform analytics and complex queries on your data using other AWS services such as Athena, AWS Glue.

upvoted 11 times

8935983 Most Recent 8 months, 3 weeks ago

Selected Answer: C

If you see gaming think Accelerator

upvoted 2 times

Omshanti 1 year, 8 months ago

Selected Answer: C

test test

upvoted 2 times

sandordini 2 years, 1 month ago

Selected Answer: C

Sub-millisecond: DynamoDB (DAX), onetime query, Least operational overhead: Athena

upvoted 6 times

Uzbekistan 2 years, 3 months ago

Selected Answer: C

Dynamo DB + DAX = low latency.

upvoted 7 times

fabiomarrocolo 2 years, 4 months ago

Scusate io ho pagato contributor perchè vedo ancora + votati invece di vedere solo la risposta corretta? Grazie.Fabio

upvoted 2 times

LoXoL 2 years, 3 months ago

Vedrai sempre e comunque sia la risposta della community ("Most Voted") che la risposta degli admin (rettangolo verde). Occhio perche' la risposta degli admin non sempre e' corretta.

upvoted 1 times

Mikado211 2 years, 6 months ago

Sub-millisecond latency == DAX

upvoted 5 times

Mikado211 2 years, 6 months ago

So C !

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: C

Amazon DynamoDB with DynamoDB Accelerator (DAX) is a fully managed, in-memory caching solution for DynamoDB. DAX can improve the performance of DynamoDB by up to 10x. This makes it a good choice for data that needs to be accessed with sub-millisecond latency.
DynamoDB table export allows you to export data from DynamoDB to an S3 bucket. This can be useful for running one-time queries on historical data.
Amazon Athena is a serverless, interactive query service that makes it easy to analyze data in Amazon S3. Athena can be used to run one-time queries on the data in the S3 bucket.

upvoted 6 times

aaroncelestin 2 years, 9 months ago

A NoSQL isn't even mentioned in the question and yet we are supposed to just imagine this fictional customer is using a NoSql DB

upvoted 4 times

marufxplorer 2 years, 11 months ago

C
Amazon DynamoDB with DynamoDB Accelerator (DAX): DynamoDB is a fully managed NoSQL database service provided by AWS. It is designed for low-latency access to frequently accessed data. DynamoDB Accelerator (DAX) is an in-memory cache for DynamoDB that can significantly reduce read latency, making it suitable for achieving sub-millisecond read times.

upvoted 4 times

lucdt4 3 years ago

Selected Answer: C

C is correct
A don't meets a requirement (LEAST operational overhead) because use script
B: Not regarding to require
D: Kinesis for near-real-time (Not for read)
-> C is correct

upvoted 3 times

DagsH 3 years, 2 months ago

Selected Answer: C

Agreed C will be best because of DynamoDB DAX

upvoted 2 times

BeeKayEnn 3 years, 2 months ago

Option C will be the best fit.
As they would like to retrieve the data with sub-millisecond, DynamoDB with DAX is the answer.
DynamoDB supports some of the world's largest scale applications by providing consistent, single-digit millisecond response times at any scale. You can build applications with virtually unlimited throughput and storage.

upvoted 3 times

Grace83 3 years, 2 months ago

C is the correct answer

upvoted 2 times

KAUS2 3 years, 3 months ago

Selected Answer: C

Option C is the right one. The questions clearly states "sub-millisecond latency "

upvoted 3 times

smgsi 3 years, 3 months ago

Selected Answer: C

https://aws.amazon.com/dynamodb/dax/?nc1=h_ls

upvoted 4 times

Question #362 Topic 1

A company uses a payment processing system that requires messages for a particular payment ID to be received in the same order that they were sent. Otherwise, the payments might be processed incorrectly.

Which actions should a solutions architect take to meet this requirement? (Choose two.)

A. Write the messages to an Amazon DynamoDB table with the payment ID as the partition key.

B. Write the messages to an Amazon Kinesis data stream with the payment ID as the partition key. Most Voted

C. Write the messages to an Amazon ElastiCache for Memcached cluster with the payment ID as the key.

D. Write the messages to an Amazon Simple Queue Service (Amazon SQS) queue. Set the message attribute to use the payment ID.

E. Write the messages to an Amazon Simple Queue Service (Amazon SQS) FIFO queue. Set the message group to use the payment ID. Most Voted

Correct Answer: BE

Community vote distribution

BE (85%) Other

Comments

Ashkan_10 Highly Voted 3 years, 2 months ago

Selected Answer: BE

Option B is preferred over A because Amazon Kinesis Data Streams inherently maintain the order of records within a shard, which is crucial for the given requirement of preserving the order of messages for a particular payment ID. When you use the payment ID as the partition key, all messages for that payment ID will be sent to the same shard, ensuring that the order of messages is maintained.

On the other hand, Amazon DynamoDB is a NoSQL database service that provides fast and predictable performance with seamless scalability. While it can store data with partition keys, it does not guarantee the order of records within a partition, which is essential for the given use case. Hence, using Kinesis Data Streams is more suitable for this requirement.

As DynamoDB does not keep the order, I think BE is the correct answer here.

upvoted 33 times

awsgeek75 Highly Voted 2 years, 4 months ago

I don't understand the question. The only requirement is: " system that requires messages for a particular payment ID to be received in the same order that they were sent"

SQS FIFO (E) meets this requirement.

Why would you "write the message" to Kinesis or DynamoDB anymore. There is no streaming or DB storage requirement in the question. Between A/B, B is better logically but it doesn't meet any stated requirement.

Happy to understand what I'm missing

upvoted 9 times

MatAlves 1 year, 9 months ago

Instead of "what actions...", the question should say "what are the alternatives/options that meet this requirement".

upvoted 5 times

Manimgh Most Recent 1 year, 1 month ago

Selected Answer: BE

same order = FIFO first in first out

upvoted 1 times

FlyingHawk 1 year, 6 months ago

Selected Answer: BE

Both B and E can ensure the messages being processed in the order they were received, however SQS FIFO queues (E) are simpler, more cost-effective, and better aligned with the needs of this use case compared to Kinesis. Unless you have additional real-time analytics or high-throughput requirements, E is the superior choice, the big reason why KDS can preserve the order is due to sequence number assigned to data record when it was written to shard, Dynamo DB does not have a sequence number associated with the record.
https://docs.aws.amazon.com/streams/latest/dev/key-concepts.html

upvoted 2 times

pentium75 2 years, 5 months ago

Selected Answer: BE

Both Kinesis and SQS FIFO queue guarantee the order, other answers don't.

upvoted 4 times

meowruki 2 years, 6 months ago

Option B (Write the messages to an Amazon Kinesis data stream with the payment ID as the partition key): Kinesis can provide ordered processing within a shard
Write the messages to an Amazon Simple Queue Service (Amazon SQS) FIFO queue. Set the message group to use the payment ID.

SQS FIFO (First-In-First-Out) queues preserve the order of messages within a message group.

upvoted 4 times

TariqKipkemei 2 years, 7 months ago

Selected Answer: BE

Technically both B and E will ensure processing order, but SQS FIFO was specifically built to handle this requirement.
There is no ask on how to store the data so A and C are out.

upvoted 2 times

Pritam228 2 years, 8 months ago

https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/HowItWorks.Partitions.html

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: DE

options D and E are better because they mimic a real-world queue system and ensure that payments are processed in the correct order, just like customers in a store would be served in the order they arrived. This is crucial for a payment processing system where order matters to avoid mistake in payment processing.

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Amazon Kinesis Data Streams Overkill for Ordering
Overkill for Ordering: While Kinesis can maintain order within a partition key, it might be seen as overkill for a scenario where your primary concern is maintaining the order of payments. SQS FIFO queues (option E) are specifically designed for this purpose and provide an easier and more cost-effective solution.

upvoted 1 times

omoakin 3 years ago

AAAAAAAAA EEEEEEEEEEEEE

upvoted 3 times

Konb 3 years ago

Selected Answer: AE

IF the question would be "Choose all the solutions that fulfill these requirements" I would chosen BE.

But it is:
"Which actions should a solutions architect take to meet this requirement? "

For this reason I chose AE, because we don't need both Kinesis AND SQS for this solution. Both choices complement to order processing: order stored in DB, work item goes to the queue.

upvoted 3 times

Smart 2 years, 10 months ago

Incorrect, AWS will clarify it by using the phrase - "combination of actions".

upvoted 2 times

luisgu 3 years, 1 month ago

Selected Answer: BE

E --> no doubt
B --> see https://docs.aws.amazon.com/streams/latest/dev/key-concepts.html

upvoted 2 times

kruasan 3 years, 1 month ago

Selected Answer: BE

1) SQS FIFO queues guarantee that messages are received in the exact order they are sent. Using the payment ID as the message group ensures all messages for a payment ID are received sequentially.
2) Kinesis data streams can also enforce ordering on a per partition key basis. Using the payment ID as the partition key will ensure strict ordering of messages for each payment ID.

upvoted 3 times

kruasan 3 years, 1 month ago

The other options do not guarantee message ordering. DynamoDB and ElastiCache are not message queues. SQS standard queues deliver messages in approximate order only.

upvoted 3 times

mrgeee 3 years, 1 month ago

Selected Answer: BE

BE no doubt.

upvoted 2 times

nosense 3 years, 1 month ago

Selected Answer: BE

Option A, writing the messages to an Amazon DynamoDB table, would not necessarily preserve the order of messages for a particular payment ID

upvoted 2 times

MssP 3 years, 2 months ago

Selected Answer: BE

I don´t unsderstand A, How you can guaratee the order with DynamoDB?? The order is guarantee with SQS FIFO and Kinesis Data Stream in 1 shard..

upvoted 5 times

pentium75 2 years, 5 months ago

If it really means "combination of actions" than A+E would work, because you'd use the FIFO queue (E) to guarantee the order. Then the order in the database doesn't matter. If they want to alternative solutions then obviously B and E would work while A alone doesn't.

upvoted 2 times

Grace83 3 years, 2 months ago

AE is the answer

upvoted 2 times

Question #363 | Topic 1

A company is building a game system that needs to send unique events to separate leaderboard, matchmaking, and authentication services concurrently. The company needs an AWS event-driven system that guarantees the order of the events.

Which solution will meet these requirements?

A. Amazon EventBridge event bus

B. Amazon Simple Notification Service (Amazon SNS) FIFO topics **Most Voted**

C. Amazon Simple Notification Service (Amazon SNS) standard topics

D. Amazon Simple Queue Service (Amazon SQS) FIFO queues

Correct Answer: B

Community vote distribution

B (79%) | D (16%) | 5%

Comments

bella Highly Voted 3 years, 1 month ago

Selected Answer: B

I don't honestly / can't understand why people go to ChapGPT to ask for the answers.... if I recall correctly they only consolidated their DB until 2021...

upvoted 19 times

LeonSauveterre 1 year, 6 months ago

Yes, it used to be just like that. Fortunately things have changed as of today - Nov 29 2024 in my timezone. Personally I find ChatGPT + Amazon Q to be quite awesome for SAA tests

upvoted 2 times

aaroncelestin 2 years, 9 months ago

Yup, ChatGPT doesn't //know// anything about AWS services. It only repeats what other people have said about it, which could be nonsense or hyperbole or some combination thereof.

upvoted 5 times

LazyTs Highly Voted 2 years, 9 months ago

Selected Answer: B

The answer is B la. SNS FIFO topics queue should be used combined with SQS FIFO queue in this case. The question asked for correct order to different event, so asking for SNS fan out here to send to individual SQS.
https://docs.aws.amazon.com/sns/latest/dg/fifo-example-use-case.html

upvoted 14 times

Po_chih 2 years, 8 months ago

The best answer!

upvoted 1 times

dkw2342 2 years, 3 months ago

B is correct, but this is not about SNS -> SQS fan-out, it's not necessary. Just SNS FIFO for ordered pub/sub messaging.

upvoted 3 times

FlyingHawk Most Recent 1 year, 6 months ago

Selected Answer: B

Base on this doc https://docs.aws.amazon.com/sns/latest/dg/fifo-topic-message-ordering.html, the answer should B and D, however, if you can only select one, then B.

upvoted 1 times

MatAlves 1 year, 9 months ago

Selected Answer: B

SNS can have many-to-many relations, while SQS supports only one consumer at a time (many-to-one).

upvoted 5 times

1e22522 1 year, 10 months ago

First time in my life that the answer is actually SNS

upvoted 6 times

richiexamaws 2 years ago

Selected Answer: D

AWS does not currently offer FIFO topics for SNS. SNS only supports standard topics, which do not guarantee message order.

upvoted 1 times

elmyth 1 year, 9 months ago

I'm creating a topic right now and I have both types to choose.

upvoted 3 times

Darshan07 2 years, 3 months ago

Selected Answer: B

Even chat gpt said B

upvoted 2 times

awsgeek75 2 years, 4 months ago

Selected Answer: B

Yes, you can technically do this with SQS FIFO partitioned queue by giving separate group ID's to leaderboard, matchmaking etc but this is not as useful as SNS FIFO and is overkill as no need for storage etc. B is more elegant and concise solution,

upvoted 5 times

foha2012 2 years, 5 months ago

Guys, ChatGPT sucks !. Try removing [most voted] from choice B and it will choose D. And if you put [most voted] in front of A, it will select A. LOL !

upvoted 3 times

Marco_St 2 years, 5 months ago

Selected Answer: B

just know SNS FIFO also can send events or messages cocurrently to many subscribers while maintaining the order it receives. SNS fanout pattern is set in standard SNS which is commonly used to fan out events to large number of subscribers and usually for duplicated messages.

upvoted 2 times

Mikado211 2 years, 6 months ago

Selected Answer: B

SQS looks like a good idea first, but since we have to send the same message to multiple destination, even if SQS could do it, SNS is much more dedicated to this kind of usage.

upvoted 5 times

sparun1607 2 years, 6 months ago

My Answer is B

https://docs.aws.amazon.com/sns/latest/dg/sns-fifo-topics.html

You can use Amazon SNS FIFO (first in, first out) topics with Amazon SQS FIFO queues to provide strict message ordering and message deduplication. The FIFO capabilities of each of these services work together to act as a fully managed service to integrate distributed applications that require data consistency in near-real time. Subscribing Amazon SQS standard queues to Amazon SNS FIFO topics provides best-effort ordering and at least once delivery.

upvoted 4 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

bbbbbbbbbbbbbbb

upvoted 2 times

jaydesai8 2 years, 11 months ago

Selected Answer: D

SQS FIFO maintains the order of the events - Answer is D

upvoted 2 times

jayce5 3 years ago

Selected Answer: B

It should be the fan-out pattern, and the pattern starts with Amazon SNS FIFO for the orders.
upvoted 3 times

danielklein09 3 years ago

Selected Answer: D

Answer is D. You are so lazy because instead of searching in documentation or your notes, you are asking ChatGPT. Do you really think you will take this exam ? Hint: ask ChatGPT
upvoted 5 times

lucdt4 3 years ago

Selected Answer: D

D is correct (SQS FIFO)
Because B can't send event concurrently though it can send in the order of the events
upvoted 1 times

Question #364 Topic 1

A hospital is designing a new application that gathers symptoms from patients. The hospital has decided to use Amazon Simple Queue Service (Amazon SQS) and Amazon Simple Notification Service (Amazon SNS) in the architecture.

A solutions architect is reviewing the infrastructure design. Data must be encrypted at rest and in transit. Only authorized personnel of the hospital should be able to access the data.

Which combination of steps should the solutions architect take to meet these requirements? (Choose two.)

A. Turn on server-side encryption on the SQS components. Update the default key policy to restrict key usage to a set of authorized principals.

B. Turn on server-side encryption on the SNS components by using an AWS Key Management Service (AWS KMS) customer managed key. Apply a key policy to restrict key usage to a set of authorized principals. **Most Voted**

C. Turn on encryption on the SNS components. Update the default key policy to restrict key usage to a set of authorized principals. Set a condition in the topic policy to allow only encrypted connections over TLS.

D. Turn on server-side encryption on the SQS components by using an AWS Key Management Service (AWS KMS) customer managed key. Apply a key policy to restrict key usage to a set of authorized principals. Set a condition in the queue policy to allow only encrypted connections over TLS. **Most Voted**

E. Turn on server-side encryption on the SQS components by using an AWS Key Management Service (AWS KMS) customer managed key. Apply an IAM policy to restrict key usage to a set of authorized principals. Set a condition in the queue policy to allow only encrypted connections over TLS.

Correct Answer: BD

Community vote distribution

BD (68%) CD (24%) 7%

Comments

awsgeek75 Highly Voted 1 year, 10 months ago

My god! Every other question is about SQS! I thought this was AWS Solution Architect test not "How to solve any problem in AWS using SQS" test!

upvoted 17 times

allanm 10 months, 1 week ago

You make it sound like it's a bad thing...

upvoted 1 times

fkie4 Highly Voted 2 years, 9 months ago

Selected Answer: BD

read this:
https://docs.aws.amazon.com/sns/latest/dg/sns-server-side-encryption.html
https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-server-side-encryption.html

upvoted 15 times

Gooniegoogoo 2 years, 5 months ago

good call.. that confirms on that page:

Important
All requests to topics with SSE enabled must use HTTPS and Signature Version 4.

For information about compatibility of other services with encrypted topics, see your service documentation.

Amazon SNS only supports symmetric encryption KMS keys. You cannot use any other type of KMS key to encrypt your service resources. For help determining whether a KMS key is a symmetric encryption key, see Identifying asymmetric KMS keys.

upvoted 4 times

LucV3 9 months, 1 week ago

Doesn't "All requests to topics with SSE enabled must use HTTPS and Signature Version 4." mean you have to enable TLS?

upvoted 1 times

mahesuma Most Recent 1 week, 5 days ago

Selected Answer: CD

Encryption at Rest and transit => this is core of this question

upvoted 1 times

LucV3 9 months, 1 week ago

Selected Answer: CD

SSL/TLS covered for both

upvoted 1 times

pentium75 1 year, 11 months ago

Selected Answer: BD

A and C involve 'updating the default key policy' which is not something you. Either you create a key policy, OR AWS assigns THE "default key policy'
E 'applies an IAM policy to restrict key usage to a set of authorized principals' which is not how IAM policies work. You can 'apply an IAM policy to restrict key usage', but it would be restricted to the principals who have the policy attached; you can't specify them in the policy.

Leaves B and D. That B lacks the TLS statement is irrelevant because "all requests to topics with SSE enabled must use HTTPS" anyway.

upvoted 7 times

dkw2342 1 year, 9 months ago

Yes, BD is correct.

"All requests to queues with SSE enabled must use HTTPS and Signature Version 4." -> valid for SNS and SQS alike:
https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-server-side-encryption.html
https://docs.aws.amazon.com/sns/latest/dg/sns-server-side-encryption.html

"Set a condition in the queue policy to allow only encrypted connections over TLS." refers to the "aws:SecureTransport" condition, but it's actually redundant.

upvoted 3 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: CD

Its only options C and D that covers encryption on transit, encryption at rest and a restriction policy.

upvoted 3 times

Lalo 2 years, 6 months ago

Answer is BD
SNS: AWS KMS, key policy, SQS: AWS KMS, Key policy

upvoted 5 times

luisgu 2 years, 7 months ago

Selected Answer: BD

"IAM policies you can't specify the principal in an identity-based policy because it applies to the user or role to which it is attached"

reference: https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/security_iam_service-with-iam.html

that excludes E

upvoted 2 times

imvb88 2 years, 7 months ago

Selected Answer: CD

Encryption at transit = use SSL/TLS -> rule out A,B
Encryption at rest = encryption on components -> keep C, D, E
KMS always need a key policy, IAM is optional -> E out

-> C, D left, one for SNS, one for SQS. TLS: checked, encryption on components: checked

upvoted 5 times

Lalo 2 years, 6 months ago

Answer is BD
SNS: AWS KMS, key policy, SQS: AWS KMS, Key policy

upvoted 4 times

imvb88 2 years, 7 months ago
https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-data-encryption.html

You can protect data in transit using Secure Sockets Layer (SSL) or client-side encryption. You can protect data at rest by requesting Amazon SQS tc encrypt your messages before saving them to disk in its data centers and then decrypt them when the messages are received.

https://docs.aws.amazon.com/kms/latest/developerguide/key-policies.html

A key policy is a resource policy for an AWS KMS key. Key policies are the primary way to control access to KMS keys. Every KMS key must have exactly one key policy. The statements in the key policy determine who has permission to use the KMS key and how they can use it. You can also use IAM policies and grants to control access to the KMS key, but every KMS key must have a key policy.
upvoted 2 times

MarkGerwich 2 years, 8 months ago
CD
B does not include encryption in transit.
upvoted 3 times

MssP 2 years, 8 months ago
in transit is included in D. With C, not include encrytion at rest.... Server-side will include it.
upvoted 1 times

Bofi 2 years, 8 months ago
That was my objection toward option B. CD cover both encryption at Rest and Server-Side_Encryption
upvoted 1 times

Maximus007 2 years, 8 months ago
ChatGPT returned AD as a correct answer)
upvoted 1 times

cegama543 2 years, 9 months ago
Selected Answer: BE
B: To encrypt data at rest, we can use a customer-managed key stored in AWS KMS to encrypt the SNS components.

E: To restrict access to the data and allow only authorized personnel to access the data, we can apply an IAM policy to restrict key usage to a set of authorized principals. We can also set a condition in the queue policy to allow only encrypted connections over TLS to encrypt data in transit.
upvoted 3 times

Karlos99 2 years, 9 months ago
Selected Answer: BD
For a customer managed KMS key, you must configure the key policy to add permissions for each queue producer and consumer.
https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-key-management.html
upvoted 4 times

Question #365 *Topic 1*

A company runs a web application that is backed by Amazon RDS. A new database administrator caused data loss by accidentally editing information in a database table. To help recover from this type of incident, the company wants the ability to restore the database to its state from 5 minutes before any change within the last 30 days.

Which feature should the solutions architect include in the design to meet this requirement?

A. Read replicas

B. Manual snapshots

C. Automated backups **Most Voted**

D. Multi-AZ deployments

Correct Answer: C

Community vote distribution

C (100%)

Comments

Uzbekistan Highly Voted 1 year, 9 months ago

Selected Answer: C

Amazon RDS provides automated backups, which can be configured to take regular snapshots of the database instance. By enabling automated backups and setting the retention period to 30 days, the company can ensure that it retains backups for up to 30 days. Additionally, Amazon RDS allows for point-in-time recovery within the retention period, enabling the restoration of the database to its state from any point within the last 30 days, including 5 minutes before any change. This feature provides the required capability to recover from accidental data loss incidents.

upvoted 5 times

Guru4Cloud Most Recent 2 years, 3 months ago

Selected Answer: C

Automated backups allow you to recover your database to any point in time within your specified retention period, which can be up to 35 days. The recovery process creates a new Amazon RDS instance with a new endpoint, and the process takes time proportional to the size of the database. Automated backups are enabled by default and occur daily during the backup window. This feature provides an easy and convenient way to recover from data loss incidents such as the one described in the scenario.

upvoted 4 times

elearningtakai 2 years, 8 months ago

Selected Answer: C

Option C, Automated backups, will meet the requirement. Amazon RDS allows you to automatically create backups of your DB instance. Automated backups enable point-in-time recovery (PITR) for your DB instance down to a specific second within the retention period, which can be up to 35 days By setting the retention period to 30 days, the company can restore the database to its state from up to 5 minutes before any change within the last 30 days.

upvoted 4 times

joechen2023 2 years, 5 months ago

I selected C as well, but still don't know how the automatic backup could have a copy 5 minutes before any change. AWS doc states "Automated backups occur daily during the preferred backup window. "
https://docs.aws.amazon.com/AmazonRDS/latest/UserGuide/USER_WorkingWithAutomatedBackups.html.
I think the answer maybe A, as read replica will be kept sync and then restore from the read replica. could an expert help?

upvoted 2 times

awsgeek75 1 year, 11 months ago

"the company wants the ability to restore the database to its state from 5 minutes before any change"
The automatic backup takes a backup every 5 minutes. This means it can restore the database to 5 minutes in the past.

upvoted 2 times

TheFivePips 1 year, 9 months ago

Automated backups enable point-in-time recovery (PITR) for your DB instance down to a specific second within the retention period, which can be up to 35 days

upvoted 2 times

gold4otas 2 years, 8 months ago

Selected Answer: C

C: Automated Backups

https://aws.amazon.com/rds/features/backup/

upvoted 3 times

WherecanIstart 2 years, 8 months ago

Selected Answer: C

Automated Backups...

upvoted 3 times

Question #366 *Topic 1*

A company's web application consists of an Amazon API Gateway API in front of an AWS Lambda function and an Amazon DynamoDB database. The Lambda function handles the business logic, and the DynamoDB table hosts the data. The application uses Amazon Cognito user pools to identify the individual users of the application. A solutions architect needs to update the application so that only users who have a subscription can access premium content.

Which solution will meet this requirement with the LEAST operational overhead?

A. Enable API caching and throttling on the API Gateway API.

B. Set up AWS WAF on the API Gateway API. Create a rule to filter users who have a subscription.

C. Apply fine-grained IAM permissions to the premium content in the DynamoDB table.

D. Implement API usage plans and API keys to limit the access of users who do not have a subscription. **Most Voted**

Correct Answer: D

Community vote distribution

D (86%) 14%

Comments

Guru4Cloud **Highly Voted** 2 years, 3 months ago

Selected Answer: D

Implementing API usage plans and API keys is a straightforward way to restrict access to specific users or groups based on subscriptions. It allows you to control access at the API level and doesn't require extensive changes to your existing architecture. This solution provides a clear and manageable way to enforce access restrictions without complicating other parts of the application

upvoted 13 times

Rcosmos **Most Recent** 1 year, 4 months ago

Selected Answer: C

Segurança robusta:

A lógica de controle de acesso é aplicada no nível do banco de dados, garantindo que usuários não autorizados não consigam acessar os dados premium, mesmo que tentem burlar o aplicativo.

upvoted 1 times

Uzbekistan 1 year, 9 months ago

Selected Answer: C

Chat GPT said:
Option C, "Apply fine-grained IAM permissions to the premium content in the DynamoDB table," would likely involve the least operational overhead.
Here's why:
Granular Control: IAM permissions allow you to control access at a very granular level, including specific actions (e.g., GetItem, PutItem) on individual resources (e.g., DynamoDB tables).
Integration with Cognito: IAM policies can be configured to allow access based on the identity of the user authenticated through Cognito. You can create IAM roles or policies that grant access to users with specific attributes or conditions, such as having a subscription.
Minimal Configuration Changes: This solution primarily involves configuring IAM policies for access control in DynamoDB, which can be done with minimal changes to the existing application architecture.

upvoted 2 times

awsgeek75 1 year, 11 months ago

Selected Answer: C

C is correct as per the link and doc:
https://docs.aws.amazon.com/apigateway/latest/developerguide/api-gateway-api-usage-plans.html#apigateway-usage-plans-best-practices

D: API keys cannot be used to limit access and this can only be done via methods defined in above link

upvoted 2 times

awsgeek75 1 year, 10 months ago

I had to chose D but must have clicked C incorrectly. It is D as my explanation is about D not C! C is the wrong answer.

upvoted 2 times

awsgeek75 1 year, 11 months ago

Also, option A is for performance and not for security
option B, WAF cannot control access based on subscription without massive custom coding which will be a big operational overhead

upvoted 2 times

lipi0035 2 years ago

In the same document https://docs.aws.amazon.com/apigateway/latest/developerguide/api-gateway-api-usage-plans.html if you scroll down, it says `Don't use API keys for authentication or authorization to control access to your APIs. If you have multiple APIs in a usage plan, a user with a valid API key for one API in that usage plan can access all APIs in that usage plan. Instead, to control access to your API, use an IAM role, a Lambda authorizer, or an Amazon Cognito user pool.`

In the same document at the bottom, it says "If you're using a developer portal to publish your APIs, note that all APIs in a given usage plan are subscribable, even if you haven't made them visible to your customers."

I go with C

upvoted 2 times

awsgeek75 1 year, 11 months ago

https://docs.aws.amazon.com/apigateway/latest/developerguide/api-gateway-api-usage-plans.html#apigateway-usage-plans-best-practices

Correct link

upvoted 2 times

TariqKipkemei 2 years, 1 month ago

Selected Answer: D

After you create, test, and deploy your APIs, you can use API Gateway usage plans to make them available as product offerings for your customers. You can configure usage plans and API keys to allow customers to access selected APIs, and begin throttling requests to those APIs based on defined limits and quotas. These can be set at the API, or API method level.
https://docs.aws.amazon.com/apigateway/latest/developerguide/api-gateway-api-usage-plans.html#:~:text=Creating%20and%20using-,usage%20plans,-with%20API%20keys

upvoted 2 times

marufxplorer 2 years, 5 months ago

D
Option D involves implementing API usage plans and API keys. By associating specific API keys with users who have a valid subscription, you can control access to the premium content.

upvoted 2 times

kruasan 2 years, 7 months ago

Selected Answer: D

A. This would not actually limit access based on subscriptions. It helps optimize and control API usage, but does not address the core requirement.
B. This could work by checking user subscription status in the WAF rule, but would require ongoing management of WAF and increases operational overhead.
C. This is a good approach, using IAM permissions to control DynamoDB access at a granular level based on subscriptions. However, it would require managing IAM permissions which adds some operational overhead.
D. This option uses API Gateway mechanisms to limit API access based on subscription status. It would require the least amount of ongoing management and changes, minimizing operational overhead. API keys could be easily revoked/changed as subscription status changes.

upvoted 4 times

imvb88 2 years, 7 months ago

CD both possible but D is more suitable since it mentioned in https://docs.aws.amazon.com/apigateway/latest/developerguide/api-gateway-api-usage-plans.html

A,B not relevant.

upvoted 3 times

elearningtakai 2 years, 8 months ago

Selected Answer: D

The solution that will meet the requirement with the least operational overhead is to implement API Gateway usage plans and API keys to limit access to premium content for users who do not have a subscription.
Option A is incorrect because API caching and throttling are not designed for authentication or authorization purposes, and it does not provide access control.
Option B is incorrect because although AWS WAF is a useful tool to protect web applications from common web exploits, it is not designed for authorization purposes, and it might require additional configuration, which increases the operational overhead.
Option C is incorrect because although IAM permissions can restrict access to data stored in a DynamoDB table, it does not provide a mechanism for limiting access to specific content based on the user subscription. Moreover, it might require a significant amount of additional IAM permissions configuration, which increases the operational overhead.

upvoted 3 times

klayytech 2 years, 8 months ago

Selected Answer: D

To meet the requirement with the least operational overhead, you can implement API usage plans and API keys to limit the access of users who do not have a subscription. This way, you can control access to your API Gateway APIs by requiring clients to submit valid API keys with requests. You can associate usage plans with API keys to configure throttling and quota limits on individual client accounts.

upvoted 3 times

techhb 2 years, 9 months ago

answer is D ,if looking for least overhead
https://docs.aws.amazon.com/apigateway/latest/developerguide/api-gateway-api-usage-plans.html
C will achieve it but operational overhead is high.

upvoted 3 times

quentin17 2 years, 9 months ago

Selected Answer: D

Both C&D are valid solution
According to ChatGPT:
"Applying fine-grained IAM permissions to the premium content in the DynamoDB table is a valid approach, but it requires more effort in managing IAM policies and roles for each user, making it more complex and adding operational overhead."

upvoted 2 times

Karlos99 2 years, 9 months ago

Selected Answer: D

https://docs.aws.amazon.com/apigateway/latest/developerguide/api-gateway-api-usage-plans.html

upvoted 4 times

Question #367 | *Topic 1*

A company is using Amazon Route 53 latency-based routing to route requests to its UDP-based application for users around the world. The application is hosted on redundant servers in the company's on-premises data centers in the United States, Asia, and Europe. The company's compliance requirements state that the application must be hosted on premises. The company wants to improve the performance and availability of the application.

What should a solutions architect do to meet these requirements?

A. Configure three Network Load Balancers (NLBs) in the three AWS Regions to address the on-premises endpoints. Create an accelerator by using AWS Global Accelerator, and register the NLBs as its endpoints. Provide access to the application by using a CNAME that points to the accelerator DNS. **Most Voted**

B. Configure three Application Load Balancers (ALBs) in the three AWS Regions to address the on-premises endpoints. Create an accelerator by using AWS Global Accelerator, and register the ALBs as its endpoints. Provide access to the application by using a CNAME that points to the accelerator DNS.

C. Configure three Network Load Balancers (NLBs) in the three AWS Regions to address the on-premises endpoints. In Route 53, create a latency-based record that points to the three NLBs, and use it as an origin for an Amazon CloudFront distribution. Provide access to the application by using a CNAME that points to the CloudFront DNS.

D. Configure three Application Load Balancers (ALBs) in the three AWS Regions to address the on-premises endpoints. In Route 53, create a latency-based record that points to the three ALBs, and use it as an origin for an Amazon CloudFront distribution. Provide access to the application by using a CNAME that points to the CloudFront DNS.

Correct Answer: A

Community vote distribution

A (98%) 2%

Comments

Guru4Cloud Highly Voted 2 years, 9 months ago

Selected Answer: A

NLBs allow UDP traffic (ALBs don't support UDP)
Global Accelerator uses Anycast IP addresses and its global network to intelligently route users to the optimal endpoint
Using NLBs as Global Accelerator endpoints provides improved availability and DDoS protection.

upvoted 12 times

lucdt4 Highly Voted 3 years ago

Selected Answer: A

C - D: Cloudfront don't support UDP/TCP
B: Global accelerator don't support ALB
A is correct

upvoted 6 times

JA2018 Most Recent 1 year, 6 months ago

Selected Answer: C

CloudFront's origin can be on-premises sources.

Check this out:
https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/DownloadDistS3AndCustomOrigins.html#concept_CustomOrigin

"A custom origin is an HTTP server, for example, a web server. The HTTP server can be an Amazon EC2 instance or an HTTP server that you host somewhere else. "

upvoted 1 times

sandordini 2 years, 1 month ago

Selected Answer: A
Non-HTTP, Massive performance: NLB, UDP: AWS Global Accelerator
upvoted 4 times

pentium75 2 years, 5 months ago
Selected Answer: A
Neither ALB (B+D) nor CloudFront (C+D) do support UDP.
upvoted 5 times

TariqKipkemei 2 years, 7 months ago
Selected Answer: A
UDP = NLB and Global Accelerator
upvoted 5 times

live_reply_developers 2 years, 11 months ago
Selected Answer: A
NLB + GA support UDP/TCP
upvoted 4 times

Gooniegoogoo 2 years, 11 months ago
good reference https://blog.cloudcraft.co/alb-vs-nlb-which-aws-load-balancer-fits-your-needs/
upvoted 3 times

SkyZeroZx 3 years, 1 month ago
Selected Answer: A
UDP = NBL
UDP = GLOBAL ACCELERATOR
UPD NOT WORKING WITH CLOUDFRONT
ANS IS A
upvoted 4 times

MssP 3 years, 2 months ago
Selected Answer: A
More discussions at: https://www.examtopics.com/discussions/amazon/view/51508-exam-aws-certified-solutions-architect-associate-saa-c02/
upvoted 1 times

Grace83 3 years, 2 months ago
Why is C not correct - does anyone know?
upvoted 2 times

MssP 3 years, 2 months ago
It could be valid but I think A is better. Uses the AWS global network to optimize the path from users to applications, improving the performance of TCP and UDP traffic
upvoted 2 times

Shrestwt 3 years, 1 month ago
Latency based routing is already using in the application, so AWS global network will optimize the path from users to applications.
upvoted 2 times

FourOfAKind 3 years, 3 months ago
Selected Answer: A
UDP == NLB
Must be hosted on-premises != CloudFront
upvoted 4 times

imvb88 3 years, 1 month ago
actually CloudFront's origin can be on-premises. Source:
https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/DownloadDistS3AndCustomOrigins.html#concept_CustomOrigin

"A custom origin is an HTTP server, for example, a web server. The HTTP server can be an Amazon EC2 instance or an HTTP server that you host somewhere else. "
upvoted 2 times

Question #368 Topic 1

A solutions architect wants all new users to have specific complexity requirements and mandatory rotation periods for IAM user passwords.

What should the solutions architect do to accomplish this?

A. Set an overall password policy for the entire AWS account. Most Voted

B. Set a password policy for each IAM user in the AWS account.

C. Use third-party vendor software to set password requirements.

D. Attach an Amazon CloudWatch rule to the Create_newuser event to set the password with the appropriate requirements.

Correct Answer: A

Community vote distribution

A (96%) 4%

Comments

lostmagnet001 Highly Voted 2 years, 4 months ago

Selected Answer: A

i get confused, the question saids "NEW" users... if you apply this password policy it would affect all the users in the AWS account....

upvoted 10 times

angel_marquina Highly Voted 2 years, 8 months ago

The question is for new users, answer A is not exact for that case.

upvoted 9 times

Rcosmos Most Recent 1 year, 4 months ago

Selected Answer: U

Por que as outras opções não são adequadas?
B. Definir uma política de senha para cada usuário do IAM:

O IAM não permite políticas de senha configuradas individualmente para usuários. As políticas de senha são aplicadas a nível da conta.

upvoted 1 times

LeonSauveterre 1 year, 6 months ago

Selected Answer: A

A: Overall password policy for the entire AWS account - affects new users immediately, but not old users immediately. Next time the old users change their passwords, the new rule will apply then.

B: Logically wrong. If you already have a user, then that user must've already have a password. So when you individually set a password policy for these users, they are only affected next time they change their passwords, just like I mentioned above about option A. Plus, this is really tiresome because this is user-wise, and there might be too many users, and the work you must do manually is increasing horribly.

upvoted 1 times

TariqKipkemei 2 years, 7 months ago

Selected Answer: A

You can set a custom password policy on your AWS account to specify complexity requirements and mandatory rotation periods for your IAM users' passwords. When you create or change a password policy, most of the password policy settings are enforced the next time your users change their passwords. However, some of the settings are enforced immediately.

https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_passwords_account-policy.html#:~:text=Setting%20an%20account-,password%20policy,-for%20IAM%20users

upvoted 4 times

klayytech 3 years, 2 months ago

Selected Answer: A

To accomplish this, the solutions architect should set an overall password policy for the entire AWS account. This policy will apply to all IAM users in the account, including new users.

upvoted 4 times

WherecanIstart 3 years, 2 months ago

Selected Answer: A

Set overall password policy ...

upvoted 3 times

kampatra 3 years, 2 months ago

Selected Answer: A

A is correct

upvoted 2 times

Question #369 Topic 1

A company has migrated an application to Amazon EC2 Linux instances. One of these EC2 instances runs several 1-hour tasks on a schedule. These tasks were written by different teams and have no common programming language. The company is concerned about performance and scalability while these tasks run on a single instance. A solutions architect needs to implement a solution to resolve these concerns.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use AWS Batch to run the tasks as jobs. Schedule the jobs by using Amazon EventBridge (Amazon CloudWatch Events). **Most Voted**

B. Convert the EC2 instance to a container. Use AWS App Runner to create the container on demand to run the tasks as jobs.

C. Copy the tasks into AWS Lambda functions. Schedule the Lambda functions by using Amazon EventBridge (Amazon CloudWatch Events).

D. Create an Amazon Machine Image (AMI) of the EC2 instance that runs the tasks. Create an Auto Scaling group with the AMI to run multiple copies of the instance.

Correct Answer: A

Community vote distribution

A (71%) C (17%) 10%

Comments

fkie4 Highly Voted 2 years, 9 months ago

Selected Answer: C

question said "These tasks were written by different teams and have no common programming language", and key word "scalable". Only Lambda can fulfil these. Lambda can be done in different programming languages, and it is scalable

upvoted 10 times

FourOfAKind 2 years, 9 months ago

But the question states "several 1-hour tasks on a schedule", and the maximum runtime for Lambda is 15 minutes, so it can't be A.

upvoted 32 times

FourOfAKind 2 years, 9 months ago

can't be C

upvoted 9 times

smgsi 2 years, 9 months ago

It's not because time limit of lambda is 15 minutes

upvoted 13 times

JTruong 1 year, 11 months ago

Lambda can only execute job under 15 mins* so C can't be the answer

upvoted 6 times

wsdasdasdqwdaw 2 years, 1 month ago

AWS Batch - As a fully managed service, AWS Batch helps you to run batch computing workloads of any scale. AWS Batch automatically provisions compute resources and optimizes the workload distribution based on the quantity and scale of the workloads. With AWS Batch, there's no need to install or manage batch computing software, so you can focus your time on analyzing results and solving problems.
https://docs.aws.amazon.com/batch/latest/userguide/what-is-batch.html ---> I am voting for A, C would have been OK if the time was within 15 minutes.

upvoted 9 times

pentium75 Highly Voted 1 year, 11 months ago

"Running on a schedule" = Batch
Not C due Lambda < 15 min

Not D, auto-scaling doesn't make sense for things running on a schedule
upvoted 9 times

Manimgh Most Recent 1 year, 1 month ago

Selected Answer: A

no common programming language = Batch
upvoted 2 times

foha2012 1 year, 10 months ago

Selected Answer: D

Answer = D
"performance and scalability while these tasks run on a single instance" They gave me a legacy application and want it to autoscale for performace. They dont want it to run on a single EC2 instance. Shouldn't I make an AMI and provision multiple EC2 instances in an autoscaling group ? I could put an ALB in front of it. I wont have to deal with "uncommon programming languages" inside the application... Just a thought..
upvoted 2 times

awsgeek75 1 year, 11 months ago

Selected Answer: A

AWS Batch is for jobs running at schedule on EC2. so option A
B is operational overhead
C Lambda is 15 mins max execution
D Scaling is not a requirement
upvoted 7 times

meowruki 2 years ago

Selected Answer: A

AWS Batch: AWS Batch is a fully managed service for running batch computing workloads. It dynamically provisions the optimal quantity and type of compute resources based on the volume and specific resource requirements of the batch jobs. It allows you to run tasks written in different programming languages with minimal operational overhead.
upvoted 4 times

hungta 2 years ago

Selected Answer: A

The tast working for hour but lambda function timeout is 15 minutes. So vote A.
upvoted 2 times

youdelin 2 years, 2 months ago

I know guys are stressed out trying to figure this exam out okay, but no matter what people say, with or without reasoning, at least put your mouth clean. Going like AAA is an issue, but talking shi* on him just because he didn't write down the reasoning is your fault.
upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: A

It can run heterogeneous workloads and tasks without needing to convert them to a common format.
AWS Batch manages the underlying compute resources - no need to manage containers, Lambda functions or Auto Scaling groups.
upvoted 6 times

zjcorpuz 2 years, 4 months ago

AWS Lambda function can only be run for 15 mins
upvoted 2 times

jaydesai8 2 years, 5 months ago

Selected Answer: A

maximum runtime for Lambda is 15 minutes, hence A
upvoted 3 times

antropaws 2 years, 6 months ago

Selected Answer: A

I also go with A.
upvoted 2 times

omoakin 2 years, 6 months ago

C. Copy the tasks into AWS Lambda functions. Schedule the Lambda functions by using Amazon EventBridge (Amazon CloudWatch Events)
upvoted 1 times

ruqui 2 years, 6 months ago

wrong, Lambda maximum runtime is 15 minutes and the tasks run for an hour
upvoted 3 times

KMohsoe 2 years, 6 months ago

Selected Answer: A

B and D out!
A and C let's think!
AWS Lambda functions are time limited.
So, Option A

upvoted 2 times

lucdt4 2 years, 6 months ago

AAAAAAAAAAAAAAAA
because lambda only run within 15 minutes

upvoted 3 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: A

Answer is A.
Could have been C but AWS Lambda functions can be only configured to run up to 15 minutes per execution. While the task in question need an 1hour to run,

upvoted 5 times

luisgu 2 years, 7 months ago

Selected Answer: D

question is asking for the LEAST operational overhead. With batch, you have to create the compute environment, create the job queue, create the jol definition and create the jobs --> more operational overhead than creating an ASG

upvoted 2 times

pentium75 1 year, 11 months ago

Things 'running on a schedule' = Batch, not autoscaling

upvoted 2 times

Question #370 Topic 1

A company runs a public three-tier web application in a VPC. The application runs on Amazon EC2 instances across multiple Availability Zones. The EC2 instances that run in private subnets need to communicate with a license server over the internet. The company needs a managed solution that minimizes operational maintenance.

Which solution meets these requirements?

A. Provision a NAT instance in a public subnet. Modify each private subnet's route table with a default route that points to the NAT instance.

B. Provision a NAT instance in a private subnet. Modify each private subnet's route table with a default route that points to the NAT instance.

C. Provision a NAT gateway in a public subnet. Modify each private subnet's route table with a default route that points to the NAT gateway. Most Voted

D. Provision a NAT gateway in a private subnet. Modify each private subnet's route table with a default route that points to the NAT gateway.

Correct Answer: C

Community vote distribution

C (100%)

Comments

UnluckyDucky Highly Voted 2 years, 9 months ago

Selected Answer: C

"The company needs a managed solution that minimizes operational maintenance"

Watch out for NAT instances vs NAT Gateways.

As the company needs a managed solution that minimizes operational maintenance - NAT Gateway is a public subnet is the answer.

upvoted 10 times

lucdt4 Highly Voted 2 years, 6 months ago

C
Nat gateway can't deploy in a private subnet.

upvoted 6 times

von_himmlen Most Recent 1 year, 7 months ago

C
https://docs.aws.amazon.com/appstream2/latest/developerguide/managing-network-internet-NAT-gateway.html
...and a NAT gateway in a public subnet.

upvoted 2 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: C

This meets the requirements for a managed, low maintenance solution for private subnets to access the internet:

NAT gateway provides automatic scaling, high availability, and fully managed service without admin overhead.
Placing the NAT gateway in a public subnet with proper routes allows private instances to use it for internet access.
Minimal operational maintenance compared to NAT instances.

upvoted 3 times

Guru4Cloud 2 years, 3 months ago

No good:
NAT instances (A, B) require more hands-on management.

Placing a NAT gateway in a private subnet (D) would not allow internet access.
upvoted 4 times

TariqKipkemei 2 years, 6 months ago

Selected Answer: C

minimizes operational maintenance = NGW
upvoted 3 times

WherecanIstart 2 years, 8 months ago

Selected Answer: C

C..provision NGW in Public Subnet
upvoted 3 times

cegama543 2 years, 9 months ago

Selected Answer: C

ccccc is the best
upvoted 2 times

Question #371 Topic 1

A company needs to create an Amazon Elastic Kubernetes Service (Amazon EKS) cluster to host a digital media streaming application. The EKS cluster will use a managed node group that is backed by Amazon Elastic Block Store (Amazon EBS) volumes for storage. The company must encrypt all data at rest by using a customer managed key that is stored in AWS Key Management Service (AWS KMS).

Which combination of actions will meet this requirement with the LEAST operational overhead? (Choose two.)

A. Use a Kubernetes plugin that uses the customer managed key to perform data encryption.

B. After creation of the EKS cluster, locate the EBS volumes. Enable encryption by using the customer managed key.

C. Enable EBS encryption by default in the AWS Region where the EKS cluster will be created. Select the customer managed key as the default key. Most Voted

D. Create the EKS cluster. Create an IAM role that has a policy that grants permission to the customer managed key. Associate the role with the EKS cluster. Most Voted

E. Store the customer managed key as a Kubernetes secret in the EKS cluster. Use the customer managed key to encrypt the EBS volumes.

Correct Answer: CD

Community vote distribution

CD (55%) BD (42%) 2%

Comments

asoli Highly Voted 3 years, 2 months ago

Selected Answer: CD

https://docs.aws.amazon.com/eks/latest/userguide/managed-node-groups.html#:~:text=encrypted%20Amazon%20EBS%20volumes%20without%20using%20a%20launch%20template%2C%20encrypt%20all%20new%20Amazon%20EBS%20volumes%20created%20in%20your%20account.

upvoted 17 times

bujuman 2 years, 2 months ago

If you want to encrypt Amazon EBS volumes for your nodes, you can deploy the nodes using a launch template. To deploy managed nodes with encrypted Amazon EBS volumes without using a launch template, encrypt all new Amazon EBS volumes created in your account. For more information, see Encryption by default in the Amazon EC2 User Guide for Linux Instances.

upvoted 3 times

imvb88 Highly Voted 3 years, 1 month ago

Selected Answer: BD

Quickly rule out A (which plugin? > overhead) and E because of bad practice

Among B,C,D: B and C are functionally similar > choice must be between B or C, D is fixed

Between B and C: C is out since it set default for all EBS volume in the region, which is more than required and even wrong, say what if other EBS volumes of other applications in the region have different requirement?

upvoted 13 times

NSA_Poker 2 years ago

(C) is correct; the EBS volumes of other applications in the region will not be affected bc an IAM role will limit the encryption key to the EKS cluster.

upvoted 2 times

JustEugen Most Recent 9 months ago

Selected Answer: CD

D is easy because you have to add permission to use the KMS key
C is better than anything else because it is recommended practice to secure your volumes

https://aws.amazon.com/blogs/compute/must-know-best-practices-for-amazon-ebs-encryption/?utm_source=chatgpt.com
upvoted 1 times

srcntpc 1 year ago

Selected Answer: CD

Why C is Correct:
Enabling EBS encryption by default in the Region ensures that all newly created EBS volumes are encrypted automatically.
You can specify a customer managed KMS key as the default encryption key.

Why D is Correct:
The EKS cluster (and its nodes) will need IAM permissions to use the customer managed KMS key for encryption/decryption.
Associating the correct IAM role with the EKS cluster ensures this access.
upvoted 2 times

Yak_Yeti 1 year, 1 month ago

Selected Answer: CD

D is clear and as to B vs C? The trade-off is the potential for encrypting other volumes in the Region, but the manual effort and complexity of Option B outweigh this concern when considering operational efficiency.
upvoted 1 times

babayomi 1 year, 7 months ago

CD
This options are wrongly stated in my understanding. B is wrong because you can not encrypt an unencrypted created ebs volume. You need to take a snapshot of the volume, encrypt the snapshot ,creat new ebs volume from snapshot. You can also enable encryption during creation. So the statement is wrong. The next available option can only be C, except that in that account all ebs volumes created would be encrypted, this is also questionable. Because if another person create a new ebs volumes it's automatically encrypted.
upvoted 2 times

scaredSquirrel 1 year, 9 months ago

Selected Answer: CD

A and E are obvious nos. D is a shoo-in.
The difference between B&C is basicually EBS encrption by default vs encrption. Encryption by default is by region, and encrypt everything in that region going forward, versus simple encryption is volume by volume, C is less operational overhead. Check doc & chatGPT.
upvoted 2 times

jjcode 2 years, 3 months ago

this one is going on my skip list
upvoted 7 times

Mahmouddddddddd 2 years, 2 months ago

Don't it came for me in my exam today xd
upvoted 4 times

JA2018 1 year, 6 months ago

Hi Mahmouddddddddd, can you share what were your chosen answers?
upvoted 1 times

jaswantn 2 years, 4 months ago

If question is giving a requirement related to a particular case and asking to encrypt all data at rest; it is clear that encryption is for this case only and not for other projects in entire region. so option B is more appropriate along with option D.
upvoted 2 times

frmrkc 2 years, 4 months ago

Selected Answer: CD

It says: 'The company must encrypt ALL data at rest', so there is nothing wrong with 'enabling EBS encryption by default' . C & D
upvoted 5 times

LeonSauveterre 1 year, 6 months ago

Exactly. Option B is out of the question. Not to mention option C barely has any operational overhead.
upvoted 1 times

upliftinghut 2 years, 5 months ago

Selected Answer: BD

B&D are correct. C is wrong because when you turn on encryption by defaul, AWS uses its own key while the requirement is using Customer key.

Detail is here: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/EBSEncryption.html#encryption-by-default
upvoted 3 times

pentium75 2 years, 5 months ago

Selected Answer: BD
Not A (avoid 3rd party plugins when there are native services)
Not C ("encryption by default" would impact other services)
Not E (Keys belong in KMS, not in EKS cluster)
upvoted 4 times

awsgeek75 2 years, 5 months ago
"The company must encrypt all data at rest by using a customer managed key that is stored in AWS Key Management Service (AWS KMS)."

I am just a bit concerned that the question does not put any limits on not encrypting all the EBS by default in the account. Both B and C can work. C is a hack but it is definitely LEAST operational overhead. Also, we don't know if there are other services or not that may be impacted. What do you think?
upvoted 2 times

Marco_St 2 years, 5 months ago
Selected Answer: CD
EBS encryption is set regionally. AWS account is global but it does not mean EBS encryption is enable by default at account level. default EBS encryption is a regional setting within your AWS account. Enabling it in a specific region ensures that all new EBS volumes created in that region are encrypted by default, using either the default AWS managed key or a customer managed key that you specify.
upvoted 2 times

pentium75 2 years, 5 months ago
"Enabling it in a specific region ensures that all new EBS volumes created in that region are encrypted by default" which is not what we want. We want to encrypt the EBS volumes used by this EKS cluster, NOT "all new EBS volumes created in that region."
upvoted 2 times

maudsha 2 years, 7 months ago
Selected Answer: CD
IF you need to encrypt an unencrypted volume,
• Create an EBS snapshot of the volume
• Encrypt the EBS snapshot (using copy)
• Create new EBS volume from the snapshot (the volume will also be encrypted)
so it has an operational overhead.

So assuming they won't use this account for anything else we can use C. Enable EBS encryption by default in the AWS Region where the EKS cluster will be created. Select the customer managed key as the default key.
upvoted 1 times

pentium75 2 years, 5 months ago
"Assuming they won't use this account for anything else" how could we assume that?
upvoted 2 times

TariqKipkemei 2 years, 7 months ago
Selected Answer: CD
Option D is required wither way.
Technically both option B and C would work, but with B you would have to enable encryption node by node, while with option C provides a onetime action of enabling encryption on all nodes.
The requirement is the option with LEAST operational overhead.
upvoted 3 times

pentium75 2 years, 5 months ago
B created some deployment work, but NOT "operational (!) overhead" once it's deployed. C enables encryption by default for all new EBS volumes which is not what we want.
upvoted 2 times

Guru4Cloud 2 years, 9 months ago
Selected Answer: CD
These options allow EBS encryption with the customer managed KMS key with minimal operational overhead:

C) Setting the KMS key as the regional EBS encryption default automatically encrypts new EKS node EBS volumes.

D) The IAM role grants the EKS nodes access to use the key for encryption/decryption operations.
upvoted 1 times

jaydesai8 2 years, 11 months ago
Selected Answer: CD
C - enable EBS encryption by default in a region -https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/EBSEncryption.html

D - Provides key access permission just to the EKS cluster without changing broader IAM permissions
upvoted 1 times

pentium75 2 years, 5 months ago

We're not asked to enable EBS encryption by default.

upvoted 2 times

Question #372 Topic 1

A company wants to migrate an Oracle database to AWS. The database consists of a single table that contains millions of geographic information systems (GIS) images that are high resolution and are identified by a geographic code.

When a natural disaster occurs, tens of thousands of images get updated every few minutes. Each geographic code has a single image or row that is associated with it. The company wants a solution that is highly available and scalable during such events.

Which solution meets these requirements MOST cost-effectively?

A. Store the images and geographic codes in a database table. Use Oracle running on an Amazon RDS Multi-AZ DB instance.

B. Store the images in Amazon S3 buckets. Use Amazon DynamoDB with the geographic code as the key and the image S3 URL as the value. Most Voted

C. Store the images and geographic codes in an Amazon DynamoDB table. Configure DynamoDB Accelerator (DAX) during times of high load.

D. Store the images in Amazon S3 buckets. Store geographic codes and image S3 URLs in a database table. Use Oracle running on an Amazon RDS Multi-AZ DB instance.

Correct Answer: B

Community vote distribution

B (71%) D (29%)

Comments

Wayne23Fang Highly Voted 2 years, 9 months ago

Selected Answer: B

Amazon prefers people to move from Oracle to its own services like DynamoDB and S3.

upvoted 17 times

Karlos99 Highly Voted 3 years, 3 months ago

Selected Answer: D

The company wants a solution that is highly available and scalable

upvoted 10 times

EllenLiu 1 year, 5 months ago

another thing is the design of dynamoDB, "Use Amazon DynamoDB with the geographic code as the key and the image S3 URL as the value." geographic code can not be the only key for any image

upvoted 2 times

[Removed] 3 years, 2 months ago

But DynamoDB is also highly available and scalable
https://aws.amazon.com/dynamodb/faqs/#:~:text=DynamoDB%20automatically%20scales%20throughput%20capacity,high%20availability%20and%20data%20durability.

upvoted 5 times

pbpally 3 years, 1 month ago

Yes but has a size limit at 400kb so theoretically it could store images but it's not a plausible solution.

upvoted 2 times

ruqui 3 years ago

It doesn't matter the size limit of DynamoDB!!!! The images are saved in S3 buckets. Right answer is B

upvoted 9 times

jaydesai8 2 years, 11 months ago

but would it be easy and cost-effective to migrate Oracle (relational db) to (Dynamodb)NoSQL?

upvoted 5 times

pentium75 2 years, 5 months ago
Yes because it's a single table with two records, for which Oracle or any relation database has been a bad choice in the first place.
upvoted 6 times

upliftinghut Most Recent 2 years, 4 months ago
Selected Answer: B
DynamoDB with its HA and built-in scalability. The nature of the table also resonates with NoSQL than SQL DB such as Oracle. Only 1 table so migration is just a script from Oracle to DynamoDB

D is workable but more expensive with Oracle licenses and other setups for HA and scalability
upvoted 3 times

upliftinghut 2 years, 4 months ago
HA & built-in scalability of Amazon DynamoDB :
https://aws.amazon.com/dynamodb/features/#:~:text=Amazon%20DynamoDB%20is%20a%20fully,for%20the%20most%20demanding%20applicat ons.
upvoted 3 times

awsgeek75 2 years, 5 months ago
Selected Answer: B
A puts images in Oracle, not a good idea
C DAX is not going to help with images
D It is doable but RDS on multi AZ does not give you more performance or write scalability. It gives more availability and read scalability which is not required here.
B works as Geographic code is the key in DynamoDB and S3 image URL is the data so DynamoDB can handle tens of thousands such record and S3 can scale for writing
upvoted 4 times

pentium75 2 years, 5 months ago
Selected Answer: B
They are currently using Oracle, but only for one simple table with a single key-value pair. This is a typical use case for a NoSQL database like DynamoDB (and whoever decided to use Oracle for this in the first place should be fired). Oracle is expensive as hell, so options A and D might work but are surely not cost-effective. C won't work because the images are too big for the database. Leaves B which would be the ideal solution and mee the availability and scalability requirements.
upvoted 7 times

wsdasdasdqwdaw 2 years, 7 months ago
For D - Oracle is not cheap as well. RDS with Oracle vs DynamoDB, I would go for pure AWS provided option. In each exam there is a lot of marketing
=> B
upvoted 3 times

jubolano 2 years, 7 months ago
Selected Answer: D
Cost effective, D
upvoted 2 times

awsgeek75 2 years, 4 months ago
How is Oracle more cost effective than other options?
upvoted 2 times

wsdasdasdqwdaw 2 years, 7 months ago
B or D, but the question is MOST cost-effectively DynamoDB is more expensive than RDS, I am going for D
upvoted 3 times

JA2018 1 year, 6 months ago
Key in sTEM states that the customer's current Oracle DB table is setup with a single key-value pair.

A typical use case for a NoSQL database like DynamoDB.
upvoted 1 times

gouranga45 2 years, 8 months ago
Selected Answer: B
Answer is B, DynamoDB is Highly available and scalable
upvoted 2 times

baba365 2 years, 8 months ago

A single table in a relational db can have items that are related ? e.g. 'select * from Faculty where department_id in (10, 20) and dept_name = AWS'.
In the sql query example above, * means all and Faculty is name of the table.

upvoted 2 times

Eminenza22 2 years, 9 months ago

Selected Answer: B

B option offers a cost-effective solution for storing and accessing high-resolution GIS images during natural disasters. Storing the images in Amazon S3 buckets provides scalable and durable storage, while using Amazon DynamoDB allows for quick and efficient retrieval of images based on geographic codes. This solution leverages the strengths of both S3 and DynamoDB to meet the requirements of high availability, scalability, and cost-effectiveness.

upvoted 2 times

cd93 2 years, 9 months ago

Selected Answer: B

What were the company thinking using the most expensive DB on the planet FOR ONE SINGLE TABLE???
Migrate a single table from SQL to NoSQL should be easy enough I guess...

upvoted 3 times

vini15 2 years, 10 months ago

Should be D.
the question says company wants to migrate oracle to AWS. Oracle is a relational db hence RDS makes more sense whereas Dynamodb is non relational db.

upvoted 3 times

pentium75 2 years, 5 months ago

But relational DB does not make sense for the use case. It's a single table.

upvoted 2 times

iBanan 2 years, 10 months ago

I hate these questions:) I can't choose between B and D

upvoted 6 times

ces_9999 2 years, 11 months ago

Guys the answer is B the oracle database only has one table without any relationships so why we should use a relational database in the first place, second we are storing the images in S3 not in the database why not use this alongside dynamo

upvoted 6 times

Kp88 2 years, 10 months ago

You can't do migration of Oracle to Dynmodb without SCT. I am not the DB guy but since its saying oracle I would go with D otherwise B makes more sense if a company is starting out from scratch.

upvoted 2 times

Kp88 2 years, 10 months ago

Actually now that I think about it , B sounds ok as well. Company just need to use SCT and that would be more cost effective.

upvoted 2 times

joehong 2 years, 11 months ago

Selected Answer: D

"A company wants to migrate an Oracle database to AWS"

upvoted 2 times

pentium75 2 years, 5 months ago

Yeah, per my understanding that doesn't implicate that the destination must be an Oracle database.

upvoted 2 times

secdgs 2 years, 12 months ago

D: Wrorng
if you caluate License Oracle Database, It is not cost-effectively. Multi-AZ is not scalable and if you set scalable, you need more license for Oracle database.

upvoted 2 times

Question #373 Topic 1

A company has an application that collects data from IoT sensors on automobiles. The data is streamed and stored in Amazon S3 through Amazon Kinesis Data Firehose. The data produces trillions of S3 objects each year. Each morning, the company uses the data from the previous 30 days to retrain a suite of machine learning (ML) models.

Four times each year, the company uses the data from the previous 12 months to perform analysis and train other ML models. The data must be available with minimal delay for up to 1 year. After 1 year, the data must be retained for archival purposes.

Which storage solution meets these requirements MOST cost-effectively?

A. Use the S3 Intelligent-Tiering storage class. Create an S3 Lifecycle policy to transition objects to S3 Glacier Deep Archive after 1 year.

B. Use the S3 Intelligent-Tiering storage class. Configure S3 Intelligent-Tiering to automatically move objects to S3 Glacier Deep Archive after 1 year.

C. Use the S3 Standard-Infrequent Access (S3 Standard-IA) storage class. Create an S3 Lifecycle policy to transition objects to S3 Glacier Deep Archive after 1 year.

D. Use the S3 Standard storage class. Create an S3 Lifecycle policy to transition objects to S3 Standard-Infrequent Access (S3 Standard-IA) after 30 days, and then to S3 Glacier Deep Archive after 1 year. Most Voted

Correct Answer: D

Community vote distribution

D (85%) Other

Comments

UnluckyDucky Highly Voted 2 years, 9 months ago

Selected Answer: D

Access patterns is given, therefore D is the most logical answer.

Intelligent tiering is for random, unpredictable access.

upvoted 14 times

ealpuche 2 years, 7 months ago

You are missing: <<The data must be available with minimal delay for up to 1 year. After one year, the data must be retained for archival purposes.>> You are secure that data after 1 year is not accessible anymore.

upvoted 2 times

TariqKipkemei Highly Voted 2 years, 6 months ago

Selected Answer: D

First 30 days data accessed every morning = S3 Standard
Beyond 30 days data accessed quarterly = S3 Standard-Infrequent Access
Beyond 1 year data retained = S3 Glacier Deep Archive

upvoted 5 times

Clouddon Most Recent 11 months ago

Selected Answer: A

A. Use the S3 Intelligent-Tiering storage class. Create an S3 Lifecycle policy to transition objects to S3 Glacier Deep Archive after 1 year.

Here's why this solution is the most suitable:

S3 Intelligent-Tiering: This storage class automatically moves data between two access tiers (frequent and infrequent) based on changing access patterns, without performance impact or operational overhead. It is cost-effective for data with unknown or changing access patterns.
Lifecycle Policy: By creating an S3 Lifecycle policy to transition objects to S3 Glacier Deep Archive after 1 year, you can significantly reduce storage costs for long-term archival.

upvoted 1 times

jjcode 1 year, 10 months ago

I dont get how its A

1. Each morning, the company uses the data from the previous 30 days
2. Four times each year, the company uses the data from the previous 12 months to perform analysis and train other ML models
3. The data must be available with minimal delay for up to 1 year. After 1 year, the data must be retained for archival purposes

The data ingestion happens 4 times a year, that means that after the initial 30 days it still needs to be pulled 3 more times, why would you put the data in standard infrequent if you were going to use it 3 more times and speed is a requirement? Makes more sense to put it in S3 standard, or intelligent then straight to glacier.

upvoted 2 times

upliftinghut 1 year, 10 months ago

Selected Answer: D

Clear access pattern. data in Standard-Infrequent Access is for data requires rapid access when needed

upvoted 2 times

awsgeek75 1 year, 11 months ago

Selected Answer: D

A and B, Intelligent Tiering cannot be configured. It is managed by AWS.
C SIA does not allow immediate access for "each morning"
D is best for 30 day standard access, SIA after 30 days and archive after 1 year

upvoted 4 times

pentium75 1 year, 11 months ago

Selected Answer: D

See reasoning below, just accidentally voted A

upvoted 2 times

pentium75 1 year, 11 months ago

Selected Answer: A

The data is used every day (typical use case for Standard) for 30 days, for the remaining 12 months it is used 3 or 4 times (typical use case for IA), after 12 months it is not used at all but must be kept (typical use case for Glacier Deep Archive).

upvoted 2 times

pentium75 1 year, 11 months ago

Sorry, D!!!!!!!!! Not A!!!! D!

upvoted 4 times

Guru4Cloud 2 years, 3 months ago

Selected Answer: D

This option optimizes costs while meeting the data access requirements:

Store new data in S3 Standard for first 30 days of frequent access
Transition to S3 Standard-IA after 30 days for infrequent access up to 1 year
Archive to Glacier Deep Archive after 1 year for long-term archival

upvoted 3 times

ealpuche 2 years, 7 months ago

Selected Answer: A

Option A meets the requirements most cost-effectively. The S3 Intelligent-Tiering storage class provides automatic tiering of objects between the S3 Standard and S3 Standard-Infrequent Access (S3 Standard-IA) tiers based on changing access patterns, which helps optimize costs. The S3 Lifecycle policy can be used to transition objects to S3 Glacier Deep Archive after 1 year for archival purposes. This solution also meets the requirement for minimal delay in accessing data for up to 1 year. Option B is not cost-effective because it does not include the transition of data to S3 Glacier Deep Archive after 1 year. Option C is not the best solution because S3 Standard-IA is not designed for long-term archival purposes and incurs higher storage costs. Option D is also not the most cost-effective solution as it transitions objects to the S3 Standard-IA tier after 30 days, which is unnecessary for the requirement to retrain the suite of ML models each morning using data from the previous 30 days.

upvoted 2 times

pentium75 1 year, 11 months ago

I can't follow. The data is used every day (typical use case for Standard) for 30 days, for the remaining 12 months it is used 3 or 4 times (typical use case for IA), after 12 months it is not used at all but must be kept (typical use case for Glacier Deep Archive).

upvoted 3 times

KAUS2 2 years, 9 months ago

Selected Answer: D

Agree with UnluckyDucky , the correct option is D

upvoted 2 times

fkie4 2 years, 9 months ago

Selected Answer: D

Should be D. see this:
https://www.examtopics.com/discussions/amazon/view/68947-exam-aws-certified-solutions-architect-associate-saa-c02/

upvoted 3 times

Nithin1119 2 years, 9 months ago

Selected Answer: B

Bbbbbbbbb

upvoted 1 times

fkie4 2 years, 9 months ago

hello!!??

upvoted 2 times

Question #374 Topic 1

A company is running several business applications in three separate VPCs within the us-east-1 Region. The applications must be able to communicate between VPCs. The applications also must be able to consistently send hundreds of gigabytes of data each day to a latency-sensitive application that runs in a single on-premises data center.

A solutions architect needs to design a network connectivity solution that maximizes cost-effectiveness.

Which solution meets these requirements?

A. Configure three AWS Site-to-Site VPN connections from the data center to AWS. Establish connectivity by configuring one VPN connection for each VPC.

B. Launch a third-party virtual network appliance in each VPC. Establish an IPsec VPN tunnel between the data center and each virtual appliance.

C. Set up three AWS Direct Connect connections from the data center to a Direct Connect gateway in us-east-1. Establish connectivity by configuring each VPC to use one of the Direct Connect connections.

D. Set up one AWS Direct Connect connection from the data center to AWS. Create a transit gateway, and attach each VPC to the transit gateway. Establish connectivity between the Direct Connect connection and the transit gateway. Most Voted

Correct Answer: D

Community vote distribution

D (100%)

Comments

TariqKipkemei Highly Voted 2 years, 1 month ago

Selected Answer: D

AWS Transit Gateway connects your Amazon Virtual Private Clouds (VPCs) and on-premises networks through a central hub. This connection simplifies your network and puts an end to complex peering relationships. Transit Gateway acts as a highly scalable cloud router—each new connection is made only once.

https://aws.amazon.com/transit-gateway/#:~:text=AWS-,Transit%20Gateway,-connects%20your%20Amazon

upvoted 6 times

Guru4Cloud Highly Voted 2 years, 3 months ago

Selected Answer: D

This option leverages a single Direct Connect for consistent, private connectivity between the data center and AWS. The transit gateway allows each VPC to share the Direct Connect while keeping the VPCs isolated. This provides a cost-effective architecture to meet the requirements.

upvoted 5 times

upliftinghut Most Recent 1 year, 10 months ago

Selected Answer: D

AWS Direct connect is costly but the saving comes from less data transfer cost with Direct Connect and Transit gateway

upvoted 4 times

alexandercamachop 2 years, 6 months ago

Selected Answer: D

Transit GW, is a hub for connecting all VPCs.
Direct Connect is expensive, therefor only 1 of them connected to the Transit GW (Hub for all our VPCs that we connect to it)

upvoted 4 times

KMohsoe 2 years, 6 months ago

Selected Answer: D

Option D

upvoted 4 times

Sivasaa 2 years, 7 months ago
Can someone tell why option C will not work here
upvoted 3 times

Guru4Cloud 2 years, 3 months ago
Using multiple Site-to-Site VPNs (A) or Direct Connects (C) incurs higher costs without providing significant benefits.
upvoted 2 times

jdamian 2 years, 7 months ago
cost-effectiveness, 3 DC are more than 1 (more expensive). There is no need to connect more than 1 DC.
upvoted 2 times

pentium75 1 year, 11 months ago
And besides the cost, C does not allow the applications "to communicate between VPCs".
upvoted 2 times

SkyZeroZx 2 years, 7 months ago
Selected Answer: D
cost-effectiveness
D
upvoted 2 times

WherecanIstart 2 years, 8 months ago
Selected Answer: D
Transit Gateway will achieve this result..
upvoted 4 times

Karlos99 2 years, 9 months ago
Selected Answer: D
maximizes cost-effectiveness
upvoted 3 times

Question #375 Topic 1

An ecommerce company is building a distributed application that involves several serverless functions and AWS services to complete order-processing tasks. These tasks require manual approvals as part of the workflow. A solutions architect needs to design an architecture for the order-processing application. The solution must be able to combine multiple AWS Lambda functions into responsive serverless applications. The solution also must orchestrate data and services that run on Amazon EC2 instances, containers, or on-premises servers.

Which solution will meet these requirements with the LEAST operational overhead?

A. Use AWS Step Functions to build the application. **Most Voted**

B. Integrate all the application components in an AWS Glue job.

C. Use Amazon Simple Queue Service (Amazon SQS) to build the application.

D. Use AWS Lambda functions and Amazon EventBridge events to build the application.

Correct Answer: A

Community vote distribution

A (100%)

Comments

kinglong12 Highly Voted 2 years, 2 months ago

Selected Answer: A

AWS Step Functions is a fully managed service that makes it easy to build applications by coordinating the components of distributed applications and microservices using visual workflows. With Step Functions, you can combine multiple AWS Lambda functions into responsive serverless applications and orchestrate data and services that run on Amazon EC2 instances, containers, or on-premises servers. Step Functions also allows for manual approvals as part of the workflow. This solution meets all the requirements with the least operational overhead.

upvoted 15 times

Guru4Cloud Highly Voted 1 year, 9 months ago

Selected Answer: A

AWS Step Functions allow you to easily coordinate multiple Lambda functions and services into serverless workflows with visual workflows. Step Functions are designed for building distributed applications that combine services and require human approval steps.

Using Step Functions provides a fully managed orchestration service with minimal operational overhead.

upvoted 7 times

TariqKipkemei Most Recent 1 year, 7 months ago

Selected Answer: A

involves several serverless functions and AWS services, require manual approvals as part of the workflow, combine the Lambda functions into responsive serverless applications, orchestrate data and services = AWS Step Functions

upvoted 5 times

capino 1 year, 10 months ago

Selected Answer: A

Serverless && workflow service that need human approval::::step functions

upvoted 4 times

BeeKayEnn 2 years, 2 months ago

Key: Distributed Application Processing, Microservices orchestration (Orchestrate Data and Services)

A would be the best fit.

AWS Step Functions is a visual workflow service that helps developers use AWS services to build distributed applications, automate processes, orchestrate microservices, and create data and machine learning (ML) pipelines.

Reference: https://aws.amazon.com/step-functions/#:~:text=AWS%20Step%20Functions%20is%20a,machine%20learning%20(ML)%20pipelines.

upvoted 5 times

COTIT 2 years, 2 months ago

Selected Answer: A

Approval is explicit for the solution. -> "A common use case for AWS Step Functions is a task that requires human intervention (for example, an approval process). Step Functions makes it easy to coordinate the components of distributed applications as a series of steps in a visual workflow called a state machine. You can quickly build and run state machines to execute the steps of your application in a reliable and scalable fashion. (https://aws.amazon.com/pt/blogs/compute/implementing-serverless-manual-approval-steps-in-aws-step-functions-and-amazon-api-gateway/)"

upvoted 6 times

ktulu2602 2 years, 3 months ago

Selected Answer: A

Option A: Use AWS Step Functions to build the application.
AWS Step Functions is a serverless workflow service that makes it easy to coordinate distributed applications and microservices using visual workflows. It is an ideal solution for designing architectures for distributed applications that involve multiple AWS services and serverless functions, as it allows us to orchestrate the flow of our application components using visual workflows. AWS Step Functions also integrates with other AWS services like AWS Lambda, Amazon EC2, and Amazon ECS, and it has built-in error handling and retry mechanisms. This option provides a serverless solution with the least operational overhead for building the application.

upvoted 5 times

Question #376 Topic 1

A company has launched an Amazon RDS for MySQL DB instance. Most of the connections to the database come from serverless applications. Application traffic to the database changes significantly at random intervals. At times of high demand, users report that their applications experience database connection rejection errors.

Which solution will resolve this issue with the LEAST operational overhead?

A. Create a proxy in RDS Proxy. Configure the users' applications to use the DB instance through RDS Proxy. **Most Voted**

B. Deploy Amazon ElastiCache for Memcached between the users' applications and the DB instance.

C. Migrate the DB instance to a different instance class that has higher I/O capacity. Configure the users' applications to use the new DB instance.

D. Configure Multi-AZ for the DB instance. Configure the users' applications to switch between the DB instances.

Correct Answer: A

Community vote distribution

A (100%)

Comments

TariqKipkemei Highly Voted 1 year, 7 months ago

Selected Answer: A

database connection rejection errors = RDS Proxy

upvoted 7 times

ktulu2602 Highly Voted 2 years, 3 months ago

Selected Answer: A

The correct solution for this scenario would be to create a proxy in RDS Proxy. RDS Proxy allows for managing thousands of concurrent database connections, which can help reduce connection errors. RDS Proxy also provides features such as connection pooling, read/write splitting, and retries. This solution requires the least operational overhead as it does not involve migrating to a different instance class or setting up a new cache layer. Therefore, option A is the correct answer.

upvoted 5 times

Guru4Cloud Most Recent 1 year, 9 months ago

Selected Answer: A

RDS Proxy provides a proxy layer that pools and shares database connections to improve scalability. This allows the proxy to handle connection spikes to the database gracefully.

Using RDS Proxy requires minimal operational overhead - just create the proxy and reconfigure applications to use it. No code changes needed.

upvoted 4 times

antropaws 2 years ago

Wait, why not B?????

upvoted 2 times

Guru4Cloud 1 year, 9 months ago

ElastiCache (B) and larger instance type (C) help performance but don't resolve connection issues.

upvoted 4 times

live_reply_developers 1 year, 11 months ago

Amazon ElastiCache tends to have a lower operational overhead compared to Amazon RDS Proxy. BUT we already have " Amazon RDS for MySQL DB instance"

upvoted 2 times

Guru4Cloud 1 year, 9 months ago

ElastiCache (B) and larger instance type (C) help performance but don't resolve connection issues.

upvoted 2 times

roxx529 2 years ago

To reduce application failures resulting from database connection timeouts, the best solution is to enable RDS Proxy on the RDS DB instances

upvoted 2 times

COTIT 2 years, 2 months ago

Selected Answer: A

Many applications, including those built on modern serverless architectures, can have a large number of open connections to the database server and may open and close database connections at a high rate, exhausting database memory and compute resources. Amazon RDS Proxy allows applications to pool and share connections established with the database, improving database efficiency and application scalability. (https://aws.amazon.com/pt/rds/proxy/)

upvoted 4 times

Question #377 Topic 1

A company recently deployed a new auditing system to centralize information about operating system versions, patching, and installed software for Amazon EC2 instances. A solutions architect must ensure all instances provisioned through EC2 Auto Scaling groups successfully send reports to the auditing system as soon as they are launched and terminated.

Which solution achieves these goals MOST efficiently?

A. Use a scheduled AWS Lambda function and run a script remotely on all EC2 instances to send data to the audit system.

B. Use EC2 Auto Scaling lifecycle hooks to run a custom script to send data to the audit system when instances are launched and terminated. **Most Voted**

C. Use an EC2 Auto Scaling launch configuration to run a custom script through user data to send data to the audit system when instances are launched and terminated.

D. Run a custom script on the instance operating system to send data to the audit system. Configure the script to be invoked by the EC2 Auto Scaling group when the instance starts and is terminated.

Correct Answer: B

Community vote distribution

B (100%)

Comments

ktulu2602 Highly Voted 3 years, 3 months ago

Selected Answer: B

The most efficient solution for this scenario is to use EC2 Auto Scaling lifecycle hooks to run a custom script to send data to the audit system when instances are launched and terminated. The lifecycle hook can be used to delay instance termination until the script has completed, ensuring that all data is sent to the audit system before the instance is terminated. This solution is more efficient than using a scheduled AWS Lambda function, which would require running the function periodically and may not capture all instances launched and terminated within the interval. Running a custom script through user data is also not an optimal solution, as it may not guarantee that all instances send data to the audit system. Therefore, option B is the correct answer.

upvoted 12 times

COTIT Highly Voted 3 years, 2 months ago

Selected Answer: B

Amazon EC2 Auto Scaling offers the ability to add lifecycle hooks to your Auto Scaling groups. These hooks let you create solutions that are aware of events in the Auto Scaling instance lifecycle, and then perform a custom action on instances when the corresponding lifecycle event occurs. (https://docs.aws.amazon.com/autoscaling/ec2/userguide/lifecycle-hooks.html)

upvoted 6 times

LeonSauveterre Most Recent 1 year, 6 months ago

Selected Answer: B

C: a **launch** configuration CANNOT handle instance termination automatically.

upvoted 1 times

TariqKipkemei 2 years, 7 months ago

Selected Answer: B

Use EC2 Auto Scaling lifecycle hooks to run a custom script to send data to the audit system when instances are launched and terminated

upvoted 2 times

Guru4Cloud 2 years, 9 months ago

Selected Answer: B

EC2 Auto Scaling lifecycle hooks allow you to perform custom actions as instances launch and terminate. This is the most efficient way to trigger the auditing script execution at instance launch and termination.

upvoted 5 times

WherecanIstart 3 years, 2 months ago

Selected Answer: B

https://docs.aws.amazon.com/autoscaling/ec2/userguide/lifecycle-hooks.html

upvoted 4 times

fkie4 3 years, 3 months ago

it is B. read this:
https://docs.aws.amazon.com/autoscaling/ec2/userguide/lifecycle-hooks.html

upvoted 3 times

Question #378 *Topic 1*

A company is developing a real-time multiplayer game that uses UDP for communications between the client and servers in an Auto Scaling group. Spikes in demand are anticipated during the day, so the game server platform must adapt accordingly. Developers want to store gamer scores and other non-relational data in a database solution that will scale without intervention.

Which solution should a solutions architect recommend?

A. Use Amazon Route 53 for traffic distribution and Amazon Aurora Serverless for data storage.

B. Use a Network Load Balancer for traffic distribution and Amazon DynamoDB on-demand for data storage. **Most Voted**

C. Use a Network Load Balancer for traffic distribution and Amazon Aurora Global Database for data storage.

D. Use an Application Load Balancer for traffic distribution and Amazon DynamoDB global tables for data storage.

Correct Answer: B

Community vote distribution

B (100%)

Comments

TariqKipkemei Highly Voted 2 years ago

Selected Answer: B

UDP = NLB
Non-relational data = Dynamo DB

upvoted 17 times

fruto123 Highly Voted 2 years, 3 months ago

Selected Answer: B

key words - UDP, non-relational data
answers - NLB for UDP application, DynamoDB for non-relational data

upvoted 6 times

8935983 Most Recent 8 months, 3 weeks ago

Selected Answer: B

the question talk about real time which mainly has something to do with the internet or streaming so NLB will work better as compared to ALB

upvoted 1 times

Manimgh 1 year, 1 month ago

Selected Answer: B

non-relational data = DynamoDB

upvoted 1 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: B

This option provides the most scalable and optimized architecture for the real-time multiplayer game:

Network Load Balancer efficiently distributes UDP gaming traffic to the Auto Scaling group of game servers.
DynamoDB On-Demand mode provides auto-scaling non-relational data storage for gamer scores and other game data. DynamoDB is optimized for fast, high-scale access patterns seen in gaming.
Together, the Network Load Balancer and DynamoDB On-Demand provide an architecture that can smoothly scale up and down to match spikes in gaming demand.

upvoted 5 times

elearningtakai 2 years, 2 months ago

Selected Answer: B

Option B is a good fit because a Network Load Balancer can handle UDP traffic, and Amazon DynamoDB on-demand can provide automatic scaling without intervention

upvoted 3 times

KAUS2 2 years, 3 months ago

Selected Answer: B

Correct option is "B"

upvoted 2 times

aragon_saa 2 years, 3 months ago

B

https://www.examtopics.com/discussions/amazon/view/29756-exam-aws-certified-solutions-architect-associate-saa-c02/

upvoted 2 times

Kenp1192 2 years, 3 months ago

B

Because NLB can handle UDP and DynamoDB is Non-Relational

upvoted 3 times

Question #379 Topic 1

A company hosts a frontend application that uses an Amazon API Gateway API backend that is integrated with AWS Lambda. When the API receives requests, the Lambda function loads many libraries. Then the Lambda function connects to an Amazon RDS database, processes the data, and returns the data to the frontend application. The company wants to ensure that response latency is as low as possible for all its users with the fewest number of changes to the company's operations.

Which solution will meet these requirements?

A. Establish a connection between the frontend application and the database to make queries faster by bypassing the API.

B. Configure provisioned concurrency for the Lambda function that handles the requests. Most Voted

C. Cache the results of the queries in Amazon S3 for faster retrieval of similar datasets.

D. Increase the size of the database to increase the number of connections Lambda can establish at one time.

Correct Answer: B

Community vote distribution

B (100%)

Comments

UnluckyDucky Highly Voted 2 years, 3 months ago

Selected Answer: B

Key: the Lambda function loads many libraries

Configuring provisioned concurrency would get rid of the "cold start" of the function therefore speeding up the proccess.

upvoted 17 times

kampatra Highly Voted 2 years, 2 months ago

Selected Answer: B

Provisioned concurrency – Provisioned concurrency initializes a requested number of execution environments so that they are prepared to respond immediately to your function's invocations. Note that configuring provisioned concurrency incurs charges to your AWS account.

upvoted 11 times

TariqKipkemei Most Recent 1 year, 7 months ago

Selected Answer: B

Provisioned concurrency pre-initializes execution environments which are prepared to respond immediately to incoming function requests.

upvoted 7 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: B

Provisioned concurrency ensures a configured number of execution environments are ready to serve requests to the Lambda function. This avoids cold starts where the function would otherwise need to load all the libraries on each invocation.

upvoted 4 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: B

Provisioned concurrency ensures a configured number of execution environments are ready to serve requests to the Lambda function. This avoids cold starts where the function would otherwise need to load all the libraries on each invocation.

upvoted 2 times

elearningtakai 2 years, 2 months ago

Selected Answer: B

Answer B is correct
https://docs.aws.amazon.com/lambda/latest/dg/provisioned-concurrency.html
Answer C: need to modify the application

upvoted 5 times

elearningtakai 2 years, 2 months ago

This is relevant to "cold start" with keywords: "Lambda function loads many libraries"

upvoted 2 times

Karlos99 2 years, 3 months ago

Selected Answer: B

https://docs.aws.amazon.com/lambda/latest/dg/provisioned-concurrency.html

upvoted 4 times

Question #380 Topic 1

A company is migrating its on-premises workload to the AWS Cloud. The company already uses several Amazon EC2 instances and Amazon RDS DB instances. The company wants a solution that automatically starts and stops the EC2 instances and DB instances outside of business hours. The solution must minimize cost and infrastructure maintenance.

Which solution will meet these requirements?

A. Scale the EC2 instances by using elastic resize. Scale the DB instances to zero outside of business hours.

B. Explore AWS Marketplace for partner solutions that will automatically start and stop the EC2 instances and DB instances on a schedule.

C. Launch another EC2 instance. Configure a crontab schedule to run shell scripts that will start and stop the existing EC2 instances and DB instances on a schedule.

D. Create an AWS Lambda function that will start and stop the EC2 instances and DB instances. Configure Amazon EventBridge to invoke the Lambda function on a schedule. **Most Voted**

Correct Answer: D

Community vote distribution

D (100%)

Comments

ktulu2602 Highly Voted 3 years, 3 months ago

Selected Answer: D

The most efficient solution for automatically starting and stopping EC2 instances and DB instances on a schedule while minimizing cost and infrastructure maintenance is to create an AWS Lambda function and configure Amazon EventBridge to invoke the function on a schedule.

Option A, scaling EC2 instances by using elastic resize and scaling DB instances to zero outside of business hours, is not feasible as DB instances cannot be scaled to zero.

Option B, exploring AWS Marketplace for partner solutions, may be an option, but it may not be the most efficient solution and could potentially adc additional costs.

Option C, launching another EC2 instance and configuring a crontab schedule to run shell scripts that will start and stop the existing EC2 instances and DB instances on a schedule, adds unnecessary infrastructure and maintenance.

upvoted 17 times

Guru4Cloud Highly Voted 2 years, 9 months ago

Selected Answer: D

This option leverages AWS Lambda and EventBridge to automatically schedule the starting and stopping of resources.

Lambda provides the script/code to stop/start instances without managing servers.
EventBridge triggers the Lambda on a schedule without cronjobs.
No additional code or third party tools needed.
Serverless, maintenance-free solution

upvoted 6 times

1e22522 Most Recent 1 year, 10 months ago

Selected Answer: D

its d but nowadays u use system manager me thinks

upvoted 3 times

TariqKipkemei 2 years, 7 months ago

Selected Answer: D

Create an AWS Lambda function that will start and stop the EC2 instances and DB instances. Configure Amazon EventBridge to invoke the Lambda function on a schedule.

upvoted 4 times

WherecanIstart 3 years, 2 months ago

Selected Answer: D

Minimize cost and maintenance...

upvoted 3 times

Question #381 Topic 1

A company hosts a three-tier web application that includes a PostgreSQL database. The database stores the metadata from documents. The company searches the metadata for key terms to retrieve documents that the company reviews in a report each month. The documents are stored in Amazon S3. The documents are usually written only once, but they are updated frequently.

The reporting process takes a few hours with the use of relational queries. The reporting process must not prevent any document modifications or the addition of new documents. A solutions architect needs to implement a solution to speed up the reporting process.

Which solution will meet these requirements with the LEAST amount of change to the application code?

A. Set up a new Amazon DocumentDB (with MongoDB compatibility) cluster that includes a read replica. Scale the read replica to generate the reports.

B. Set up a new Amazon Aurora PostgreSQL DB cluster that includes an Aurora Replica. Issue queries to the Aurora Replica to generate the reports. **Most Voted**

C. Set up a new Amazon RDS for PostgreSQL Multi-AZ DB instance. Configure the reporting module to query the secondary RDS node so that the reporting module does not affect the primary node.

D. Set up a new Amazon DynamoDB table to store the documents. Use a fixed write capacity to support new document entries. Automatically scale the read capacity to support the reports.

Correct Answer: B

Community vote distribution

B (96%) 2%

Comments

Guru4Cloud Highly Voted 2 years, 8 months ago

Selected Answer: B

The key reasons are:

Aurora PostgreSQL provides native PostgreSQL compatibility, so minimal code changes would be required.
Using an Aurora Replica separates the reporting workload from the main workload, preventing any slowdown of document updates/inserts.
Aurora can auto-scale read replicas to handle the reporting load.
This allows leveraging the existing PostgreSQL database without major changes. DynamoDB would require more significant rewrite of data access code.
RDS Multi-AZ alone would not fully separate the workloads, as the secondary is for HA/failover more than scaling read workloads.

upvoted 13 times

TariqKipkemei Highly Voted 3 years ago

Selected Answer: B

Load balancing = Read replica
High availability = Multi AZ

upvoted 8 times

BillaRanga 2 years, 4 months ago

No Modifications allowerd = Read Replica

upvoted 4 times

JosephKY Most Recent 7 months, 1 week ago

Selected Answer: B

This is a perfect example of the kind of ambiguous requirement that an actual person in an actual job would push back on. What exactly do you wan the report to return? Are you wanting a report of meta data, or the actual files themselves? Are you upset that it's currently a relational query?

upvoted 1 times

terminator69 1 year, 9 months ago
How in the bloody hell it's D?????
upvoted 1 times

TruthWS 2 years, 2 months ago
B is correct
upvoted 2 times

ExamGuru727 2 years, 2 months ago
Selected Answer: B
We also have a requirement for the Least amount of change to the code.
Since our DB is PostgreSQL, A & D are immediately out.
Multi-AZ won't help with offloading read requests, hence the answer is B ;)
upvoted 5 times

Buck12345 2 years, 3 months ago
It is B
upvoted 2 times

Cyberkayu 2 years, 5 months ago
Selected Answer: C
D. Reporting process Must not prevent = allow modification and addition of new document.

all read replica were wrong.
upvoted 1 times

pentium75 2 years, 5 months ago
How would 'issuing queries to the read replica' prevent modifications or updates?
upvoted 2 times

KMohsoe 3 years ago
Selected Answer: A
Why not A? :(
upvoted 1 times

wRhlH 2 years, 11 months ago
"The reporting process takes a few hours with the use of RELATIONAL queries."
upvoted 4 times

Murtadhaceit 2 years, 6 months ago
DocumentDB (For MongoDB) is no SQL. DynamoDB is also No SQL. Therefore, options A and D are out.
upvoted 5 times

lexotan 3 years, 1 month ago
Selected Answer: B
B is the right one. why admin does not correct these wrong answers?
upvoted 4 times

imvb88 3 years, 1 month ago
Selected Answer: B
The reporting process queries the metadata (not the documents) and use relational queries-> A, D out
C: wrong since secondary RDS node in MultiAZ setup is in standby mode, not available for querying
B: reporting using a Replica is a design pattern. Using Aurora is an exam pattern.
upvoted 5 times

WherecanIstart 3 years, 2 months ago
Selected Answer: B
B is right..
upvoted 2 times

Maximus007 3 years, 2 months ago
Selected Answer: B
While both B&D seems to be a relevant, ChatGPT suggest B as a correct one
upvoted 2 times

cegama543 3 years, 3 months ago
Selected Answer: B

Option B (Set up a new Amazon Aurora PostgreSQL DB cluster that includes an Aurora Replica. Issue queries to the Aurora Replica to generate the reports) is the best option for speeding up the reporting process for a three-tier web application that includes a PostgreSQL database storing metadata from documents, while not impacting document modifications or additions, with the least amount of change to the application code.

upvoted 3 times

UnluckyDucky 3 years, 3 months ago

Selected Answer: B

"LEAST amount of change to the application code"

Aurora is a relational database, it supports PostgreSQL and with the help of read replicas we can issue the reporting proccess that take several hours to the replica, therefore not affecting the primary node which can handle new writes or document modifications.

upvoted 2 times

Ashukaushal619 3 years, 3 months ago

its D only ,recorrected

upvoted 2 times

Murtadhaceit 2 years, 6 months ago

DynamoDB is no SQL. A and D are out!

upvoted 2 times

Ashukaushal619 3 years, 3 months ago

Selected Answer: B

bbbbbbbb

upvoted 2 times

Question #382 *Topic 1*

A company has a three-tier application on AWS that ingests sensor data from its users' devices. The traffic flows through a Network Load Balancer (NLB), then to Amazon EC2 instances for the web tier, and finally to EC2 instances for the application tier. The application tier makes calls to a database.

What should a solutions architect do to improve the security of the data in transit?

A. Configure a TLS listener. Deploy the server certificate on the NLB. **Most Voted**

B. Configure AWS Shield Advanced. Enable AWS WAF on the NLB.

C. Change the load balancer to an Application Load Balancer (ALB). Enable AWS WAF on the ALB.

D. Encrypt the Amazon Elastic Block Store (Amazon EBS) volume on the EC2 instances by using AWS Key Management Service (AWS KMS).

Correct Answer: A

Community vote distribution

A (100%)

Comments

fruto123 Highly Voted 2 years, 3 months ago

Selected Answer: A

Network Load Balancers now support TLS protocol. With this launch, you can now offload resource intensive decryption/encryption from your application servers to a high throughput, and low latency Network Load Balancer. Network Load Balancer is now able to terminate TLS traffic and set up connections with your targets either over TCP or TLS protocol.

https://docs.aws.amazon.com/elasticloadbalancing/latest/network/create-tls-listener.html

https://exampleloadbalancer.com/nlbtls_demo.html

upvoted 24 times

imvb88 Highly Voted 2 years, 1 month ago

Selected Answer: A

security of data in transit -> think of SSL/TLS. Check: NLB supports TLS
https://docs.aws.amazon.com/elasticloadbalancing/latest/network/create-tls-listener.html

B (DDoS), C (SQL Injection), D (EBS) is for data at rest.

upvoted 18 times

TariqKipkemei Most Recent 1 year, 7 months ago

Selected Answer: A

secure data in transit = TLS

upvoted 2 times

Guru4Cloud 1 year, 9 months ago

Selected Answer: A

TLS provides encryption for data in motion over the network, protecting against eavesdropping and tampering. A valid server certificate signed by a trusted CA will provide further security.

upvoted 6 times

klayytech 2 years, 2 months ago

Selected Answer: A

To improve the security of data in transit, you can configure a TLS listener on the Network Load Balancer (NLB) and deploy the server certificate on it This will encrypt traffic between clients and the NLB. You can also use AWS Certificate Manager (ACM) to provision, manage, and deploy SSL/TLS certificates for use with AWS services and your internal connected resources1.

You can also change the load balancer to an Application Load Balancer (ALB) and enable AWS WAF on it. AWS WAF is a web application firewall that helps protect your web applications from common web exploits that could affect application availability, compromise security, or consume excessive resources3.

the A and C correct without transit but the need to improve the security of the data in transit? so he need SSL/TLS certificates

upvoted 4 times

Maximus007 2 years, 2 months ago

Selected Answer: A

agree with fruto123

upvoted 4 times